Foreword

Education is not just knowledge transfer, but a spark for curiosity and critical thinking. In line with NEP 2020, NCF 2023, and SCF–AP, this Teacher Handbook empowers teachers in this transformative journey. Teachers today are facilitators of learning, shaping future-ready students with values, skills, and competencies. This handbook offers practical guidance for lesson planning, assessment, pedagogy, and reflection.

It includes curricular goals, learning outcomes, and innovative teaching strategies. Resources like TLMs, science labs, clubs, and joyful learning techniques are featured. It aligns with the Academic Calendar for structured year-round teaching. This is your professional companion encouraging creativity, collaboration, and growth. Let classrooms be inclusive, engaging, and empowering learning spaces. Teachers, lead with purpose and inspire lifelong learning in your students. May this handbook support meaningful classroom transformation?

The Department of School Education, Andhra Pradesh acknowledges the unwavering commitment of educators, academic experts, and resource persons who contributed to the development of this Teacher Handbook

Together, let's shape the future of education in Andhra Pradesh.

– Department of School Education, Andhra Pradesh

Government of Andhra Pradesh, Amaravati
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EXAMINATION SHEDULE									
	FA1	FA2	SA1	FA3	FA4	SA2	Pre Final 1 For X Class	Pre Final 2 For X Class	Public Exams For X Class
Syllabus	Jun,July	Aug,Sep	June-Oct	Nov,Dec	Jan				
Time Slot	Aug-25 4-7	Oct-25 13-16	Nov-25 10-19	Jan-26 5-8	FEB-26 9-12	April-26 6-15	FEB-26 18-27	March-26 2-12	16.03.26 to 02.04.26

No of Working Days

MONTH	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	JANUARY	FEBRUARY	MARCH	APRIL	TOTAL
No of working days For General	15	26	21	18	23	24	25	19	23	21	18	233
No of Holidays For General	4	5	10	12	8	6	6	12	5	10	5	83
No of working days For Minority	15	26	21	21	23	24	20	21	23	21	18	233
No of Holidays For Minority	4	5	10	9	8	6	11	10	5	10	5	83

For General Schools

Dasara Holidays	24.09.2025 to 02.10.2025
Pongal Holidays	10.01.2026 to 18.01.2026

For Minority Schools

Dasara Holidays	24.09.2025 to 02.10.2025
Pongal Holidays	10.01.2026 to 15.01.2026
Christmas Holidays	21.12.2025 to 28.12.2025

YEAR PLAN

S.NO	NAME OF THE CHAPTER	Month	No. of Instructional Periods Allocated	No. of Non-Instructional Periods Allocated	Total No of Periods
1	Motion	Jun	9	4	**3-30**
2	Matter in or surroundings	July	16	7	**31-65**
3	Force and Laws of Motion	Aug	11	4	**66-91**
4	Is matter Around us pure	Sep	11	5	**92-118**
5	Gravitation	Oct	12	4	**119-148**
6	Atoms and Molecules	Nov	9	3	**149-176**
7	Work and Energy	Dec	14	4	**177-214**
8	**Structure of atom**	Jan	10	3	**215-238**
9	Sound	Feb	10	6	**239-275**

June 2025

No bag Day

Cluster Complex

Teacher Resources

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1 Sunday	2	3	4	5	6	7 Bakr Eid
8 Sunday	9	10	11	NI 12 Reopening day	13 NI	14 Second Saturday
15 Sunday	16 1.1	17 1.2	18 1.3	19 1.4	20 1.5	21 International Yoga Day No Bag Day
22 Sunday	23 1.6	24 1.7	25 1.8	26 1.9	27 NI	28 No Bag Day Mega PTM
Sunday 29 National Statistics Day	30 NI					

Week1:

Week 2:

Week 3: 1.1 – 1.5

Week 4: 1.6 – 1.9

Week 5:

↓ TEACHER'S NOTES ↓

Chapter: 1. Motion

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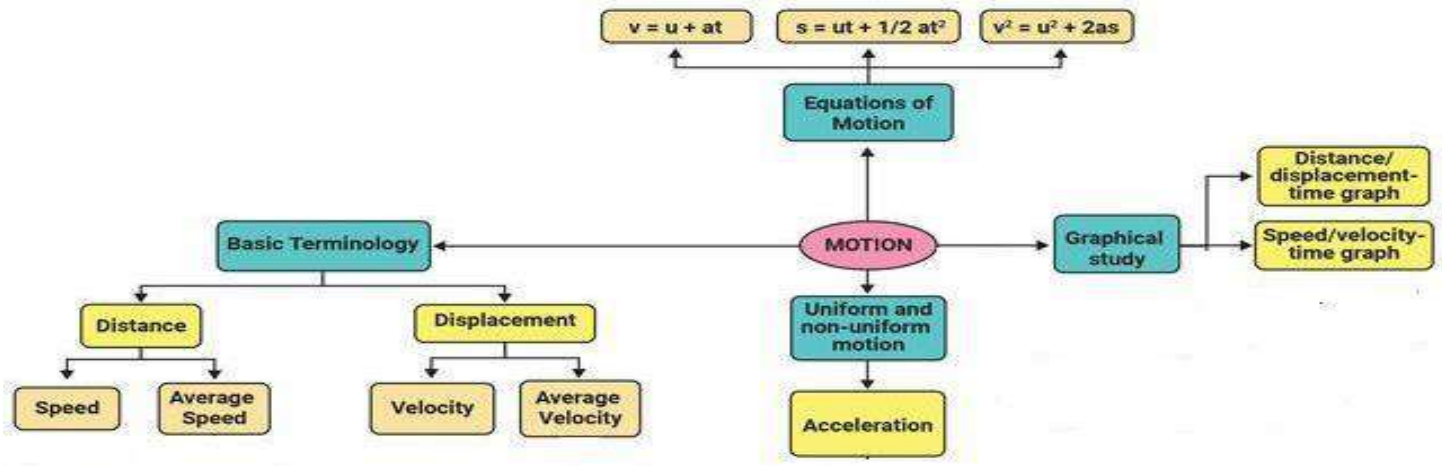
Learning Outcomes:

1. Differentiate distance, displacement and speed, velocity.
2. Calculate distance, displacement, speed and velocity from the given data.
3. Analyse the velocity - time graphs.
4. Interpret graphs and figures to compute distance, speed, and acceleration of the objects in motion.
5. Derive formula and equations of motion.

Prior Concepts/Skills:

1. How many types of motions are based on the path taken by the bodies in motion?
2. What is the rest of the object?
3. What is the S.I unit of speed?
4. How much distance from your home to school?

Concept Map



Synopsis

- **Types of motion:** Uniform motion and Non-uniform motion
- **Uniform and Non-uniform motion:** If an object covers equal distance in equal intervals of time.
- **Non-uniform motion:** If an object covers unequal distance in equal intervals of time (or) If an object covers equal distance in unequal intervals of time.
- **Uniform and Non-uniform motion:** The motion of an object could be uniform or non-uniform depending on whether its velocity is constant or changing
- **Speed:** The speed of an object is the distance covered per unit time
- **Velocity:** velocity is the displacement per unit time.
- **Average speed:** The total distance travelled by the body divided by the total time taken to cover distance.
- **Average velocity:** The total displacement travelled by the body divided by the total time taken to cover displacement.
- **Acceleration:** The acceleration of an object is the change in velocity per unit time.
- **Graphical representation:** Uniform and non-uniform motions of objects can be shown through graphs.
- **Equation of motion:** 1) $v = u + at$ 2) $s = ut + \frac{1}{2} at^2$ 3) $2as = v^2 - u^2$
- **Uniform motion:** If an object moves in a circular path with uniform speed, its motion is called uniform circular motion

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
1.1	1.1 Describing motion 1.1.1 Motion along a straight line	explains the different motions along a straight line	
1.2	1.1. 2 Uniform and Non uniform motion	differentiate uniform and non-uniform motions	
1.3	1.2 Measuring the rate of motion	measures the speed of a body	

1.4	1.2.1 Speed with direction	calculates the velocity of a body	
1.5	1.3 Rate of change of velocity	understand the concept of acceleration	
1.6	1.4,1 Distance-time graphs	interprets distance time graphs	
1.7	1.4.2 Velocity-time graphs	analyse velocity time graphs	
1.8	1.5 Equations of motion	derive formulas of equations of motion	
1.9	1.6 Uniform circular motion	understand the concept of uniform circular motion	

PERIOD Number: 1.1

TOPIC: 1.1 Describing Motion : 1.1.1 Motion along a straight line

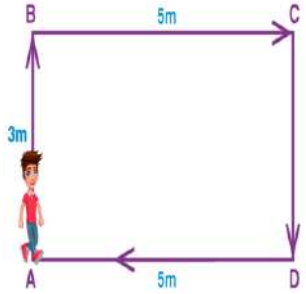
Learning Outcome:

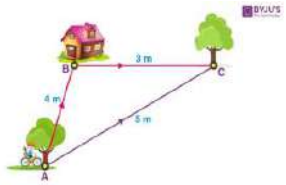
Students will be able to explains the different motions along a straight line

Teaching Learning Process

I DO	WE DO	YOU DO
Motion An object may appear to be moving for one person and stationary for some other. For the passengers in a moving bus, the roadside trees appear to be moving backwards. A person standing on the road–side perceives the bus along with the passengers as moving. However, a passenger inside the bus sees his fellow passengers to be at rest. What do these observations indicate? Origin	• Is the life possible without moving objects? • Imagine how would be the life without moving objects? • Can you give some examples for moving objects? • What do you think about motion? • Is the earth in motion or at rest? ✔ Let us assume that a school in a village is 2 km north of the railway station. ✔ We have specified the position of the school with respect to the railway station. ✔ In this example, the railway station is the reference point.	1. What are the moving objects? 2. What are the non-moving objects? ✔ Therefore, to describe the position of an object we need to specify a reference point called the origin.

We describe the location of an object by specifying a reference point. Let us understand this by an example. Let us assume that a school in a village is 2 km north of the railway station. We have specified the position of the school with respect to the railway station. In this example, the railway station is the reference point. We could have also chosen other reference points according to our convenience. Therefore, to describe the position of an object we need to specify a reference point called the origin.





✔ We could have also chosen other reference points according to our convenience.

Q1. Can you give some examples for the objects in motion and at rest?

Q2. Is our class room in motion or at rest? Explain your answer?

Q3. How does a fish move ?

Q4. Is the motion of a fan and motion of a bicycle is same ?

Q5. Does the bus move in a line?

A----------B-----------*C----------D*

In the above path of motion

Q6. What is the total length from A to D ?

Students observe that

" The total length of the path from initial to final position is called distance."

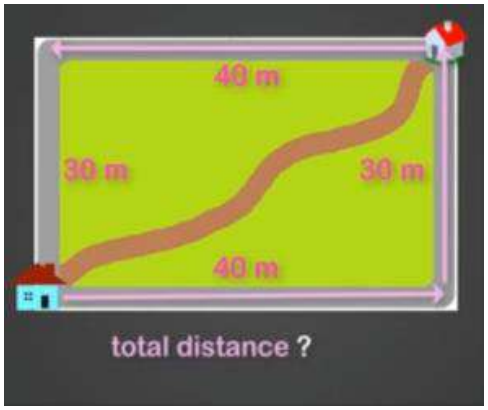
Activity:

- One of the student is asked to do an activity, to measure the distance in feet's while

  Walking beside the wall of your class room from one side to remaining three walls and observe the actual distance of the path and shortest distance of that path also the shortest length of the path is termed as displacement:

Q1. What do you think about distance?

Q2. What do you know about displacement?



Q3. What is the distance in the above path?

Q4. What is the displacement in the above path?

Q5. Throw a ball into the air and observe its positions, draw the paths of distance and displacements?

Q6. Are the distance and displacement same in the given path point "A and C "?

Q7. When does the displacement

	Q5. What is the distance between point A and C? Q6. What is the displacement in this path from A to C? ”Distance never become zero.” When does displacement become zero?	becomes zero? Q8. What are the differences between distance and displacement? Q9. Is the direction play make any difference in the distance or Displacement?

Summary	Factual Questions	CFU	Home Work
Students will be able to explain the different motions along a straight line.	1. An object has moved through a distance. Can it have zero displacement? If yes, support your answer with an example.	1. What is rest? 2. What is motion? 3. What type of motions?	1.A farmer moves along the boundary of a square field of side 10 m in 40 s. What will be the magnitude of displacement of the farmer at the end of 2 minutes 20 seconds from his initial position? 2. Which of the following is true for displacement? (a) It cannot be zero. (b) Its magnitude is greater than the distance travelled by the object.

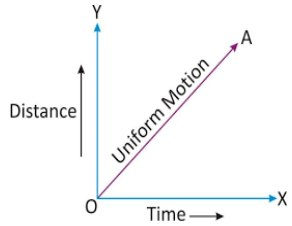
PERIOD Number: 1.2

TOPIC: 1.1.2 Uniform and Non-uniform motion

Learning Outcome: Students will be able to differentiate uniform and non-uniform motions

Teaching Learning Process

I DO	WE DO	YOU DO
Uniform Motion Consider an object moving along a straight line. Let it travel 5 m in the first second, 5 m more in the next second, 5 m in the third	Recall the previous period knowledge. Q1. What is the motion? Q2. What is the difference between motion of an earth and motion of a car?	Q1. What do you know about uniform and non uniform motion?

second and 5 m in the fourth second. In this case, the object covers 5 m in each second. As the object covers equal distances in equal intervals of time, it is said to be in uniform motion. The time interval in this motion should be small. Non-Uniform Motion In our day-to-day life, we come across motions where objects cover unequal distances in equal intervals of time, for example, when a car is moving on a crowded street or a person is jogging in a park. These are some instances of non-uniform motion.	Q3. When do the distance and displacement become equal? Q4. When does the displacement become zero? Q5. Does a bus move with same speed from starting to ending in it's journey? Q6. Do you think that speed of a running fan changes time to time? Teacher show the image and ask the following questions.  Q.What do you think about uniform motion? Q.What is the distance covered by to object A, B in same time of interval? Q.Is that distance is changing with same interval of time? Activity :	Q2. When you travel in a bus, what do you observe about its distance covered in same intervals of time? Q3. How is the speed of an object calculated

Time	9.30 am	9.45 am	10.00 am	10.15 am	10.30 am
Distance travelled By “A”	1	20	30	40	50
Distance travelled by “B”	1	19	23	35	37

Q.What is the distance of object “A “at 9.45am?

Q.What is the distance of object B” at 9.45am?

Q. What do you observe in the distance covered object A and B in the same interval of time?

Q.What do you observe that the difference between distance travelled by A and B in that table.

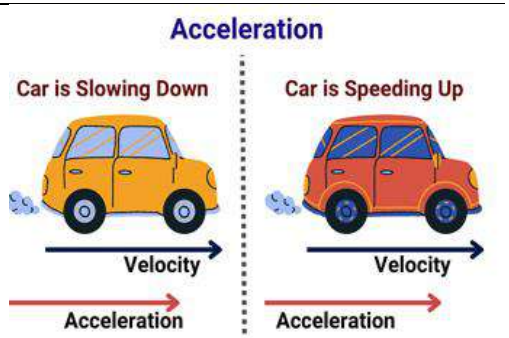
Q.What is the speed of object B?

The average speed of an object is obtained by dividing the total distance travelled by the total time taken.

Q.What is the relation between distance and time?

Q.What is the unit SI unit of speed?

Application: A bus travels a distance 200 km in 5 hours. What is the speed of that bus?



Q4. What is the relation between distance, speed and time?

Q4. What is the average speed of an object?

Q5. Calculate the speed of a train if it covers 50km in first hour and 100 km in one hour 30 minutes?

Q6. From the given table, identify that is it uniform motion or non-uniform motion.

Distance	20m	40m	60m	80m
Time	5s	10s	15s	20s

	Sol. Students apply the formula of speed for calculating as discussed as above they get speed 40km/h as answer.	

Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate uniform and non-uniform motions	A particle moves over three quarters of a circle of radius r. What is the magnitude of its displacement?	1. What is the SI unit of displacement? 2. Can the displacement be greater than the distance travelled by an object?	1. What is meant by uniform motion? Give an example

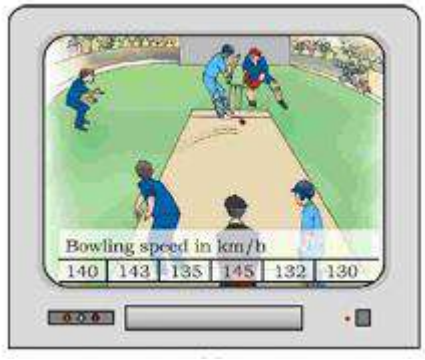
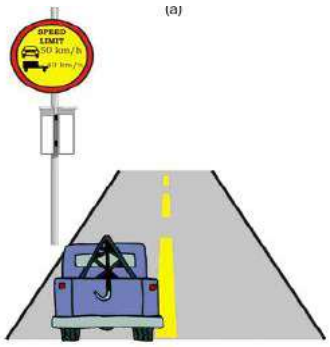
PERIOD Number: 1.3

TOPIC: 1.2 Measuring the rate of motion

Learning Outcome:

Students will be able to measure the speed of a body

Teaching Learning Process

I DO	WE DO	YOU DO
Different objects may take different amounts of time to cover a given distance. Some of them move fast and some move slowly. The rate at which objects move can be different. Also, different objects can move at the same rate. One of the ways of measuring the rate of motion of an object is to find out the distance travelled by the object in unit time. This quantity is referred	  ✔ Different objects may take different amounts of time to	First fig. shows the bowling speed is 143 km h^{-1} What do you understand from the signboard?

to as speed. The SI unit of speed is metre per second. This is represented by the symbol m s–1 or m/s.The other units of speed include centimetre per second (cm s–1) and kilometre per hour (km h–1). To specify the speed of an object, we require only its magnitude. The speed of an object need not be constant. In most cases, objects will be in non-uniform motion. Therefore, we describe the rate of motion of such objects in terms of their average speed. The average speed of an object is obtained by dividing the total distance travelled by the total time taken. That is, average speed = Total distance travelled Total time taken	cover a given distance. ✔ Some of them move fast and some move slowly. ✔ The rate at which objects move can be different. ✔ Also, different objects can move at the same rate. ✔ One of the ways of measuring the rate of motion of an object is to find out the distance travelled by the object in unit time. ✔ This quantity is referred to as speed. ✔ The SI unit of speed is metre per second. T ✔ his is represented by the symbol m s^{-1} or m/s. ✔ The other units of speed include centimetre per second (cm s^{-1}) and kilometre per hour (km h^{-1}). ✔ To specify the speed of an object, we require only its magnitude.., ✔ we describe the rate of motion of such objects in terms of their average speed. ✔ The average speed of an object is obtained by dividing the total distance travelled by the total time taken $\text{average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$ $v = \frac{s}{t}$	• The speed of an object need not be constant. • In most cases, objects will be in non-uniform motion.

Summary	Factual Questions	CFU	Home Work
Students will be able to	An object travels 16 m in 4 s and	1.Define the term average	Define the term speed. What are its

the speed of a body	then another 16 m in 2 s. What is the average speed of the object?	speed? 2. Give one example of Ono-uniform motion?	various units? Is it a scalar or vector quantity?

PERIOD Number: 1.4

TOPIC: 1.2.1 Speed with direction

Learning Outcome:

Students will be able to calculates the velocity of acceleration

Teaching Learning Process

I DO	WE DO	YOU DO
The rate of motion of an object can be more comprehensive if we specify its direction of motion along with its speed. The quantity that specifies both these aspects is called velocity. **Example 1.2** The odometer of a car reads 2000 km at the start of a trip and 2400 km at the end of the trip. If the trip took 8 h, calculate the average speed of the car in km h^{-1} and m s^{-1}. **Example 7.3** Usha swims in a 90 m long pool. She covers 180 m in one minute by	• Velocity is the speed of an object moving in a definite direction. • The velocity of an object can be uniform or variable. • It can be changed by changing the object's speed, direction of motion or both. • When an object is moving along a straight line at a variable speed, we can express the magnitude of its rate of motion in terms of average velocity. • It is calculated in the same way as we calculate average speed. • In case the velocity of the object is changing at a uniform rate, then average velocity is given by the arithmetic mean of initial velocity and final velocity for a given period of time.	What is velocity? What is formula of average velocity? What is the SI Unit of average velocity? $v_{av} = \frac{s}{t} = \frac{400\ km}{8\ h}$ $= 50\ km\ h^{-1}$ $= 50\ \frac{km}{h} \times \frac{1000\ m}{1\ km} \times \frac{1\ h}{3600\ s}$ $= 13.9\ m\ s^{-1}$

swimming from one end to the other and back along the same straight path. Find the average speed and average velocity of Usha.	$\text{average velocity} = \frac{\text{initial velocity} + \text{final velocity}}{2}$ Mathematically, $v_{av} = \frac{u + v}{2}$ Distance, s = 2400 km – 2000 km = 400 km Time elapsed, t = 8 h Total distance covered by Usha in 1 min is 180 m. Displacement of Usha in 1 min = 0 m	$\text{Average speed} = \frac{\text{Total distance covered}}{\text{Total time taken}}$ $= \frac{180\,\text{m}}{1\,\text{min}} = \frac{180\ \text{m}}{1\,\text{min}} \times \frac{1\,\text{min}}{60\,\text{s}}$ $= 3\ \text{m s}^{-1}$ $\text{Average velocity} = \frac{\text{Displacement}}{\text{Total time taken}}$ $= \frac{0\,\text{m}}{60\ \text{s}}$ $= 0\ \text{m s}^{-1}$

Summary	Factual Questions	CFU	Home Work
ents will be able to calculates elocity of acceleration	1.What does the odometer of an automobile measure? 2. What does the path of an object look like when it is in uniform motion?	Under what condition(s) is the magnitude of average velocity of an object equal to its average speed?	1. Distinguish between speed and velocity. 2. During an experiment, a signal from a spaceship reached the ground station in five minutes. What was the distance of the spaceship from the ground station? The signal travels at the speed of light, that is, $3x10^8$ m s^{-1}

PERIOD Number: 1.5

TOPIC: 1.3 Rate of change of velocity

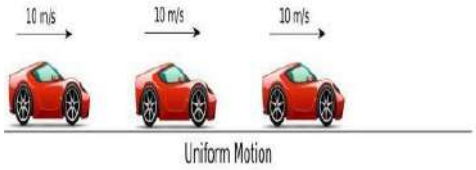
Learning Outcome:

Students will be able to understand the concept of acceleration

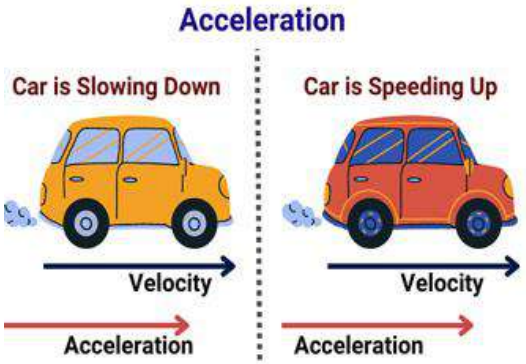
Teaching Learning Process

I DO	WE DO	YOU DO
Recall the previous period Knowledge. Q. Do you remember uniform motion and non uniform motion? Q. What is the velocity change in uniform motion? Q. In non uniform motion, does the velocity keep constant? Q. What is the unit of speed in SI sytem?	Presentation of key concept Q. What do you say about change of velocity? Q. When you are travelling in bus or a car, you observe the driver place near by his legs, there are three peddles at the legs of driver, do you know what are they? Q. Do you observe how does a driver change the speed of vehicle? Q. When you go by a bike with your parents ,frequently speed of bike changes do you observe , how it happens? Q. Wat do you mean by rate of change of velocity? Q. If the change of velocity per second is 50m/s,what is its acceleration? Q. What is the ratio of change of velocity to the time taken? Q. Do you derive the formula of acceleration?	Q1. What do you about rate change of displacement? And the rate of change of velocity? Q2. What is the unit of acceleration? Q3. Does the acceleration of freely falling body become zero? Q4.When does acceleration becomes negative? Q5. What is the relation between v, u and t? Q6. What is the acceleration of an object if it starts at appoint with velocity “p” and after time “x” it has got velocity ‘Q”.

Acceleration







Q. What type of acceleration is observed when a car arriving to stopping position?

Q. What happens to the acceleration when a car is getting increased its speed?



Uniform and non uniform accelerated motion

Q. When a ball thrown up ,what about its acceleration?

Q. When does an acceleration becomes zero?

Q. What is retardation?

Q. What is the unit of acceleration?

Q. Is the acceleration a vector or scalar?

Q7. Give some examples where the acceleration get positive value?

Q8. Mention some examples for non-uniform acceleration?

Q9. What do you mean by the term retardation or negative acceleration? give two examples of such a motion?

Q10. What is the acceleration of a body moving with uniform velocity?

Q11. Differentiate acceleration from velocity?

Summary	Factual Questions	CFU	Home Work
Students will be able to understand the concept of acceleration	State which of the following situations are possible and give an example for each of these: (a) an object with a constant acceleration but with zero velocity (b) an object moving with an acceleration but with uniform speed.	When will you say a body is in (i) uniform acceleration? (ii) non-uniform acceleration?	1. Define acceleration? Write its formula and SI unit. 2. A bus decreases its speed from 80 km h^{-1} to 60 km h^{-1} in 5 s. Find the acceleration of the bus.

PERIOD Number: 1.6

TOPIC: 1.4.1 Distance-Time graphs

Learning Outcome:

Students will be able to interprets distance time graphs

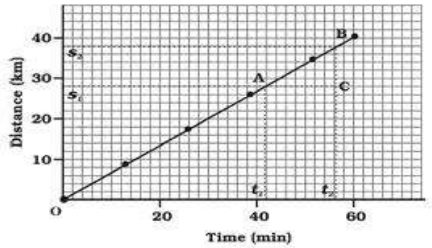
Teaching Learning Process

I DO	WE DO	YOU DO
Graphs provide a convenient method to present basic information about a variety of events.	The change in the position of an object with time can be represented on the distance-time graph adopting a convenient scale of choice. ✔ In this graph, time is taken along the x–axis ✔ In this graph, distance is taken along the y-axis. ✔ Distance-time graphs can be employed under various conditions where objects move with uniform speed, non-uniform speed, remain at rest etc.	

The change in the position of an object with time can be represented on the distance-time graph adopting a convenient scale of choice. In this graph, time is taken along the *x*–axiS and distance is taken along the *y*-axis. Distance-time graphs can be employed under various conditions where objects move with uniform speed, non-uniform speed, remain

at rest etc

Explanation of Distance – Time graphs



What is uniform speed?

Shows that distance travelled by the object is directly proportional to time taken.

Distance-Time graph:

Above is uniform speed, a graph of distance travelled against time is a straight line

- The portion OB of the graph shows that the distance is increasing at a uniform rate.
- Note that, you can also use the term uniform velocity in place of uniform speed if you take the magnitude of displacement equal to the distance travelled by the object along the y-axis.
- We can use the distance-time graph to determine the speed of an object.
- To do so, consider a small part AB of the distance-time graph.
- Draw a line parallel to the x-axis from point A and another line parallel to the y-axis from point B.
- These two lines meet each other at point C to form a

when an object travels equal distances in equal intervals of time, it moves with uniform speed.

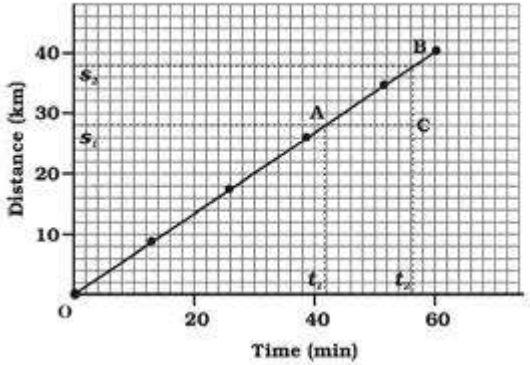


Table 7.2: Distance travelled by a car at regular time intervals

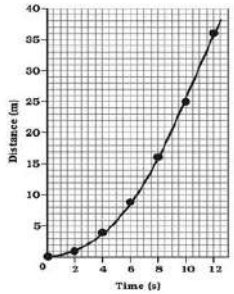
Time in seconds	Distance in metres
0	0
2	1
4	4
6	9
8	16
10	25
12	36

triangle ABC.

- On the graph, AC denotes the time interval ($t_2 - t_1$) while BC corresponds to the distance ($s_2 - s_1$). We can see from the graph that as the object moves from the point A to B, it covers a distance ($s_2 - s_1$) in time ($t_2 - t_1$). The speed, v of the object, therefore can be represented as

$$v = \frac{s_2 - s_1}{t_2 - t_1}$$

How plot the distance-time graph for accelerated motion



The nature of this graph shows non linear variation of the distance travelled by the car with time. Thus, the graph represents motion with non-uniform speed.

Taken

Time values on x-axis

Distance values on y-axis

Scale: X -axis (1 cm =2 units)

Y-axis (1 cm=5 units)

Summary	Factual Questions	CFU	Home Work
Students will be to interprets distance time graphs	What is nature of the displacement – time graph of a body moving with constant acceleration?	What is distance-time graph of a body? Draw distance – time graphs for s stationary body	1. What is the nature of the distance-time graphs for uniform and non-uniform motion of an object?

PERIOD Number: 1.7

TOPIC: 1.4.2 Velocity-Time graphs

Learning Outcome: Students will be able to analyse velocity time graphs

Teaching Learning Process

I DO	WE DO	YOU DO
Velocity-time graph	**Introduction** ● What do you know about Velocity of a body? ● What is meant by Uniform Velocity? The variation in velocity with time for an object moving in a straight line can be represented by a velocity-time graph. ✔ In this graph, time is represented along the x-axis. ✔ In this graph, velocity is represented along the y-axis. ✔ If the object moves at uniform velocity, the height of its velocity-time graph will not change with time. ✔ It will be a straight line parallel to the x-axis. ✔ shows the velocity-time graph for a car moving with uniform velocity of 40 km h^{-1}. ✔ We know that the product of velocity and time give displacement of an object moving with uniform velocity.	So, the distance s moved by the car in time ($t_2 - t_1$) can be expressed as $s = AC \times CD$

Velocity-time graph explanation

- ✔ The area enclosed by velocity-time graph and the time axis will be equal to the magnitude of the displacement.
- ✔ To know the distance moved by the car between time t_1 and t_2 , draw perpendiculars from the points corresponding to the time t_1 and t_2 on the graph.
- ✔ The velocity of 40 km h^{-1} is represented by the height AC or BD and the time $(t_2 - t_1)$ is represented by the length AB.

Table 7.3: Velocity of a car at regular instants of time

Time (s)	Velocity of the car ($m\ s^{-1}$)	($km\ h^{-1}$)
0	0	0
5	2.5	9
10	5.0	18
15	7.5	27
20	10.0	36
25	12.5	45
30	15.0	54

s = area ABCDE
= area of the rectangle ABCD + area of the triangle ADE

$$= AB \times BC + \frac{1}{2}(AD \times DE)$$

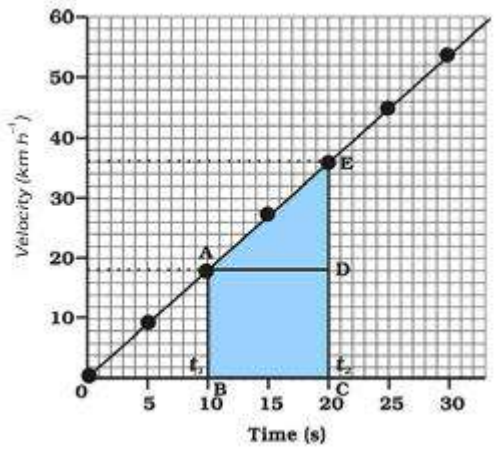
$= [(40\ km\ h^{-1}) \times (t_2 - t_1)\ h]$

$= 40\ (t_2 - t_1)\ km$

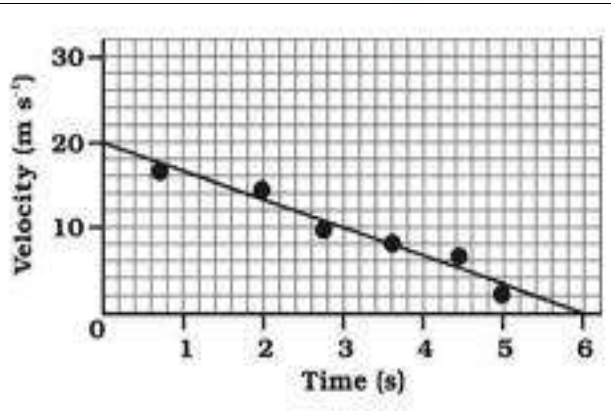
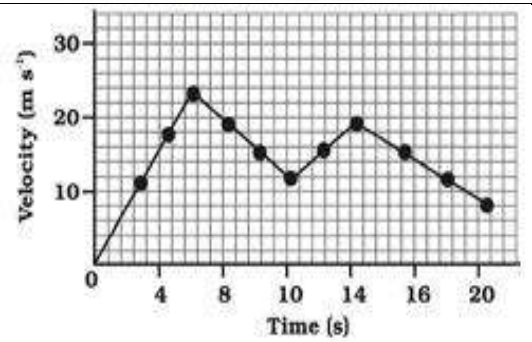
= area of the rectangle ABDC (shaded)

x -axis □ time

y-axis □ velocity



We can also study about uniformly accelerated motion by plotting its velocity time graph.

velocity-time graph that represents the motion of an object whose velocity is decreasing with time .		 velocity-time graph representing the non-uniform variation of velocity of the object with time

Summary	Factual Questions	CFU	Home Work
ents will be able to analyse velocity graphs	1. What can you say about the motion of an object if its speed time graph is a straight line parallel to the time axis?	1. What is the quantity which is measured by the area occupied below the velocity-time graph?	1. Mention the use of a velocity - time graph. 2. Draw velocity-time graphs for the object is at rest.

PERIOD Number: 1.8

TOPIC: Equation of motion

Learning Outcome: Students will be able to derive formulas of equation of motion

Teaching Learning Process

I DO	WE DO	YOU DO
Symbols of physical quantities	Initial velocity	u
	Final velocity	v
	Time taken	t
	Distance travelled	s
	Uniform acceleration	a
	When an object moves along a straight line with uniform acceleration, it is possible to relate its velocity, acceleration during motion and the distance covered by it in a certain time interval by a set of equations known as the equations of motion.	velocity-time relation position-time relation relation between the position and the velocity,
	$v = u + at$ --------(1)	
	$s = ut + \frac{1}{2} at^2$ -------(2)	
	$2as = v^2 - u^2$ -------(3)	
Example 1.5	u = 0	
A train starting from rest attains a velocity of 72 km h^{-1} in 5 minutes. Assuming that the acceleration is uniform, find (i) the acceleration and (ii) the distance travelled by the train	v = 72 km h^{-1} = 20 m s^{-1}	
	t = 5 minutes = 300 s.	

for attaining this velocity.

Example 1.6

A car accelerates uniformly from 18 km h^{-1} to 36 km h^{-1} in 5 s. Calculate (i) the acceleration and (ii) the distance covered by the car in that time.

$$a = \frac{(v-u)}{t}$$

$$= \frac{20\ m\ s^{-1} - 0\ m\ s^{-1}}{300\ s}$$

$$= \frac{1}{15}\ m\ s^{-2}$$

u = 18 km h^{-1} = 5 m s^{-1}
v = 36 km h^{-1} = 10 m s^{-1} and
t = 5 s .

$$a = \frac{v-u}{t}$$

$$= \frac{10\ m\ s^{-1} - 5\ m\ s^{-1}}{5\ s}$$

$$= 1\ m\ s^{-2}$$

$2as = v^2 - u^2 = v^2 - 0$
Thus,

$$s = \frac{v^2}{2a}$$

$$= \frac{(20\ m\ s^{-1})^2}{2 \times (1/15)\ m\ s^{-2}}$$

= 3000 m
= 3 km

$$s = ut + \frac{1}{2}at^2$$

$$= 5\ m\ s^{-1} \times 5\ s + \frac{1}{2} \times 1\ m\ s^{-2} \times (5\ s)^2$$

= 25 m + 12.5 m

= 37.5 m

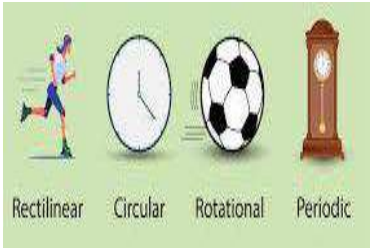
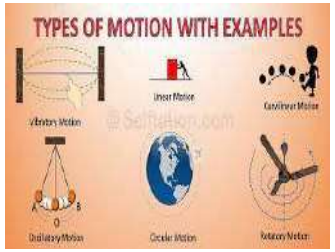
Summary	Factual Questions	CFU	Home Work
ents will be able to derive ulas of equation of motion	A trolley, while going down an inclined plane, has an acceleration of 2 cm $s^{-2.}$ What will be its velocity 3 s after the start.	1. Write the equations of motion.	A motorboat starting from rest on a lake accelerates in a straight line at a constant rate of 3.0 m s^{-2} for 8.0 s. How far does the boat travel during this time?

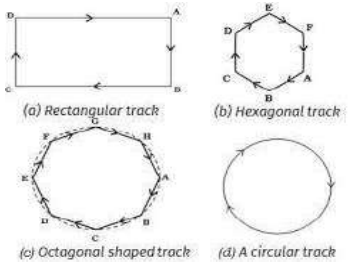
PERIOD Number: 1.9

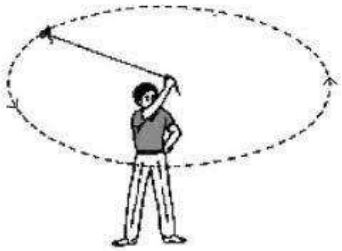
TOPIC: 1.6 Uniform circular motion

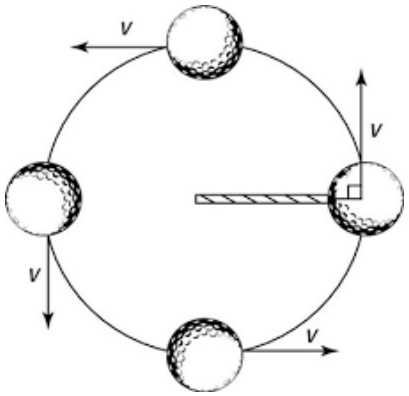
Learning Outcome: Students will be able to understand the concept of uniform circular motion

Teaching Learning Process

I DO	WE DO	YOU DO
    	• **Introductory activity** • Teachers shows different kinds of motions and asks the students to identify the type of motion. • What is the difference between speed and velocity? • The quantity which has only magnitude is called? • The quantity which has both magnitude and direction is called? • Which quantity is scalar or vector among Speed and Velocity? • If the velocity of an object changes , then body acquires which quantity? • Is the change of velocity is due to change in its magnitude or the direction or both? • Do you state one example in which object can not change magnitude of velocity but only its direction? • We will know about the motion of an object whose magnitude of velocity does not change but changes its direction continuously ,which we call as **Uniform Circular Motion.**	







- Teacher display a closed rectangular track ABCD on IFP.
- Let us assume that athlete runs at a uniform speed on the straight parts AB,BC,CD and DA of the track.
- Teacher may ask a student to run on a closed path drawn on the floor in the class room /out side the room
- Is he need to change his direction at any points on ABCD track?
- To complete one round on the ABCD track, how many times he changes his direction ?
- Suppose instead of a rectangular track, if the athlete runs along a hexagonal shaped track ABCDEF (as shown on IFP), how many times the athlete has to change his direction?
- Identify the points where direction of person changes ?
- If the track is a rectangular Octagon then?
- From these examples , we can conclude that as the number of sides of the track increases , the athlete has to take turns more and more often.
- What happen if we go on increasing the number of sides of the track indefinitely?
- The shape of the track approaches the shape of a circle and the length of each of the sides will decrease to a point.
- If the athlete moves with the velocity of constant magnitude along the circular path, the only change in its

Q: A cyclist goes around a track for every 2 minutes. If the radius of the track is 105 m, find his speed?

Q: A particle is moving in a circular path of radius r. Its displacement after moving through half the circle would be :

A .Zero

B. r

C. 2r

D. r2

	velocity is due to the change in the direction of motion. ● The motion of the athlete moving along a circular path is therefore, an example of an accelerated motion. ● If the radius of the circle is "r", what is the circumference of the circle? ● If the athlete takes 't' seconds to go once around the circular path of radius 'r', what is the distance travelled by the person? ● What is the relation between Speed, Velocity and Time? ● Speed = Distance covered / time $V = 2\pi r / t$ ● When an object moves in a circular path with uniform speed, its motion is called Uniform Circular Motion.	Q: If a particle moves in a circle with constant speed, its velocity : a particle moves in a circle with constant speed, its velocity : A. remains constant B. changes in magnitude C. changes direction changes both in magnitude and direction

Summary	Factual Questions	CFU	Home Work
Students will be able understand the concept of uniform circular motion	Why is the motion of an athlete moving along the circular path, with a constant speed, an accelerated motion?	1.Define uniform circular motion. 2. Give examples of uniform circular motion.	An artificial satellite is moving in a circular orbit of radius 42250 km. Calculate its speed if it takes 24 hours to revolve around the earth.

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson: 1. Motion		Class: 9	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
1.1	1.1 Describing motion 1.1.1 Motion along a straight line				
1.2	1.1. 2 Uniform and Non-uniform motion				
1. 3	1.2 Measuring the rate of motion				
1. 4	1.2.1 Speed with direction				
1. 5	1.3 Rate of change of velocity				
1. 6	1.4,1 Distance-time graphs				
1. 7	1.4.2 Velocity-time graphs				
1. 8	1.5 Equations of motion				
1.9	1.6 Uniform circular motion				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?
Head Teacher's Signature	Teacher's Signature
Head Teacher's Suggestions:	

Teacher Notes:

July 2025

No bag Day

Cluster Complex

Teacher Resources

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		1 NI	2 2.1	3 2.2	4 NI	5 No Bag Day
6 Sunday Muharram	7 2.3	8 2.4	9 2.5	10 2.6	11 NI	12 Second Saturday
13 Sunday	14 2.7	15 2.8	16 2.9	17 2.9	18 NI	19 No Bag Day Cluster meeting
20 Sunday	21 2.10	22 2.11	23 2.11	24 2.12	24 NI	26 No Bag Day SMC Meeting
27 Sunday	28 2.13	29 2.14	30 NI	31 NI		

Week1: 2.1 – 2.2

Week 2: 2.3 – 2.6

Week 3: 2.7 – 2.9

Week 4: 2.10 – 2.12

Week 5: 2.13 – 2.14

↓ TEACHER'S NOTES ↓

Chapter -2

Chapter: 2. Matter In Our Surroundings

Learning Outcomes:

- Classify the matter based on their states as solids, liquids and gases.
- Conduct simple investigations to seek answers to queries like, "what is the effect of temperature on state of substances"?
- Draw a flowchart of interconversion of the three States of matter.
- Calculate using the given data such as conversion of Celsius scale to Kelvin scale and vice versa.
- Establish cause and effect relationship among evaporation with cooling effect.

Prior Concepts/Skills:

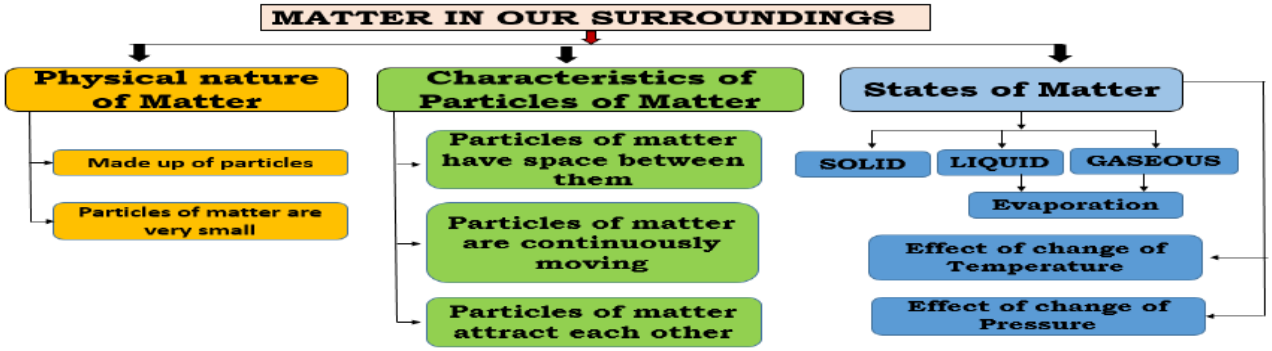
1. Give examples of matter in our surroundings?

2. What is the "Panch Tatva"?

3. In how many states does matter exist?

4. What is the hidden phenomena in dying wet clothes in outside?

Concept Map



Synopsis

- **Physical Nature of Matter**: Matter is made up particles, The particles of matter are very small
- **Characteristics of particles of matter:** Particles of matter have space between them, Particles of matter are continuously moving, Particles of matter attract each other.
- **States of matter:** The matter around us exists in three states—solid, liquid and gas.
- **Sublimation:** Sublimation is the change of solid state directly to gaseous state without going through liquid state.
- **Evaporation:** Evaporation is a surface phenomenon. Particles from the surface gain enough energy to overcome the forces of attraction present in the liquid and change into the vapour state.
- **Factors affecting evaporation:** The rate of evaporation depends upon the surface area exposed to the atmosphere, the temperature, the humidity and the wind speed.

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
2.1	2.1 Physical nature of matter 2.1.1 Matter is made up of particles Activity 2.1	classify matter based on physical state	
2.2	2.1.2 How small are these particles of matter? Activity 2.2	explain how small the particles of matter	
2.3	2.2. characterestics of particles of matter 2.2.1Particles of matter have space between them Activity 2.3 2.2.2 Particles of matter are continously moving Activity 2.4	understand the characterestics of particles of matter	

2.4	Activity 2.5	draws to a conclusion that the particles of matter are continously in motion.	
2.5	2.2.3 Particles of matter attract each other Activity 2.6 , 2.7 , 2.8	understand that the particles of matter attract each other	
2.6	2.3 States of matter 2,3.1 The solid state Activity 2.9	draw the flow chart of three states of matter	
2.7	2.3.2 The liquid state Activity 2.10	concludes that liquids have no definite shape , but they get the shape of container	
2.8	2.3.3 The gaseous state Activity 2.11	understand that gases are highly compressible	
2.9	2.4 Can matter change its state? 2.4.1 Effect of change of temperature Activity 2.12	explain the effect of temparature on states of matter	
2.10	Melting point , Boiling point , Latent heat of fusion	exaplain how boiling takes place and what is the boiling temperature	
2.11	Activity 2.13 Sublimation , Deposition	give examples for sublimation	
2.12	2.4.2 Effect of change of pressure	explain the effect of pressure on state of matter	
2.13	2.5 evaporation 2.5.1 Factors affecting evaporation	understand the factors depends on which evaporation depends	
2.14	2.5.2 How does evaporation cause cooling	establish cause and effect relation between evaporation with cooling effect	

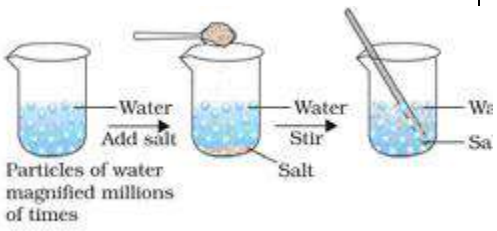
PERIOD Number: 2.1

TOPIC: 2.1 Physical nature of matter 2.1.1 Matter is made up of particles Activity 2.1

Learning Outcome: Classify matter based on physical state

Teaching Learning Process

I DO	WE DO	YOU DO
Introduction of matter concept . 1). What are their physical states? 2). What changes do you observe when you add lemon juice to water? 3). What change do you observe when you add sugar/ salt to water? 4). Does the taste change why? MATTER IS MADE UP OF PARTICLES -Activity 2.1	Teacher give a chalk piece (or any solid material), a bottle of water and a closed empty bottle to three groups of children and ask them to discuss about their observation on physical states of these three objects. 1). How do you make lemonade? What are the ingredients? ✔ Early Indian philosophers classified matter in the form of five basic elements — the “Panch Tatva” — air, earth, fire, sky and water. ✔ According to them everything, living or non-living, was made up of these five basic elements. ✔ Ancient Greek philosophers had arrived at a similar classification of matter. ✔ Modern day scientists have evolved two types of classification of matter based on their physical properties and chemical nature. **MATTER IS MADE UP OF PARTICLES -Activity 2.1** **Procedure:** ● Take a 100 mL beaker. ● Fill half the beaker with water and mark its level	1) How the different materials of matter are made up of?

	• Dissolve some salt/ sugar and stirrwith a glass rod. • Observe any change in water level. • What do you think has happened to the salt? • Where does it disappear? • Does the level of water change?	Salt dissolves in water There is no change in water level
Physical Nature of Matter	**Observation:** matter is made up of particles.	

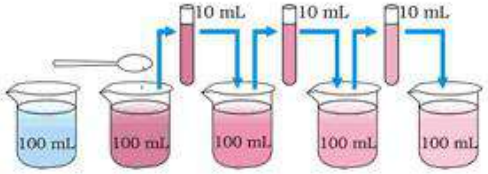
Summary	**Factual Questions**	**CFU**	**Home Work**
Students are able to Classify matter based on physical state. Like solids, liquids and gases	In what ways air can be considered as matter?	How many types of classification of matter by the modern scientists? What Panch Tatva?	Explain matter is matter is made up of particles with suitable activity.

PERIOD Number: 2.2 TOPIC: 2.1.2 How small are these particles of matter? Activity 2.2

Learning Outcome: Explain how small the particles of matter

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions. How small are these particles of matter?	What is the physical nature of matter? How many types of matter? **Activity-2.2** 1) Take 2–3 crystals of potassium permanganate and dissolve	Matter is made up of particles

	them in 100 ml of water. 2) Take out approximately 10 mL of this solution and put it into 90 ml of clear water. 3) Take out 10 ml of this solution and put it into another 90 mL of clear water. 4) Keep diluting the solution like this 5 to 8 times. 5) Is the water still coloured ? ✔ This experiment shows that just a few crystals of potassium permanganate can colour a large volume of water (about 1000 l). ✔ So we conclude that there must be millions of tiny particles in just one crystal of potassium permanganate, which keep on dividing themselves into smaller and smaller particles. ✔ The same activity can be done using 2 ml of Dettol instead of potassium permanganate. ✔ The smell can be detected even on repeated dilution. The particles of matter are very small they are small beyond our imagination!	The water remains coloured till the last dilution What is dilution?

Summary	**Factual Questions**	**CFU**	**Home Work**
Students are able to Explain how small the particles of matter, by conducting activity with their own words.	Which property of gases help us in detecting the leakage of LPG gas?	What is state of Potassium permanganate crystals?	Explain with the help of an activity which shows that particles of matter are very small.

PERIOD Number: 2.3 TOPIC: 2.2. Characteristics of particles of matter 2.2.1 Particles of matter have space between them Activity 2.3 2.2.2 Particles of matter are continuously moving Activity 2.4

Learning Outcome: Understand the characteristics of particles of matter

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions. 1) What is matter? **The characteristics of particles of matter** Let us learnt one by one. We shall discuss and proved in previous activities Particles of matter have space between them	1. How matter is classified? 2. What happens when salt or sugar dissolved in water? 3. Tell where we can waive our hand freely i.e. in water or air? 4) Students respond and then we start the topic by raising the question why it is difficult to waive our hand freely in water? 5) How we can waive our hand freely in air? **ACTIVITY 2.3** ✔ Put an unlit incense stick at a corner of your classroom. ✔ How close you have to go near it to sense its smell? ✔ Now light the incense stick. What do you observe? ✔ Do you get the smell sitting at a distance? ✔ Record your observations? **ACTIVITY 2.4** o Take two glass beakers and fill them with water. o Put a drop of blue or red ink slowly and carefully along the sides of the first beaker and honey in the same way in the	• We have to go quite close to the incense stick to get its smell • Now the smell of incense can be felt even sitting at a distance • The particles of incense move faster at higher temperature. • The colour of ink starts spreading soon • The honey drops slowly disappear • The colour of ink spreads evenly

	second beaker. o Leave them undisturbed at the corner of the classroom. o Record your observations. o What do you observe immediately after adding a drop of honey? o How many seconds or minutes does it take for the ink to spread evenly throughout the water? o How many seconds or minutes does it take for the ink to spread evenly throughout the water?	throughout the water within a few minutes. 1. Which of the following spread faster in water? Explain? (a) honey (b)ink (c)lemon juice 2. Why copper sulphate dissolves quickly in hot water than in cold water?

Summary	**Factual Questions**	**CFU**	**Home Work**
Students Understand the characteristics of particles of matter	Why are light and sound not considered as matter?	We can get the smell of perfume sitting several meters away. Explain	How will you show that particles of matter have space between them?

PERIOD Number: 2.4 TOPIC: Activity 2.5

Learning Outcome:

Draws to a conclusion that the particles of matter are continuously in motion.

Teaching Learning Process

I DO	**WE DO**	**YOU DO**
Asking previous knowledge questions	**Activity – 2.5** ● Drop a crystal of copper sulphate or potassium permanganate in to a glass of hot water and another container with cold water.	✔ Solid crystals of potassium permangate start diffusing.

	• Do not stir the solution. Allow the crystals to settle at the bottom. • What do you observe just above the solid crystals in the glass? • What happens as time passes? • What does this suggest about the particles of solid and liquid? • Does the rate of mixing change with temperature? Why and how? From the above three activities pupil conclude that particles of matter are continuously moving i.e. They possess kinetic energy. As temperature increases particles move faster. So, we can say that increase in temperature raises the kinetic energy of particles. In the above three activities we observe that particles of matter intermix on their own as they have spaces between them. This intermixing of particles of two different types on their own is called diffusion.	✔ As time passes particles of solid diffuse through water. ✔ Particles of solid and liquid mix into each other. ✔ The mixing increases with temperature. This is because the kinetic energy of particles increases with temperature.

Summary	**Factual Questions**	**CFU**	**Home Work**
Students concluded that the particles of matter are continuously in motion.	Why is the rate of diffusion faster in gases?	Name the phenomenon in which particles of two or more substances intermix on their own?	Define Diffusion?

PERIOD Number: 2.5 **TOPIC: 2.2.3 Particles of matter attract each other** **Activity 2.6, 2.7, 2.8**

Learning Outcome:

Understand that the particles of matter attract each other

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Activity	). What happens when we lit an incense stick kept at the corner of a room?). How can we smell the cooked food items from a distance?). What happens when water falls on the floor? 4) Can we pass our hand through the book? If not why? **Activity 2.6** Prepare four groups A, B, C &D with the students.  Fig. 1.3 **Group A**: Students hold each other from the back and lock arms tightly and form human chain **Group B**: Students hold each other with hands and form human chain.	1) If we consider each student as a particle of matter then which of the groups are tough to break the human chain? 2) What is the order of strengths among the groups A,B& C 3).Which among the three particles do you think the particles are held together with greater force? 4) Can you relate the object with groups A,B and C in the Activity 2.6 Are you able to cut the surface of the water?) A diver is able to cut through water in a swimming pool, which property of matter does this observation made?

	 **Group C**: Students hold each other by touching with finger tips and from human chain **Group D**: students are free and not touching each other. Now students of group D has to run around the three groups and try to break the chain of students in to individual- one. Which human chain is tough to break? Group A human chain is tough to break because in group A the students are held hand tightly using maximum force. Which one is easy to break? Group C the students are held hands loosely with minimum force hence group C is easy to break the human chain. **ACTIVITY 2.7**	7) Identify various objects in the everyday life and try to differentiate based on their strengths and report as a project. 8)A few substances are arranged in the increasing order of 'forces of attraction' between their particles. Which one of the following represents a correct arrangement? (a) Water, air, wind (b) Air, sugar, oil (c) Oxygen, water, sugar (d) Salt, juice, air 9) Fill in the blanks:

Activity   	Take an iron nail, a cardboard and a rubber band seperately and try to break the substances using hammer, hand or stretching. What do you observe? We observe that the iron nail is difficult to break even with hammer and it is not stretchable. Can we stretch cardboard? Can we break easily rubber band? Why iron nail is difficult to break or unstreachable? the particles in iron nail are held together with maximum force. **Activity 2.8**  Take water in the beaker and try to cut the surface of the water with fingers. What do you observe? Why is it easy to insert your fingure into water? What does the above three activities suggest? Particles of matter have force acting between them. This force keeps the particles together. The force of atrraction between the particles in every matter remain same or different?	) At room temperature the forces of attraction between the particles of solid substances are———than those which exist in the gaseous state. (b) The arrangement of particles is less ordered in the ——— state. However, there is no order in the ——— state.
Activity		

	The strength of this force of attraction varies from one kind of matter to another.	

Summary	**Factual Questions**	**CFU**	**Home Work**
Students Understand that the particles of matter attract each other.	Give reasons for the following observation: The smell of hot sizzling food reaches you several metres away, but to get the smell from cold food you have to go close.	**Which of the following are matter?** *Chair, air, love, smell, hate, almonds, thought, cold, lemon water, smell of perfume*	What are the characteristics of the particles of matter?

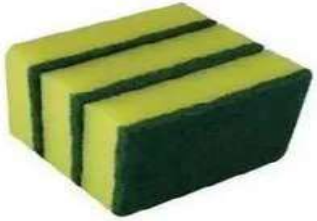
PERIOD Number: 2.6 TOPIC: 2.3 States of matter 2.3.1 The solid state Activity 2.9

Learning Outcome: Draw the flow chart of three states of matter

Teaching Learning Process

I DO	**WE DO**	**YOU DO**
Asking previous knowledge questions. Pen, Book, Needle and wooden stick are the some examples for solids.	The teacher will start the lesson with a discussion to check for prior knowledge and probing questions What are different states of matter? How solids are different from liquids and gases? Lets do an activity. Collect the following articles — a pen, a book, a needle and a piece of wooden stick. • Sketch the shape of the above articles in your notebook by moving a pencil around them.	1) Classify the following into different states of matter?

Activity





Do all these have a definite shape, distinct boundaries and a fixed volume?

- What happens if they are hammered, Pulled or dropped?
- Are these capable of diffusing into each other?
- Try compressing them by applying force. Are you able to compress them? Why solids are rigid?

Solids have a tendency to maintain their shape when subjected to outside force. Solids may break under force but it is difficult to change their shape, so they are rigid.

.) What about a rubber band, can it change its shape on stretching? Is it a solid?

?) A rubber band changes shape under force and regains the same shape when the force is removed?

If excessive force is applied, it breaks.

(3) What about a sponge?

(4) It is a solid,yet we are able to compress it. Why?

A sponge has minute holes, in which air is trapped, when we press it, the air

is expelled out and we are able to compress it.



Substance	Solid	Liquid	Gas
Bench			
Book			
Water			
Thought			
Cold			
Hate			
Pen			
Air			

2) Do an activity. Take an ice cube (solid). Press it on its sides. Is it easy or hard to press it?

3) Stretch the given ice cube?

4) Put the ice cubes a side for 10 minutes?

5) What happens to it?

6) Will it have same shape of ice?

7) Do all these have definite shape, distinct boundaries?

8) What happens if they are

Properties of solids	(c)What about sugar and salt? When kept in different jars these take the shape of the jar. Are they solids? The shape of each individual sugar or salt crystal remains fixed, whether we take it into our hand, put it in a plate or in a jar.	hammered, pulled or dropped? 9) Collect the names of different solid substances in your surroundings? 10) Name the state of matter that tends to maintain its shape when subjected to outside force? Why do particles start vibrating in solids?

Summary	**Factual Questions**	**CFU**	**Home Work**
Students Draw the flow chart of three states (solid, liquid and gas) of matter	A wooden chair is a solid at room temperature. Give reason	State one important characteristics of matter which makes it a solid?	Name the three states of matter. Give examples of each state.

PERIOD Number: 2.7 TOPIC: 2.3.2 The liquid state Activity 2.10

Learning Outcome: Concludes that liquids have no definite shape , but they get the shape of container

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Water, cooking oil and Milk are some examples for liquids. Activity 	The teacher will start the lesson with a discussion to check for prior knowledge and Probing questions. Identify the properties of solids? Definite shape Distinct boundaries Fixed volume Negligible compressibility What happens to solids when you apply outside force on them? Solids change their shape? ***Activity 2.10*** Collect the following: a) water, cooking oil, milk, juice, a Cold drink. b) containers of different shapes. ✔ Put a 50 ml mark on these Containers using a measuring Cylinder from the laboratory. ✔ What will happen if these liquids are Spilt on the floor?  	) Arrange the following in order of increasing density – Air, exhaust from chimneys, honey, water, chalk, cotton and iron.) Liquids generally have lower density as compared to solids. But you must have observed that ice floats on water. Find out why? 3) Give reasons a) a gas fills completely the Vessel in which it is kept. b) a gas exerts pressure on the Walls of the container. c) a wooden table should be called a solid. d) we can easily move our hand In air but to do the same through a solid block of wood we need a karate expert. 4) What is LPG? 5) In what form lpg is stored? 6) Give examples for gases which are compressed like LPG? 7) What are the advantages of compressing a

container into another, does it flow easily?




They take up the shape of the container in which they are kept.



✔ Measure 50 ml of any one liquid and transfer it into different containers One by one.

✔ Does the volume remain the same?

✔ Does the shape of the liquid remain the same ?

✔ When you pour the liquid from one

We observe that liquids have no fixed shape but have a fixed volume.



Liquids flow and change shape, so they are not rigid but can be called fluid.

How does aquatic life exists?

Like humanbeigns aquatic animals or plants need oxygen to respire?

gas?

8) We come to know of what is being cooked in the kitchen without even entering there, by the smell that reaches our nostrils. How does this smell reach us?

9) Choose the correct statement of the following

a) conversion of solid into vapours without passing through the liquid state is called sublimation.

b) conversion of vapours into solid without passing through the liquid state is called vapourisation..

c) conversion of vapours into solid without passing through the liquid state is called freezing.

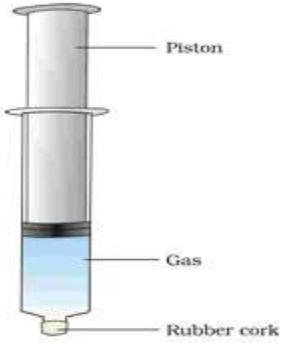
d) conversion of solid into liquid is called sublimation.

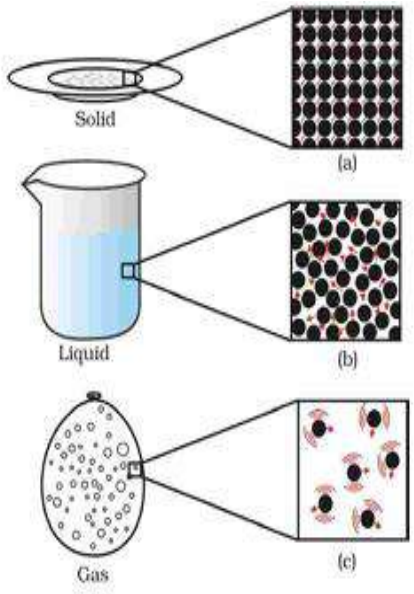
Summary	**Factual Questions**	**CFU**	**Home Work**
Students Concludes that liquids have no definite shape , but they get the shape of container.	Why do liquids have open surface?	Write two properties of liquids.	Liquids generally have lower density as compared to solids. But you must have observed that ice floats on water. Find out why

PERIOD Number: 2.8 **TOPIC: 2.3.3 The gaseous state** **Activity 2.11**

Learning Outcome: Understand that gases are highly compressible

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Activity  	1. Among gases,liquids and solids which dissolve in water quickly? 2. What is the name of this dissolution? 3. Why gases have greater diffussion than liquids and solids? **Activity -2.11** ● Take three 100 mL syringes and close their nozzles by rubber corks. ● Remove the pistons from all the syringes. ● Leaving one syringe untouched, fill water in the second and pieces of chalk in the third. ● Insert the pistons back into the syringes. You may apply some vaseline on the pistons before inserting them into the syringes for their smooth movement. ● Now, try to compress the content by pushing the piston in each syringe. ● What do you observe? ● In which case was the piston easily pushed in? ● What do you infer from your observations? We have observed that gases are highly compressible as compared to solids	

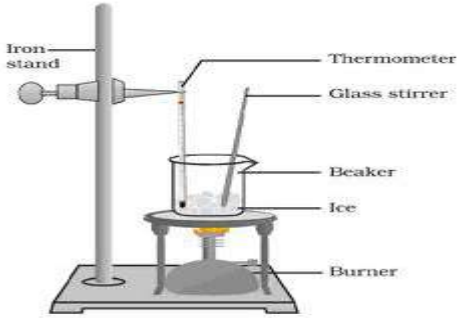
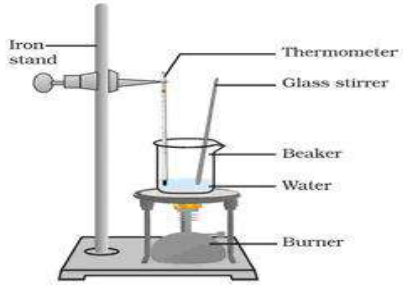
Properties of gases  	and liquids. The liquefied petroleum gas (LPG) cylinder that we get in our home for cooking or the oxygen supplied to hospitals in cylinders is compressed gas. Compressed natural gas (CNG) is used as fuel these days in vehicles. Due to its high compressibility, large volumes of a gas can be compressed into a small cylinder and transported easily. The particles of the aroma of food mix with the particles of air spread from the kitchen, reach us and even farther away. The smell of hot cooked food reaches us in seconds; compare this with the rate of diffusion of solids and liquids. Due to high speed of particles and large space between them, gases show the property of diffusing very fast into other gases. The pressure exerted by the gas is because of this force exerted by gas particles per unit area on the walls of the container.	The piston was pushed in easily in the syringe in which nothing except air was filled. Gases can be compressed easily.

Summary	**Factual Questions**	**CFU**	**Home Work**
Students Understand that gases are highly compressible than liquids and solids.	Why are liquids and gases called fluids?	Arrange the following substances in increasing order of forces of attraction between the particles—water, sugar, oxygen.	Write the differences between solids, liquids and gases.

PERIOD Number: 2.9 **TOPIC: 2.4 Can matter change its state?** **2.4.1 Effect of change of temperature** **Activity 2.12**

Learning Outcome: Explain the effect of temperature on states of matter

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions  Activity 2.12 Effect of change of temperature	t water can exist in three states of matter • solid, as ice, • liquid, as the familiar water, and • gas, as water vapour.  **Activity _____ 2.12** ✔ Take about 150 g of ice in a beaker and suspend a laboratory thermometer so that its bulb is in contact with the ice. ✔ Start heating the beaker on a low flame. ✔ Note the temperature when the ice starts melting. ✔ Note the temperature when all the ice converted into water. ✔ Record your observations for this conversion of solid to	The ice melts at 0^0C. All the ice gets converted into at 0^0C

	liquid s ✔ Now, put a glass rod in the beaker and heat while stirring till the water starts boiling. ✔ Keep a careful eye on the thermometer reading till most of the water has vaporised. ✔ Record your observations for the conversion of water in the liquid state to the gaseous state. **We conclude that:** 1) On increasing the temperature of solids, the kinetic energy of the particles increases. 2) Due to the increase in kinetic energy, the particles start vibrating with greater speed. 3) The energy supplied by heat overcomes the forces of attraction between the particles. 4) The particles leave their fixed positions and start moving more freely. 5) A stage is reached when the solid melts and is converted to a liquid.	The water starts boiling at 100^0C and gets converted into vapours at the same temperature.

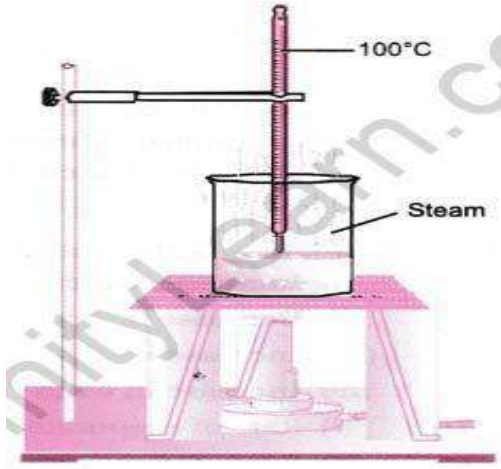
Summary	**Factual Questions**	**CFU**	**Home Work**
Students Explain the effect of temperature on change of states of matter.	For any substance, why does the temperature remain constant during the change of state?	Why is ice at 273 K more effective in cooling than water at the same temperature?	What produces more severe burns, boiling water or steam?

PERIOD Number: 2.10 **TOPIC: Melting point , Boiling point , Latent heat of fusion**

Learning Outcome: Explain how boiling takes place and what is the boiling temperature

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions	1) Give examples of solid-state substance 2) Is ice melt? 3) What is effect of temperature on the solid?	Wood, ice, Iron etc Yes Change the state
Melting point	What is melting point? • The minimum temperature at which a solid melts to become a liquid at the atmospheric pressure is called its melting point. • The melting point of a solid is an indication of the strength of the force of attraction between its particles.	Melting point of ice is 0^0C
Fusion	The melting point of ice is 273.15 K. The process of melting, that is, change of solid state into liquid state is also known as fusion. When a solid melts, its temperature remains the same.	O^0C = 273.15 K.
Latent heat	During the change of state from solid to liquid at 0^0c , a certain amount of heat energy is absorbed by ice without any rise in temperature which is called **latent heat.**	
Latent heat of fusion	The amount of heat energy that is required to change 1kg of solid into liquid at atmospheric pressure at its melting point is known as **latent heat of fusion.**	What is latest heat?



Boiling

Boiling point

1. What happens to the readings of the thermometer when heating the water gradually?

What happens to the water when the reading in the thermometer is 100^0c?

3. Why water is converted to vapour state at 100^0c?

As we supply heat energy to water, at a certain temperature (100^0c) the energy sufficient to break the forces of attraction between each other, the liquid starts changing into gas. **This process is called boiling.**

The temperature at which a liquid starts boiling at atmospheric pressure is known as **boiling point.**

1. What is the boiling point of water?
2. Can you convert 100^0c to kelvin scale?

The steam that is water vapour at 373k has more energy than water at the same temperature.

This because particles in steam have observed extra energy in the form of latent heat of vaporization.

Hence we can say that the state of matter can be changed from one state to another state by changing the temperature.



What is latest heat of fusion?

What is boiling?

What is boiling point?

What is boiling point of water?

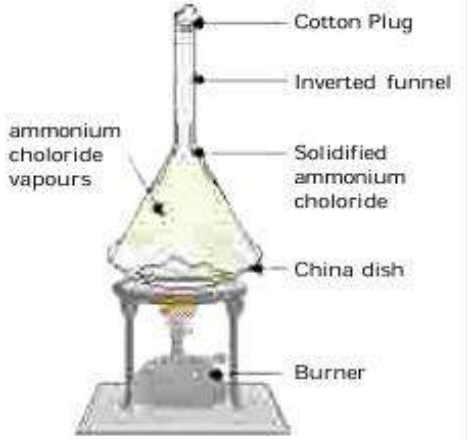
Summary	Factual Questions	CFU	Home Work
Students Explain the boiling and the boiling temperature.	What is the physical state of water at: a) 250°C b)100°C ?	What is melting point of Ice? What is boiling point of Water?	Convert the following temperatures to the Celsius scale. (a) 293 K (b) 470 K

PERIOD Number: 2.11 TOPIC: Activity 2.13 Sublimation, Deposition

Learning Outcome: Give examples for sublimation

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Sublimation Activity	Is solid state being directly change into gaseous state? If Possible, give example? What process called this? . Directly from solid state to gaseous state and vice versa without changing into the liquid state is called as **sublimation.** **Activity-2.13** ● Take some camphor or ammonium chloride. ● Crush it and put it in a china dish. ● Put an inverted funnel over the china dish. ● Put a cotton plug on the stem of the funnel.	What is sublimation?

 *Fig. 1.7: Sublimation of ammonium chloride* Deposition	• Now, heat slowly and observe. • What do you infer from the above activity? A change of state directly from solid to gas without changing into liquid state is called sublimation **Example of sublimation:** 1. Sublimation of ammonium chloride 2. Sublimation of iodine 3. Sublimation of wax What is reverse process of sublimation The direct change of gas to solid without changing into liquid is called **depositio**	Vapours of ammonium chloride get deposited on the inner stem of funnel. Solid ammonium chloride gets converted into vapours directly.

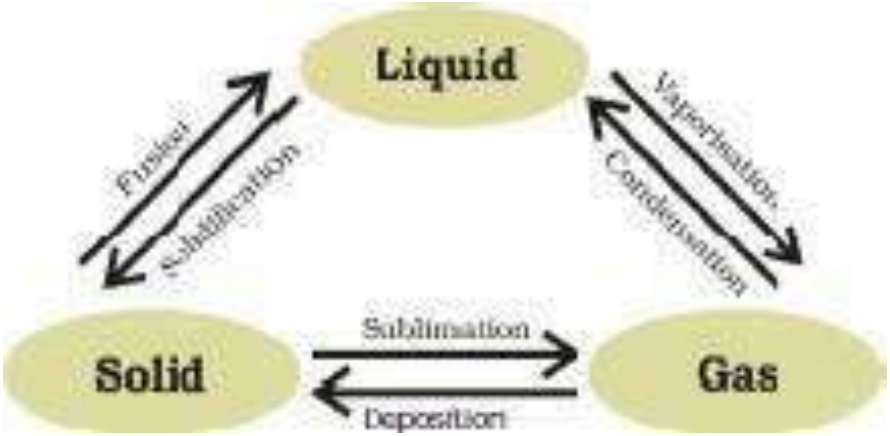
Summary	**Factual Questions**	**CFU**	**Home Work**
Students Give examples for sublimation in their own words.	State the characteristics of gas used in supplying oxygen cylinders to hospitals.	What is sublimation? What is Deposition?	Convert the following temperatures to the kelvin scale. (a) $25^{o}C$ (b) $373^{o}C$

PERIOD Number: 2.12 **TOPIC: 2.4.2 Effect of change of pressure**

Learning Outcome: Explain the effect of pressure on state of matter

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions What is pressure?	**<u>Introduction:</u>** 1. How are the particles arranged in solids, liquids and gases? 2. Do the particles change their state by changing temperature (heating or cooling)? 3. What is the effect of change of pressure on the state of gas particles? Let us take a gas in a cylinder with a piston and start applying pressure with piston. 1. What will happen when we start applying pressure? 2. Do you think that increasing or decreasing the pressure can change the state of matter?	1) What happen when high pressure is applied to the gas in the cylinder? 2) What is the phenomenon involved in storage of CNG gas? 3) What is the phenomenon involved in disappearance of camphor? A) Condensation B) sublimation C) vapourisation D)deposition 4) What will happen when we increase pressure on CO_2 gas in a cylinder? 5) What is dry ice? 6) At what temperature, is dry ice converted into gas? 7) Con version of solid to vapour

Effect of change of pressure	The gas particles in the cylinder come closer to each other and if we continue apply pressure, then the compression of gas takes place in a cylinder and converts into liquid state. **This process is called condensation.** Generally solid co_2 g**a**s is stored under high pressure. But when the pressure is decreased and equal to 1atm then the solid co_2 is converted into gaseous state directly without going into liquid state. **This process is called sublimation.**  Finally, we say that pressure determines the states of matter. I.e., matter is solid or liquid or gas.	is called sublimation. Name the term used to denote the conversion of vapour to solid? 8) Cooking gas is known as LPG (liquid petroleum gas). How can a gas to be liquefied? A) when pressure is applied to the gas B) when temperature of the gas is increased. C) when gas is mixed with a liquid substance. when the force of attraction between particles is reduced.

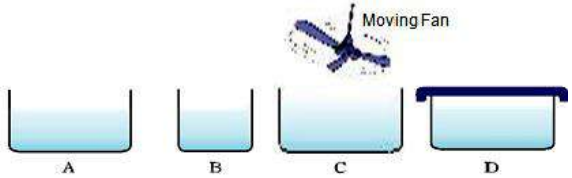
Summary	**Factual Questions**	**CFU**	**Home Work**
Students Explain the effect of pressure on state of matter	A gas exerts pressure on the walls of the container. Give reason.	What is atmospheric pressure?	Draw a diagram of interconversion of the three states of matter.

PERIOD Number: 2.13 TOPIC: 2.5 evaporation 2.5.1 Factors affecting evaporation

Learning Outcome: Understand the factors depends on which evaporation depends

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Evaporation Concept Activity	1) Do we always need to heat or change pressure for changing the state of matter? 2) Why does the level of water decreases in ponds, lakes, rivers in summer? 3) What happens to water when wet clothes dry? 4) How does common salt recovered from sea water? To know the answer for the above questions we will discuss the concept. i.e., evaporation. 5) Explain the definition of evaporation? 6) Explain the mechanism involved in the process of evaporation? 7). Why wet clothes dry faster in summer than that of in winter? 8). Why some people prefer to sip tea in saucer rather than in cup? To understand the effect of factors on evaporation we perform an activity. **Acitivity-2.14:** 1. Take 5 ml of water in a test tube and keep it under a fan. 2. Take 5 ml of water in an open china dish and keep it under a fan.	□ You want to wear your favorite shirt to a party, but the problem is that it is still wet after a wash. □ What steps would you take to dry it faster? Write the differences between evaporation and boiling. **Project:** Collect the information about the examples of evaporation and its applications in daily life. 1. Which of the following factor/s is not responsible for faster evaporation? (a) increase in surface area (b) increase in humidity (c) increase in wind speed (d) Increase in temperature. 2. Why are we prefer to sip hot tea or milk in a saucer rather than in a cup? 3. During summer, water kept in an earthen

Factors affecting of evaporation	3. Take 5 ml of water in an open china dish and keep it inside a cupboard in the class. Students can record the time taken for evaporation in all the cases. Teacher will suggest the students to do the above activity in rainy day and record the observations (assignment). Now teacher makes the class in to 4 groups and ask them to discuss how the following factors affect the rate of evaporation. ● Group-A: Discuss how evaporation increases with increase of surface area. (e.g. In our daily life, we spread out the washed clothes while drying). ● Group-B: Discuss how evaporation increases with increase of temperature (e.g., clothes dry quickly on a sunny day) ● Group-C: Discuss how evaporation increases with a decrease in humidity (e.g. Clothes dry faster under low humidity in air) ● Group-D: Discuss how evaporation increases with an increase in wind speed. (e.g. Clothes dry faster on a windy day). From this activity we conclude that temperature, surface area, humidity and wind	pot becomes cool because of the phenomenon of (a) diffusion (b) transpiration (c) osmosis (d) evaporation 4. Which condition out of the following will increase the evaporation of water? (a) Increase in temperature of water (b) Decrease in temperature of water (c) Less exposed surface area of water (d) Adding common salt to water 5. Look at figure 1.3 and suggest in which of the vessels A,B, C or D the rate of evaporation will be the highest? Explain?  Fig. 1.3 Comment on the following statements: (a) Evaporation produces cooling.

	speed affect the rate of evaporation.	Rate of evaporation of an aqueous solution decreases with increase in humidity.

Summary	**Factual Questions**	**CFU**	**Home Work**
Students Understand the factors depends on which evaporation depends	Is evaporation is cooling process?	Define evaporation?	Discuss the factors which affect evaporation.

PERIOD Number: 2.14

TOPIC: 2.5.2 How does evaporation cause cooling

Learning Outcome:

Establish cause and effect relation between evaporation with cooling effect

Teaching Learning Process

I DO	**WE DO**	**YOU DO**
Asking previous knowledge questions What is evaporation?	**<u>Introduction:</u>** 1) What kinds of clothes keep us cool? 2) Why do wet clothes make us feel cool? 3) How does sweating help in keeping our body temperature under control on a hot day? 4) Why does our palm become cool when pour acetone/sanitizer? 5) Why do people sprinkle water on the roof or open ground	Perform a group activity; ask the students to list out examples of evaporation causes cooling. Why does a desert cooler give better cooling on a hot dry day? Why do we prefer to sip hot tea or milk in a saucer rather than in a cup? 4) What type of clothes should we wear in summer? Which of the following process/es

What are the factors affecting evaporation? Is Evaporation cooling process?	after a hot sunny day? 6) Write some examples for evaporation in everyday life (water is in earthen pot gets cool due to evaporation, why pigs toil in mud water, --- etc.). **<u>Activity:</u>** Demonstrate the process of evaporation by taking one/two drops of alcohol or acetone on the palm of the hand. The palm feels cool. Explain perspiration or sweating help in keeps our body cool on a hot day?	absorb heat? (1) condensation (2) evaporation (3) freezing (4) melting. Why do dogs panting on hot summer days? Give reasons.

Summary	**Factual Questions**	**CFU**	**Home Work**
Students Establish the relation between evaporation with cooling effect.	Why should we wear cotton clothes in summer?	Give reason why we are able to sip hot tea from a saucer than from a cup?	Why does evaporation cause cooling of a liquid system?

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson: 2. Matter In Our Surroundings		Class: 9	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
2.1	2.1 Physical nature of matter 2.1.1 Matter is made up of particles Activity 2.1				
2.2	2.1.2 How small are these particles of matter? Activity 2.2				
2. 3	2.2. Characteristics of particles of matter 2.2.1 Particles of matter have space between them. Activity 2.3 2.2.2 Particles of matter are continuously moving Activity 2.4				
2. 4	Activity 2.5				
2. 5	2.2.3 Particles of matter attract each other Activity 2.6, 2.7, 2.8				
2. 6	2.3 States of matter 2,3.1 The solid state Activity 2.9				
2. 7	2.3.2 The liquid state Activity 2.10				

2. 8	2.3.3 The gaseous state Activity 2.11				
2.9	2.4 Can matter change its state? 2.4.1 Effect of change of temperature Activity 2.12				
2. 10	Melting point, Boiling point, Latent heat of fusion				
2.11	Activity 2.13 Sublimation, Deposition				
2.12	2.4.2 Effect of change of pressure				
2.13	2.5 evaporation 2.5.1 Factors affecting evaporation				
2.14	2.5.2 How does evaporation cause cooling				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?

4	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

August 2025

No bag Day

Cluster Complex

Teacher Resources

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					1 NI	2 No Bag Day
3 Sunday	4 FA - 1	5 FA - 1	6 FA - 1	7 FA - 1	Vara 8 Lakshmi Vratham 3.1	9 Second Saturday
10 Sunday	11 3.2	12 3.2	13 3.3	14 3.4	Independence day 15	Sri 16 Krishna Ashtami
17 Sunday	18 3.4	19 3.5	20 3.6	21 3.7	22 NI	No Bag Day 23 Cluster meeting
24 Sunday	25 3.8	26 3.9	Sri 27 Vinayaka Chavithi	28 NI	National Sports day 29 NI	30 No Bag Day
31 Sunday						

Week1: FA 1 – 3.1

Week 2: 3.2 – 3.4

Week 3: 3.4 – 3.7

Week 4: 3.8 – 3.9

TEACHER'S NOTES

Chapter: 3. Force and Laws of Motion

Learning Outcomes:

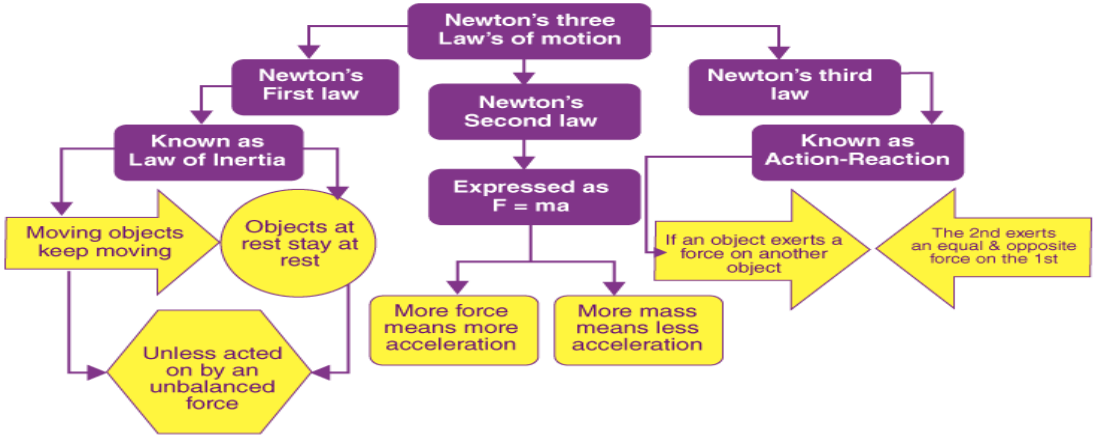
L3I2T1

- differentiate balanced and unbalanced forces
- explain first law of motion
- understand that no body dispalce with out application of force
- relates inertia with mass of the body
- formulate f=ma from second law of motion
- do the calculations regarding f=ma
- solve problems insecond law of motion
- Do calculations related to 2nd law
- establish a relation between action and reaction

Prior Concepts/Skills: (Essential concepts and skills to be checked/bridged before teaching the current concept)

1. Define the force of friction.
2. Why tie the luggage with a rope on the roof of buses?
3. What is acceleration and write its formula?
4. Express your experience when you are standing on the moving bus, suddenly the bus driver applies breaks?

Concept Map



Synopsis

- **First law of motion:** An object continues to be in a state of rest or of uniform motion along a straight line unless acted upon by an unbalanced force.
- **Inertia:** The natural tendency of objects to resist a change in their state of rest or of uniform motion is called inertia. The mass of an object is a measure of its inertia.
- **Second law of motion:** The rate of change of momentum of an object is proportional to the applied unbalanced force in the direction of the force.
- **Linear momentum:** The momentum of an object is the product of its mass and velocity and has the same direction as that of the velocity.
- **Third law of motion:** To every action, there is an equal and opposite reaction and they act on two different bodies.

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
3.1	3.1 Balanced and unbalanced forces	differentiate balanced and unbalanced forces	
3.2	3.2 First law of motion	explain first law of motion	
3.3	Activity 3.1 , Activity 3.2 , Activity 3.3	understand that no body dispalce with out application of force	
3.4	3.3 Inertia and mass	relates inertia with mass of the body	
3.5	3.4 second law of motion	formulate f=ma from second law of motion	
3.6	3.4.1 Mathematical formulation of second law of motion	do the calculations regarding f=ma	
3.7	examples 3.1 , 3.2 , 3.3 problems	solve problems insecond law of motion	
3.8	examples 3.4 , 3,5 problems	do calculations related to 2nd law	
3.9	3.5 Third law of motion	establish a relation between action and reaction	

PERIOD Number: 3.1

TOPIC: 3.1 Balanced and unbalanced forces

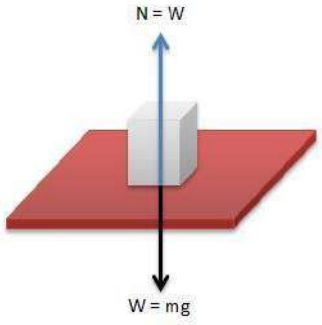
Learning Outcome: Students will be able to differentiate balanced and unbalanced forces

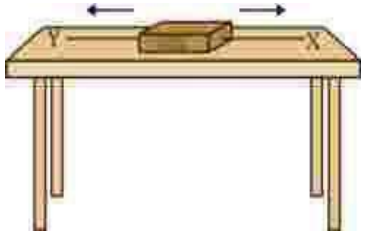
Teaching Learning Process

I DO	WE DO	YOU DO
Introductory activity (5min) : **(observe the following pictures)** What is railway porter doing? 	 What is women doing in the picture? What do you do to make a football move?  1. Why does a ball return back when thrown upwards? 2. How do you walk? 3. How does a bus move? 4. What do you do to make the moving ball move faster?	**(Previous knowledge questions)** 1. What is force? (push/pull) 2. Have you ever played the game of **tug of war?** 3. Which team wins the game? 4. What happens when two teamsPull equally hard? 1. What happens when youand your friend push the heavy box in the same direction? 2. What happens when you and your friend push the heavy box in opposite directions? 3. How do we express the strength of a force usually? 4. What is a vector? 5. Is force a vector or scalar? 6. What is net force?



- How does a goalkeeper stop a ball?
- What happens when a hockey player flicks a ball?
- What happens when you stretch a rubber band?
- In this chapter let's explore balanced and unbalanced forces
- Let's us consider wooden block placed on a table





What is the state of the block?Does any force act on the wooden block?

- Balanced forces cannot produce motion or stop a moving body, and can change the shape of the body.

✔ What are the various Forces acting on the wooden block?

✔ What is meant by balanced Force?

✔ Is the net force in such cases Zero?

✔ What is an unbalanced force?

Teacher statement: In all these examples we observe that at least two objects must interact for a force to come into play and its effects.(20 min)	Example: squeezing a rubber balloon. ● Observe the figure : ✔ Does the block move? ✔ Increase the force of push? Does the block move? ✔ Further increase the force of push .Does the block move? 2). When our applied force is greater than maximum static frictional force, box moves.(15 min) • Net force is not equal to zero. • What do we call such forces?(unbalanced force) **3).** Unbalanced force acting on an	

Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate balanced and unbalanced forces	1. How can we change the state of motion of the objects? 2. A ball is moving over a horizontal smooth surface with a constant velocity. What type of forces are acting on the ball?	(a) Can the velocity of a body change when no unbalanced force is acting on it? (b) State the various effects produced by a force.	3. An object experiences a net zero external unbalanced force. Is it possible for the object to be travelling with a non-zero velocity? If yes, state the conditions that must be placed on

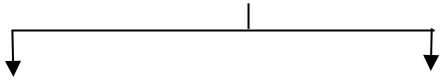
PERIOD Number: 3.2

TOPIC: 3.2 First law of motion

Learning Outcome: Students will be able to explain first law of motion

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask question to children:  ✔ Have you played carom board? ✔ Have you struck the pile of carom coins? ✔ What do you observe when you hit the base of a pile of carom coins?	**Let's do the following activity :** ✔ Make a pile of your textbooks or notebooks, one above the other. Now pull quickly the book which is placed at the bottom. What do you ✔ observe? ✔ Teacher performs the following activity in the class. **Now the teacher will call two children to come in front and perform this activity together.** 	✔ Why only the carom coin at the bottom of the pile is removed when a fast moving striker hits it? ✔ What was the state of books when piled up? ✔ Why do the books remain in the stack even though we applied force?

The first Law of Motion is also known " Law of inertia"	✔ What did you observe in this activity, write in your notebook? Why was the coin falling in the glass? ✔ Why was the coin not moving with the cardboard? ✔ Teacher will collect the children's response and move towards the law of inertia concept. The reason for the above mentioned events was Newton's first law of motion. ✔ Have you heard about it? **The first Law of Motion is also known "Law of inertia"** First Law gives a relationship between zero net forces acting on an object. **F net = O (balanced Force)**  **Rest (v=0)** **Uniform Motion (v=constant)** ✔ What is the mathematical expression for Newton's first Law of motion? $F_{net} = o$ ✔ Teacher defines Newton's first Law of motion.	✔ What is the common reason for these events? ✔ What is another name for Newton's First Law? ✔ Which property does Newton's First Law depicts about? ✔ Explain why some of the leaves may get detached from a tree if we vigorously shake its branch?

First Law gives a relationship between zero net forces acting on an object. ✔ Teacher defines inertia and its types. ✔ Teacher explains examples for static inertia.	✔ Teacher defines inertia and its types. ✔ Teacher explains examples for static inertia.	✔ When you hit a ball, why does it continue to move? ✔ When a carpet is beaten with a stick , dust comes out of it? Explain. ✔ Explain why some of the leaves get detached from the tree if we vigorously shake its branch? ✔ Why does dust come out from a doormat when beaten with a stick?

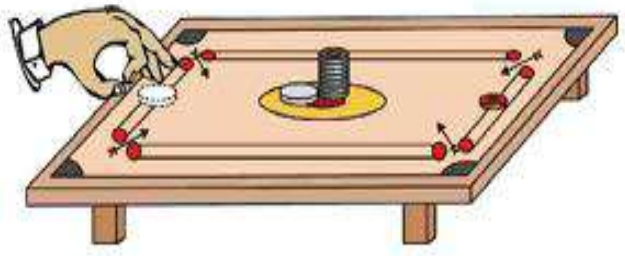
Summary	Factual Questions	CFU	Home Work
Students will be able to explain first law of motion	1. Define the term force 2. Which law of motion defines the force?	3. State Newton's first law of motion 4. Name the property of bodies to resist a change in their velocity.	1. When a carpet is beaten with a stick, dust comes out of it, Explain. 2. Why is it advised to tie any luggage kept on the roof of a bus with a rope

PERIOD Number: 3.3

TOPIC: Activity 3.1 , Activity 3.2 , Activity 3.3

Learning Outcome: Students will be able to understand that no body dispalce with out application of force

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons. **Introductory activity** Teacher will explain step by step procedure	**Activity 3.1** 1. Make a pile of similar carrom coins on a table. 2. Attempt a sharp horizontal hit at the bottom of the pile using another carom coin or the striker. 3. If the hit is strong enough, the bottom coin moves out quickly. 4. Once the lowest coin is removed, the inertia of the other coins makes them 'fall' vertically on the table. 	Record the observation

Teacher will explain step by step procedure	**Activity 3.2** 1. Set a five-rupee coin on a stiff card covering an empty glass tumbler standing on a table. 2. Give the card a sharp horizontal flick with a finger. 3. If we do it fast then the card shoots away, allowing the coin to fall vertically into the glass tumbler due to its inertia. 4. The inertia of the coin tries to maintain its state of rest even when the card flows off  **Activity 3.3** **1.** Place a water-filled tumbler on a tray. **2.** Hold the tray and turn around as fast as you can.	Observe that a groove is provided in a saucer for placing the tea cup. It prevents the cup from toppling over in case of sudden jerks.

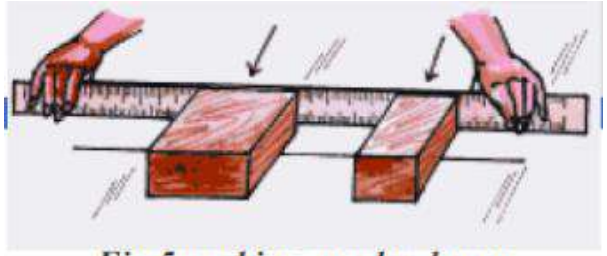
Summary	Factual Questions	CFU	Home Work
Students will be able to understand that no body displace without application of force	1. Explain the use of seat belts in car? 2. Why does an athlete always run some distance before taking a long jump?	1. What is meant by inertia? 2. Which law of motion is also called law of inertia?	1. What would happen if a fielder stops the fast moving ball suddenly? Justify your answer

PERIOD Number: 3.4

TOPIC: 3.3 Inertia and mass

Learning Outcome: Students will be able to relates inertia with mass of the body

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons. ✔ Do all the bodies have the same inertia? ✔ What factors can decide the inertia of a body? 	**Let us try to perform this activity-** Take two rectangular wooden blocks with different masses and place them on a straight line drawn on a floor. Give the same push at the same time to both blocks with the help of the wooden scale.  *Fig-5 pushing wooden boxes with same force*	✔ What is inertia? ✔ What do you think if you push a bicycle and car then which one will move easily or go farther by applying equal force? ✔ What you observed in all above examples of daily life?

	✔ What do you find? ✔ Which goes farther? Why? **Let us think about these cases.** ✔ Which is easier for you, to push a bicycle or a car? ✔ What is the measure of inertia? ✔ Why is it easier to push a bicycle or car? ✔ Which of these has more inertia? **Inertia is measured by the mass of an object.** ✔ If mass is higher, inertia will be higher and vice versa also.	✔ What is the SI unit of mass? Two objects have masses 8 Kg and 5 Kg . Which one has more inertia? Why?

Summary	**Factual Questions**	**CFU**	**Home Work**
Students will be able to relates inertia with mass of the body	1. Why do passengers in a bus tend to fall backward when it starts suddenly? 2. Give the SI unit of mass and inertia?	1. Name the factor on which the inertia of a body depends? 2. Which one is more inertia? a) 5 Kg Bag b) 2 Kg Bag c) 7 Kg Bag	1. What are the different types of inertia? 2. What is relation between mass and inertia?

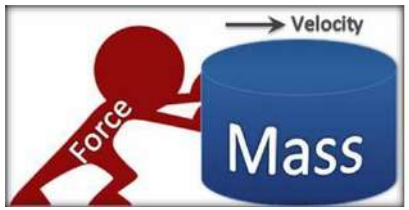
PERIOD Number: 3.5

TOPIC: 3.4 second law of motion

Learning Outcome: Students will be able to formulate f=ma from second law of motion

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons.  **Teacher's statement:** ❖ To push heavy objects we have to use more force. ❖ Force Is proportional to mass of the body	**Introductory Activity: (5mins)**  We may observe these kind of situations several times on roads when vehicles break-down. ✔ Which vehicle is easy to move and which is difficult too move by pushing. Why? 	**Previous knowledge:** ❖ When an object is at rest/stationary. And when it comes into motion?. ❖ What is an inertia?



Teacher's statement:

✔ If the mass is less acceleration is more and force is more acceleration also more.

✔ Mass is indirectly proportional to acceleration.

Force is directly proportional to acceleration.

✔ In tug of war why, the rope not moved from Centre line?

❖ If the same amount of force is applied to move a bike and a bicycle, which one accelerates more?

❖ If the same amount of force is applied to move a car and truck, which one accelerates more?

✔ Why a small mass of bullet may kill a person when fired from a gun.

✔ Why the table tennis ball hits a player it doesn't hurt, but a fast-moving cricket ball hits a spectator, it hurts him?

✔ Why a moving truck, even as low as 5m/s may kill a person.

Teacher's statement and Explanation: (15mins)

P=MV

Units of momentum is kilogram-meter/sec.

❖ What is acceleration?

❖ Can acceleration of a body become zero?

✔ If you add an external force to moving objects, what happens?

✔ Can you Find the momentum of a round stone weighing 12.05kg rolling down a hill at 8m/s.

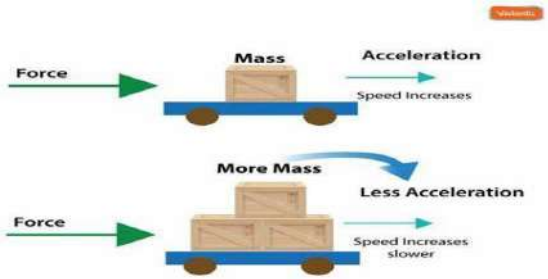
Summary	Factual Questions	CFU	Home Work
Students will be able to formulate f=ma from second law of motion	1. Name the physical quantity which is the combined effect of mass and velocity? 2. Why is glass or chinaware packed with straw?	1. State Newton's second law of motion 2. Why are shockers used in cars, scooters and motorcycles?	1. Define momentum of a body? 2. Write the SI unit of momentum.

PERIOD Number: 3.6

TOPIC: 3.4.1 Mathematical formulation of second law of motion

Learning Outcome: Students will be able to do the calculations regarding f=ma

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons.	 Teacher defines 2nd law of motion and derives the formula for force. P_1= mu	**Prerequisites :** ✔ when two same forces are applied to push a car and a bus?

EXPLANATION and DERIVING: Let us consider an object of mass m, moving along a straight line with an initial velocity of u. Let us say, after a certain time t, with a constant acceleration, the final velocity becomes v. Here we see that, the initial momentum is: The rate of change of velocities is called as?	The final momentum is: $P_2 = mv$ The change in momentum can be written as: $P_2 - P_1 = (mv-mu)$ $P_2 - P_1 = m(v-u)$ As we know, the rate of change of momentum with respect to time is proportional to the applied force. The applied force_ $F \propto m(v-u)/t$ $F \propto ma$ $F=Kma$ k is the constant of proportionality, taken as one when unit of mass and of unit acceleration	✔ Calculate the force required to impart a car a velocity of 30m/s in 10 seconds. The mass of the car is 1500 kg.

Summary	Factual Questions	CFU	Home Work
Students will be able to do the calculations regarding f=ma	1. Explain why a horse continuously applies a force in order to move a cart with a constant velocity.	1. Give some examples to illustrate Newton's second law of motion. 2. Write mathematical formula of Newton's second law of motion.	1. A truck starts from rest and rolls down a hill with a constant acceleration. It travels a distance of 400 m in 20 s. Find its acceleration. Find the force acting on it if its mass is 7 tonnes (Hint: 1 tonne = 1000 kg.)

PERIOD Number: 3.7

TOPIC: examples 3.1, 3.2, 3.3 problems

Learning Outcome: Students will be able to solve problems in second law of motion

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons. **Example 8.1** A constant force acts on an object of mass 5 kg for a duration of 2 s. It increases the object's velocity from 3 m s^{-1} to 7 m s^{-1}. Find the magnitude of the applied force. Now, if the force was applied for a duration of 5 s, what would be the final velocity of the object?	u = ? v = ? t = ? m = ? m_1 = 2 kg a_1 = 5 m s^{-2} m_2 = 4 kg a_2 = 2 m s^{-2}. $F_1 = m_1 a_1$ = 2 kg x 5 m s^{-2} = 10 N $F_2 = m_2 a_2$ = 4 kg x 2 m s^{-2} = 8 N F1 > F2	$F = \frac{m(v-u)}{t}$ Substitution of values in this relation gives F = 5 kg (7 m s^{-1} – 3 m s^{-1})/2 s = 10 N. t=5s $v = u + \frac{Ft}{m}$ v = 13 m s^{-1}.
Example 8.2 Which would require a greater force — accelerating a 2 kg mass at 5 m s^{-2} or a 4 kg mass at 2 m s^{-2} ?	accelerating a 2 kg mass at 5 m s^{-2} would require a greater force	F = ma $F = \frac{m(v-u)}{t}$
Example 8.3	u = 108 km/h = 108 x 1000 m/(60 60 s) = 30 m s^{-1}	

A motorcar is moving with a velocity of 108 km/h and it takes 4 s to stop after the brakes are applied. Calculate the force exerted by the brakes on the motorcar if its mass along with the passengers is 1000 kg	$v = 0\ m\ s^{-1}$ Total mass with its passengers= 1000kg t= 4s F= m(v – u)/t. $F = 1000\ kg\ (0 – 30)\ m\ s^{-1}/4\ s$ $= – 7500\ kg\ m\ s^{-2}$ or – 7500 N.	The negative sign tells us that the force exerted by the brakes is opposite to the direction of motion of the motorcar

Summary	Factual Questions	CFU	Home Work
Students will be able to solve problems in second law of motion	1. An automobile vehicle has a mass of 1500 kg. What must be the force between the vehicle and road if the vehicle is to be stopped with a negative acceleration of $1.7\ m\ s^{-2}$?	1. A force of 10N accelerate an object from 15 m/s to 25m/s in 20 s. Calculate the mass of the object.	1. Calculate the force required to impact to a car a velocity of 30 m/s in 10s. The mass of the car is 1500 kg.

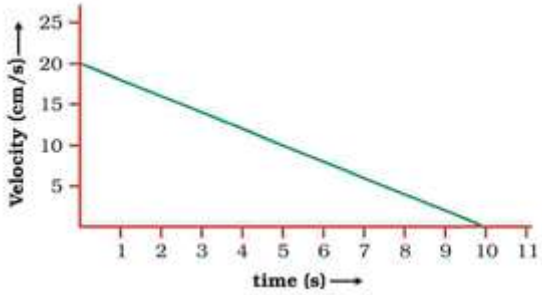
PERIOD Number: 3.8

TOPIC: examples 3.4, 3,5 problems

Learning Outcome: Students will be able to do calculations related to 2nd law

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons.	$m_1 = F/a_1$ $m_2 = F/a_2$ $a_1 = 10\ m\ s^{-2}$	If the two masses were tied together

Example 8.4 A force of 5 N gives a mass m_1 , an acceleration of 10 m s^{-2} and a mass m_2 , an acceleration of 20 m s^{-2}. What acceleration would it give if both the masses were tied together?	$a_2 = 20$ m s^{-2} F = 5 N $m_1 = 5$ N/10 m $s^{-2} = 0.50$ kg $m_2 = 5$ N/20 m $s^{-2} = 0.25$ kg. The total mass, m = 0.50 kg + 0.25 kg = 0.75 kg. The acceleration, a produced in the combined mass by the 5 N force would be, a = F/m = 5 N/0.75 kg = 6.67 m s^{-2}	the velocity of the ball decreases down to zero in 10 s.
Example 8.5 The velocity-time graph of a ball of mass 20 g moving along a straight line on a long table is given in Fig. 3.9.  How much force does the table exert on the ball to bring it to rest?	u = 20 cm s^{-1} u = 20 cm s^{-1} v = 0 cm s^{-1} t = 10 s. $a = \frac{v-u}{t}$ acceleration, a= (0 cm s^{-1} – 20 cm s^{-1})/10 s = –2 cm s^{-2} = –0.02 m $s^{-2.}$ F = ma = (20/1000) kg x (– 0.02 m s^{-2}) = – 0.0004 N.	The negative sign implies that the frictional force exerted by the table is opposite to the direction of motion of the ball

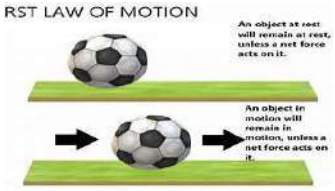
Summary	Factual Questions	CFU	Home Work
Students will be able to do calculations related to 2nd law	1. How much momentum will a dumb-bell of mass 10 kg transfer to the floor if it falls from a height of 80 cm? Take its downward acceleration to be 10 m $s^{-2.}$	1. Using a horizontal force of 200 N, we intend to move a wooden cabinet across a floor at a constant velocity. What is the friction force that will be exerted on the cabinet?	1. For how much time should the force of 400N be exerted on a body of mass 8 kg to increase its velocity from 150 m/s to 300 m/s?

PERIOD Number: 3.9

TOPIC: 3.5 Third law of motion

Learning Outcome: Students will be able to establish a relation between action and reaction

Teaching Learning Process

I DO	WE DO	YOU DO	
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons. **Teacher defines newton's 2nd law of motion.** **Demonstration and explanation:** **Example:**	**Introductory part:** By the first law of motion we can understand, how the force can be changing the state of motion.  By the second law of motion, provide us with a method of determining the force.	**Prerequisites** ✔ What implies the first law of motion? ✔ What implies the second law of motion? ✔ How many types of forces you know? ✔ In first two cases the force acts on how	

A dog sits on the ground ...

- ✔ the cat pulls the Earth up
- ✔ the Earth pulls the cat down

These forces are equal in size and opposite in direction.

Car tyre on a road...

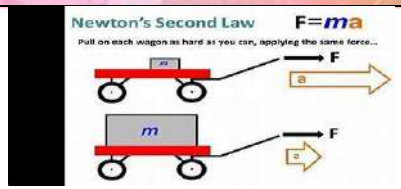
- ✔ the tyre pushes the road backwards
- ✔ the road pushes the tyre forwards

These forces are equal in size and opposite in direction.

A satellite in Earth orbit

- ✔ the Earth pulls the satellite
- ✔ the satellite pulls Earth

These forces are equal in size and opposite in direction.



EXAMPLE:

Consider two spring balances A and B connected together. And the fixed end of B is attached to a rigid support like wall.

When a force is applied through the free end of spring balance A, it is observed that both the spring balances show the same readings on their scales.

That means the force exerted by spring A on the spring balance B is equal but opposite in direction to the force exerted by the balance B on balance A.

These two forces can be called as ACTION – REACTION forces.

Newton's Third Law of Motion states that:

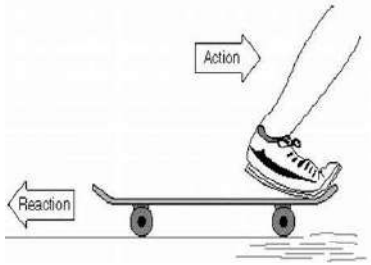
"To every action, there is an equal and opposite reaction."

many objects and what type of forces they are?

- ✔ If the forces act on two different objects What happens?

What type of forces you are observing here?

- ✔ Explain how does a bird fly in the sky?
- ✔ How do we walk?



- ✔ Observe the following picture and mention the action and reaction forces.



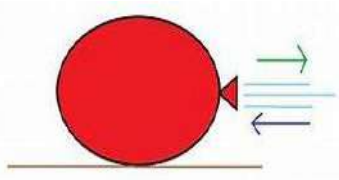


✔ These forces are pair forces.

✔ The action and reaction forces are always equal in magnitude, these forces may not produce accelerations of equal magnitude. This is because of each force acts on different object that may have different mass.

✔ How do we swim?

What are the forces acting on the balloon?



Summary	Factual Questions	CFU	Home Work
Students will be able to establish a relation between action and reaction	1. Why does a gun man get a jerk on firing a bullet? 2. Name the physical quantity whose SI unit is newton.	1. Name the principle on which a rocket works. 2. Why does a boat move backward, when a man jumps out of a boat to the bank of a river?	1. State and explain Newton's third law of motion.

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson: 3. Force and Laws of Motion		Class: 9	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
3.1	3.1 Balanced and unbalanced forces				
3.2	3.2 First law of motion				
3. 3	Activity 3.1 , Activity 3.2 , Activity 3.3				
3. 4	3.3 Inertia and mass				
3. 5	3.4 second law of motion				
3. 6	3.4.1 Mathematical formulation of the second law of motion				
3. 7	Examples 3.1 , 3.2 , 3.3 problems				
3. 8	Examples 3.4 , 3,5 problems				
3.9	3.5 Third law of motion				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

September 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 4.1	2 4.2	3 4.3	4 NI	5 Milad -Un -Nabi National Teacher's Day	6 No Bag Day
7 Sunday	8 International Literacy NI	9 4.4	10 4.4	11 4.5	12 4.6	13 Second Saturday
14 Sunday Hindi Diwas	15 International Day of Democracy NI	16 4.7	17 4.8	18 4.9	19 NI	20 No Bag Day Cluster and SMC meeting
21 Sunday	22 4.10	23 NI	24 Dussera Holiday	25 Dussera Holiday	26 Dussera Holiday	27 Dussera Holiday
28 Sunday	29 Dussera Holiday	30 Durgaa shtami				

E8W4U1 No bag Day

H6H1Q4 Cluster Complex

K5M8W3 Teacher Resources

Dasara Holidays 24.09.25 to 02.10.25
Dasara Holidays for Minority Institutions
27.09.25 to 02.10.25

Week1: **4.1 – 4.3**

Week 2: **4.4 – 4.6**

Week 3: **4.7 – 4.9**

Week 4: **4.10**

Week 5:

↓ TEACHER'S NOTES ↓

Chapter: 4. Is Matter Around Us Pure?

C6U3F2

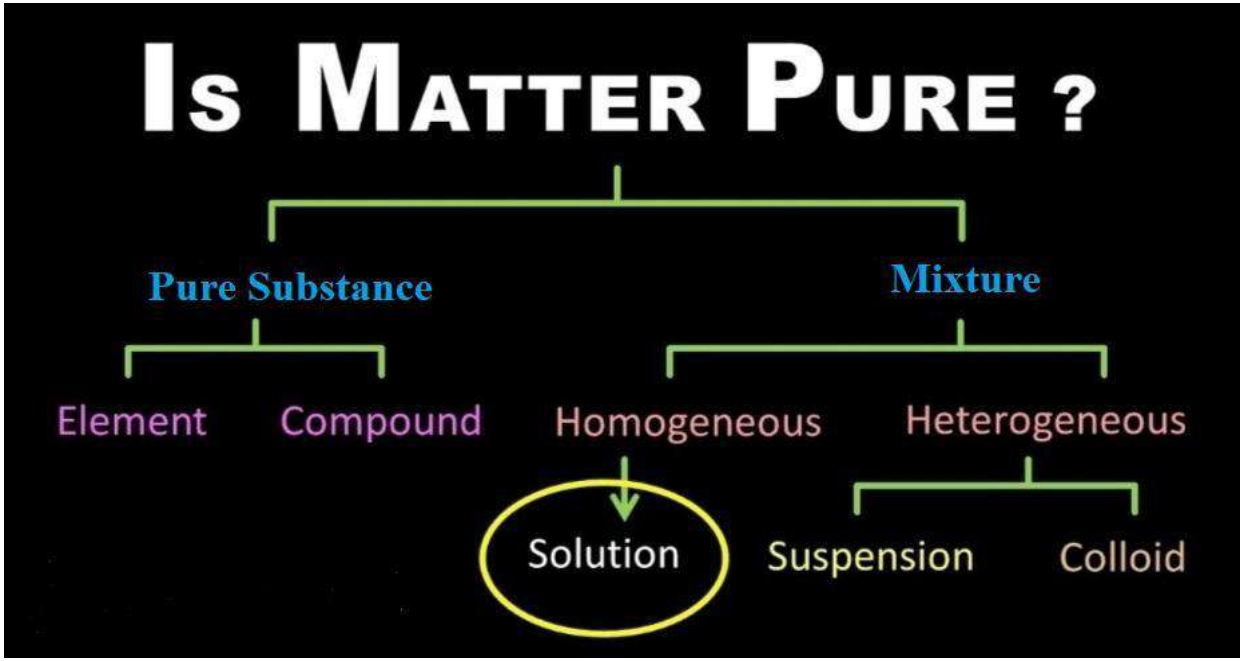
Learning Outcomes:

1. Differentiate element, compound and mixtures based on their properties.
2. Effectively communicate the findings and conclusions on the concentration of mixtures.
3. Apply scientific principles in solving problems of day to day life like separation of mixers.
4. Classify solutions, suspensions and colloids based on the properties.
5. Draw a flowchart on States of matter.

Prior Concepts/Skills:(Essential concepts and skills to be checked/bridged before teaching the current concept)

1. Name the method by which you can separate butter from milk.
2. Which method of separation is used for husk from wheat flour?
3. What is air called, if it is a combination of some gases?

Concept Map



Synopsis

- **Mixture:** A mixture contains more than one substance (element and/or compound) mixed in any proportion.
- **Types of mixtures:** homogeneous mixture, heterogeneous mixture
- **Solution:** appropriate separation techniques. • A solution is a homogeneous mixture of two or more substances.
- **Concentration of a solution:** The concentration of a solution is the amount of solute present per unit volume or per unit mass of the solution.
- **Suspension:** A suspension is a heterogeneous mixture.
- **Colloidal solution:** Colloids are heterogeneous mixtures in which the particle size is too small to be seen with the naked eye, but is big enough to scatter light
- **Tyndall effect:** Particles can easily scatter a beam of visible light as observed. This scattering of a beam of light is called the Tyndall effect
- **Types of pure substances** Pure substances can be elements or compounds.
- **Elements:** An element is a form of matter that cannot be broken down by chemical reactions into simpler substances.
- **Compounds** A compound is a substance composed of two or more different types of elements, chemically combined in a fixed proportion.

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
4.1	Introduction 4.1 What is a mixture 4.1.1 types of mixtures	divide the mixtures into different types	
4.2	Activity 4.1 , Activity 4.2	divide the given mixtures into compounds	
4.3	4.2 what is a solution? Properties of a solution	understand what is asolution and its types	
4.4	4.2.1 Concentration of a solution Activity 4.3	explain the concentration of solution	

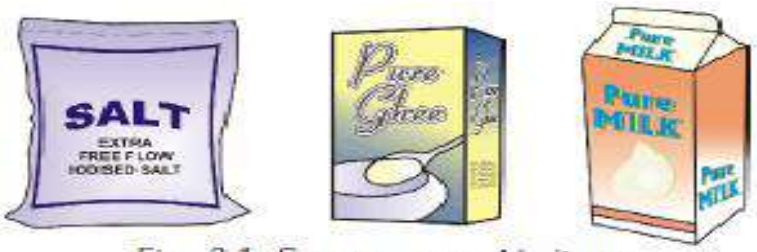
4.5	4.2.2 What is a suspension? Properties of suspensions	differentiate suspensions from mixtures	
4.6	4.2.3 What is a colloidal solution? Properties of colloid	explain the properties of colloidals	
4.7	4.3 Physical and chemical changes	classify the changes as physical and chemical	
4.8	4.4 What are the types of pure substances? 4.4.1 Elements	differentiate elements and compounds	
4.9	4.4.2 Compounds Activity 4.4	understand the properties of compounds	
4.10	Diffences between mixtures and compounds	differentiate mixtures and compounds	

PERIOD Number: 4.1

TOPIC: Introduction : 4.1 What is a mixture : 4.1.1 types of mixtures

Learning Outcome: Students will be able to divide the mixtures into different types

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Pure Substance	1. What is Matter? 2. How matter is made?  Fig. 2.1: Some consumable items	What is pure substance?

Mixture Explanation Separation of mixtures	3. How can we classify the matter? By showing the above pictures asks the following questions. And try to get the answers or answered by the teacher. 4. Answer the label of each packet shown. Does the pure milk is really pure substance? ✔ But, for a scientist all these things are actually mixtures of different substances and hence not pure. ✔ For example, milk is actually a mixture of water, fat, proteins, etc. ✔ When a scientist says that something is pure, it means that all the constituent particles of that substance are the same in their chemical nature. ✔ A pure substance consists of a single type of particle. In other words, a substance is a pure single form of matter. ✔ As we look around, we can see that most of the matter around us exists as mixtures of two or more pure components. ✔ for example, sea water, minerals, soil, etc., are all mixtures. **Mixture: -** ● Mixtures are constituted by more than one kind of pure form of matter. ● We know that dissolved sodium chloride can be separated from water by the physical process of evaporation. ● sodium chloride is itself a pure substance and cannot be separated by physical process into its chemical constituents. ● sugar is a substance which contains only one kind of pure matter and its composition is the same throughout.	Give examples of pure substances? What is mixture? Give examples of mixtures? How to separate the mixtures?

Types of mixtures	• Soft drink and soil are not single pure substances. Whatever the source of a pure substance may be, it will always have the same characteristic properties. • Therefore, we can say that a mixture contains more than one pure substance. **Types of Mixtures** Depending upon the nature of the components that form a mixture, we can have different types of mixtures. They are 1. heterogeneous mixtures. 2. heterogeneous mixtures.	How many types of mixtures?

Summary	Factual Questions	CFU	Home Work
Mixture contain more than one element or compound mixed in any proportion .	1. What is meant by a substance?	1. List the points of differences between homogeneous and heterogeneous mixtures.	Classify each of the following as a homogeneous or heterogeneous mixture. *soda water, wood, air, soil, vinegar, filtered tea.*

PERIOD Number: 4.2

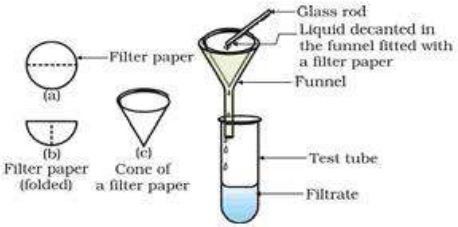
TOPIC: Activity 4.1 , Activity 4.2

Learning Outcome: Students will be able to divide the given mixtures into compounds

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions **Activity - 4.1**	• Let us divide the class into groups A, B, C and D. • Group A takes a beaker containing 50 mL of water and	

Explanation	one spatula full of copper sulphate powder. • Group B takes 50 mL of water and two spatula full of copper sulphate powder in a beaker. • Groups C and D can take different amounts of copper sulphate and potassium permanganate or common salt (sodium chloride) and mix the given components to form a mixture. • Report the observations on the uniformity in colour and texture. • Groups A and B have obtained a mixture which has a uniform composition throughout. Such mixtures are called homogeneous mixtures or solutions. Some other examples of such mixtures are: • salt dissolved in water and • sugar dissolved in water. • Compare the colour of the solutions of the two groups. Though both the groups have obtained copper sulphate solution but the intensity of colour of the solutions is different. This shows that a homogeneous mixture can have a variable composition. • Groups C and D have obtained mixtures, which contain physically distinct parts and have non-uniform compositions. • Such mixtures are called heterogeneous mixtures. Mixtures of sodium chloride and iron filings, salt and sulphur, and oil and water are examples of heterogeneous **Activity - 4.2** • Let us again divide the class into four groups— A, B,	Explain how mixtures are classified on the basis of their physical properties? What is homogeneous mixture?

Activity Explanation  	C and D. • Distribute the following samples to each group: - Few crystals of copper sulphate to group A. - One spatula full of copper sulphate to group B. - Chalk powder or wheat flour to group C. - Few drops of milk or ink to group D. • Each group should add the given sample in water and stir properly using a glass rod. Are the particles in the mixture visible? • Direct a beam of light from a torch through the beaker containing the mixture and observe from the front. Was the path of the beam of light visible? • Leave the mixtures undisturbed for a few minutes (and set up the filtration apparatus in the meantime). Is the mixture stable or do the particles begin to settle after some time? • Filter the mixture. Is there any residue on the filter paper? • Discuss the results and form an opinion. --- Groups A and B have got a solution. --- Group C has got a suspension. --- Group D has got a colloidal solution. we shall learn about solutions, suspensions and colloidal solutions.	What is heterogeneous mixtures? List the various types of homogeneous and heterogeneous mixtures?

Summary	Factual Questions	CFU	Home Work
Students will be able to divide the given mixtures into compounds	1. Water is a compound and not a mixture. List two reasons to justify your answer	Which separation techniques will you apply for the separation of the following? (a) Oil from water (b) Tea leaves from tea (c) Wheat grains from husk (d) Iron pins from sand	Which separation techniques will you apply for the separation of the following? (a) Sodium chloride from its solution in water (b) Small pieces of metal in the engine oil of a car (c) Different pigments from an extract of flower petals (d) (Butter from curd

PERIOD Number: 4.3

TOPIC: 4.2 What is a solution? Properties of a solution

Learning Outcome: Students will be able to understand what is a solution and its types

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Solution	What is a solution? ▪ A solution is a homogeneous mixture of two or more substances. ▪ **Examples of solution**: Lemonade, soda water, etc. ▪ Usually we think of a solution as a liquid that contains either a solid, liquid or a gas dissolved in it. ▪ But, we can also have solid solutions (alloys) and gaseous solutions (air). ▪ In a solution there is homogeneity at the particle level.	Give examples of solution. What are particles size in solution?

	For example, lemonade tastes the same throughout. This shows that particles of sugar or salt are evenly distributed in the solution. A solution has a solvent and a solute as its components.	What is solvent?
Solvent	**Solvent:** The component of the solution that dissolves the other component in it (usually the component present in larger amount) is called the solvent.	
Solute	**Solute:** The component of the solution that is dissolved in the solvent (usually present in lesser quantity) is called the solute.	What is solute?
Examples of solution	Examples: 1. A solution of sugar in water is a solid in liquid solution. In this solution, sugar is the solute and water is the solvent.	
Properties of a solution	**Properties of a Solution** ❖ A solution is a homogeneous mixture. ❖ The particles of a solution are smaller than 1 nm (10^{-9} metre) in diameter. So, they cannot be seen by naked eyes. ❖ Because of very small particle size, they do not scatter a beam of light passing through the solution. So, the path of light is not visible in a solution. ❖ The solute particles cannot be separated from the mixture by the process of filtration. The solute particles do not settle down when left undisturbed, that is, a solution is stable.	Write the solute and solvent in the solution of sugar in water? What are the properties of a solution?

Summary	Factual Questions	CFU	Home Work
Solvent: The component of the solution that dissolves the other component in it (usually the component present in larger amount) is called the solvent. **Solute:** The component of the solution that is dissolved in the solvent (usually present in lesser quantity) is called the solute.	What makes water as a universal solvent?	Explain Saturated solution and giving examples.	Write the steps you would use for making tea. Use the words solution, solvent, solute, dissolve, soluble, insoluble, filtrate and residue.

PERIOD Number: 4.4

TOPIC: 4.2.1 Concentration of a solution , Activity 4.3

Learning Outcome: Students will be able to explain the concentration of a solution

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Solubility	▪ Depending upon the amount of solute present in a solution, it can be called dilute, concentrated or saturated solution. ▪ At any particular temperature, a solution that has dissolved as much solute as it is capable of dissolving, is said to be a saturated solution. ▪ If the amount of solute contained in a solution is less than the saturation level, it is called an unsaturated solution. **What is solubility?** The amount of the solute present in the saturated solution at	What is a solution? How many types of solutions? What is solubility?

Concentration of a solution Methods **Example 4.1** A solution contains 40 g of common salt in 320 g of water. Calculate the concentration in terms of mass by mass percentage of the solution.	this temperature is called its solubility. What is concentration of a solution? The concentration of a solution is the amount (mass or volume) of solute present in a given amount (mass or volume) of solution. There are various ways of expressing the concentration of a solution, but here we will learn only three methods. (i) Mass by mass percentage of a solution (ii) Mass by volume percentage of a solution (iii) Volume by volume percentage of a solution Mass of solute (salt) = 40 g Mass of solvent (water)= 320 g	What is concentration of a solution? How express concentration of solution? $\frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$ $\frac{\text{Mass of solute}}{\text{Volume of solution}} \times 100$ $\frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$

Summary	Factual Questions	CFU	Home Work
Mass of solution = Mass of solute + Mass of solvent = 40 g + 320 g = 360 g Mass percentage of solution $= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$ $= \frac{40}{360} \times 100 = 11.1\%$	What is the effect of change of temperature on the solubility of a salt	Find the solubility of each salt at 293 K. Which salt has the highest solubility at this temperature	What mass of potassium nitrate would be needed to produce a saturated solution of potassium nitrate in 50 grams of water at 313 K?

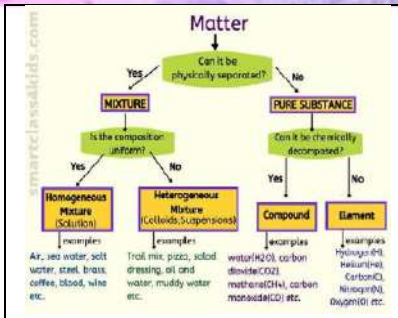
PERIOD Number: 4.5

TOPIC: 4.2.2 What is a suspension? Properties of suspensions

Learning Outcome: Students will be able to differentiate suspensions from mixtures

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions. Suspension	1) What is a heterogeneous mixture? 2) Give examples for heterogeneous mixtures?  **What is a suspension?** **(with reference to Activity-4.2)** ✔ The mixtures obtained by group A and B in activity 4.2 are homogenous mixtures. ✔ The mixture obtained by group C in activity 4.2 is a Suspension. 1. Is suspension homogeneous or heterogeneous in nature? 2. Are Non-homogeneous systems in which solids are dispersed in liquids, are called Suspensions? Explain	What is suspension?

 Explanation Properties of suspensions	3. How the solute particles present in the suspension? 4. Are the Particles of a suspension are visible to the naked eye? **Properties of a suspension:** ● Suspension is a heterogeneous mixture. ● The particles of a suspension can be seen by the naked eye. ● The particles of a suspension scatter a beam of light passing through it and make its path visible. ● The solute particles settle down when a suspension is left undisturbed, that is, a suspension is unstable. They can be separated from the mixture by the process of filtration. When the particles settle down, the suspension breaks and it does not scatter light any more.	Write the properties of Suspension. When will Suspension shows Tyndall and when does not shown?

Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate suspensions from mixtures	How are solution and suspension different from each other?	Write the properties of a suspension?	Explain Suspension and giving examples.

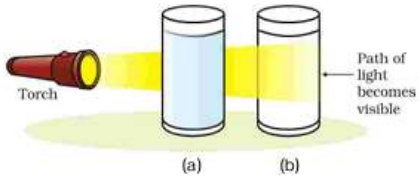
PERIOD Number: 4.6

TOPIC: 4.2.3 What is a colloidal solution? Properties of colloid

Learning Outcome: Students will be able to explain the properties of colloidal

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions	✔ The mixture obtained by group D in activity 4.2 is called a colloid or a colloidal solution.	

Colloidal solution Tyndall effect  Example of Tyndall effect	✔ The particles of a colloid are uniformly spread throughout the solution. ✔ Due to the relatively smaller size of particles, as compared to that of a suspension, the mixture appears to be homogeneous. ✔ But actually, a colloidal solution is a heterogeneous mixture. ✔ Example, milk. ✔ Because of the small size of colloidal particles, we cannot see them with naked eyes. ✔ But, these particles can easily scatter a beam of visible light as observed in activity 4.2. Tyndall effect:  ✔ This scattering of a beam of light is called the Tyndall effect ✔ Tyndall effect can also be observed when a fine beam of light enters a room through a small hole. ✔ This happens due to the scattering of light by the particles of dust and smoke in the air.	What is homogeneous mixture? What is colloidal solution? What is Tyndall effect? Give example of Tyndall effect?

Properties of a colloid	Tyndall effect can be observed when sunlight passes through the canopy of a dense forest. **Properties of a Colloid** o A colloid is a heterogeneous mixture. o The size of particles of a colloid is too small to be individually seen with naked eyes. o Colloids are big enough to scatter a beam of light passing through it and make its path visible. o They do not settle down when left undisturbed, that is, a colloid is quite stable. o They cannot be separated from the mixture by the process of filtration.	What are the properties of a colloid? What is dispersed phase in fog?

Dispersed phase	Dispersing Medium	Type	Example
Liquid	Gas	Aerosol	Fog, clouds, mist
Solid	Gas	Aerosol	Smoke, automobile exhaust
Gas	Liquid	Foam	Shaving cream
Liquid	Liquid	Emulsion	Milk, face cream
Solid	Liquid	Sol	Milk of magnesia, mud
Gas	Solid	Foam	Foam, rubber, sponge, pumice
Liquid	Solid	Gel	Jelly, cheese, butter
Solid	Solid	Solid Sol	Coloured gemstone, milky glass

Summary	Factual Questions	CFU	Home Work
The components of a colloidal solution are the dispersed phase and the dispersion medium. The solute-like component or the dispersed particles in a colloid form the dispersed phase, and the component in which the dispersed phase is suspended is known as the dispersing medium.	What is Tyndall effect?	Explain Colloid and giving examples.	Comparison properties of suspension and colloidal solution

PERIOD Number: 4.7

TOPIC: 4.3 Physical and chemical changes

Learning Outcome: Students will be able to classify the changes as physical and chemical

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Physical changes	By asking questions about (i). Rusting of Iron rod, (ii). Boiling of water to steam, (iii) Dissolving common salt in water and (iv). Blackening of Silver ornaments Teacher introduces the concept Physical and Chemical changes. 1). What happen to Iron nail when exposed to moisture and air? 2. What do you observe while boiling of water to steam? 3) Name the colour of old and new Silver ornaments. **Activity-1** By doing lab activity (If possible) or Showing the different physical states of water (Ice, Water & Water vapour), the interconversion of Ice to water at room temperature and Water to Water vapour at boiling point of water. 1) What is inter-convertible temperature of (i). ice to water? ----------	1. What is a physical change? Give two examples. 1. What is a chemical change? Give two examples. 3) How do you distinguish water from cooking oil based on their nature? 4) Which one of the following highly inflammable (a). Coconut oil (b) water. (c). Custard oil (d) Both a&b. 1. Classify the following as chemical or physical changes: • cutting of trees, • melting of butter in a pan, • rusting of metal window,

Chemical changes By doing lab activity (if possible) or asks the students to observe the boiling of water and boiling of cooking oil (sunflower oil or groundnut oil or rice bran oil) at home.	(ii). Water to water vapour? ---------- 2). How the kinetic energy of matter changes, when (i). Ice to water (increases/decreases) (ii) Water to water vapour------ (decreases/increases 3). Which type of change is the change of Ice to water? **Activity-2** Make the pupils to write"what are the differences observed like odors, burning nature of cooking oil, fire extinguish nature water etc……" 1). Do you observe any odour changes during cooking edible oil? 2). What happens to the water molecules while boiling? 3). What are the differences when boiling/cooking water/ edible oil?	• boiling of water to form steam, passing of electric current, through water and the water breaking down into hydrogen and oxygen gases, • dissolving common salt in water, • making a fruit salad with raw fruits • burning of paper and wood. Similarly, probe the following questions 4) Which type of changes takes place in the below examples 1). When cut the apple? 2). when lemon juice fall on marble floor? 3). When wood burns? By discussing the above questions and answers, Teacher discusses and explains the topic physical and chemical changes.

Summary	Factual Questions	CFU	Home Work
Students will be able to classify the changes as physical and chemical	Do the energy changes take place both in physical and chemical changes?	Write one example in which both physical and chemical change take place.	**Classify the following as chemical or physical changes:** • cutting of trees, • melting of butter in a pan, • rusting of almirah.

PERIOD Number: 4.8

TOPIC: 4.4 What are the types of pure substances?

4.4.1 Elements

Learning Outcome: Students will be able to differentiate elements and compounds

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions Compound Element History On the basis of their chemical composition, substances can be classified either as elements or compounds. ✔ Robert Boyle was the first scientist to use the term element in 1661. ✔ Antoine Laurent Lavoisier (1743–94), a French chemist, was the first to establish an experimentally useful definition of an element. ✔ He defined an element as a basic form of matter that cannot be broken down into simpler substances by chemical reactions.	1). Name the matter that contains same type of particles. 2). What are the differences do we observe while writing the formulae of hydrogen as H_2 and water as H_2O? **Properties of metals** ✔ Metals usually show some or all of the following properties: ✔ They have a lustre (shine). ✔ They have silvery-grey or golden-yellow colour. ✔ They conduct heat and electricity. ✔ They are ductile (can be drawn into wires). ✔ They are malleable (can be hammered into thin sheets). ✔ They are sonorous (make a ringing sound when hit). **Examples of metals** : gold, silver, copper, iron, sodium, potassium etc. ✔ Mercury is the only metal that is liquid at room temperature. **Properties of Non-Metals:** ✔ They display a variety of colours.	What is an element? Write the properties of metals? Give examples of metals? Write the properties of Non-Metals? Give examples of Non-Metals? What are metalloids?

✔ Elements can be normally divided into metals, non-metals and metalloids. Properties of metals Properties of Non-metals Metalloids	✔ They are poor conductors of heat and electricity. ✔ They are not lustrous, sonorous or malleable. **Examples of non-metals:** hydrogen, oxygen, iodine, carbon (coal, coke), bromine, chlorine etc. Some elements have intermediate properties between those of metals and non metals, they are called metalloids **Examples:** boron, silicon, germanium, etc.	

Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate elements and compounds	Why is Hydrogen considered an element?	Which of the following materials fall in the category of a "pure substance"? (a) Ice (b) Milk (c) Iron (d) Hydrochloric acid (e) Calcium oxide (f) Mercury (g) Brick (h) Wood (i) Air	Explain pure substance giving examples.

PERIOD Number: 4.9

TOPIC: 4.4.2 Compounds

Activity 4.4

Learning Outcome: Students will be able to understand the properties of compounds

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions.	1). What is the difference in the composition of H_2 gas and H_2O?	1). How compound different from element?
	2). How water is formed by combining atoms of which elements?	
	3). What do we get when two or more elements combined in affixed ratio?	2). Classify the following into elements, compounds and mixtures.
Compounds	Compound is a substance composed of two or more elements in a fixed ratio.	(a) Sodium (b) Soil (c) Sugar solution (d) Silver (e) Calcium carbonate (f) Tin (g) Silicon (h) Coal
	Activity 4.4:	
Activity	Divide the class in to two groups. Give 5 g of Iron filings and 3 g of Sulphur powder in a China dish to both the groups. Ask	(i) Air (j) Soap (k) Methane (l) Carbon dioxide (m) Blood
	Group-I: students to mix and crush Iron filings and Sulphur powder and	3). How would you confirm that a colorless liquid given to you is pure water?
	Group-II: Students to mix crush Iron filings and Sulphur powder and heat the contents strongly till	
	(1) red hot. Remove flame and let the contents cool. Make the two groups students to check the magnetism of the material obtained by them respectively by bring a magnet near the material, whether attracted or not attracted towards the magnet.	4). Write the differences between Mixtures and Compounds
	(2) Ask the students of two groups divide the material obtained respectively into 2 parts, to one part add carbon disulphide and add dil hydrochloric acid to the other part.	
	Group-I students observe the insolubility of material in carbon	

	disulphide and the liberation of hydrogen gas, whereas group-II students observe the solubility of material in carbon disulphide and the liberation of hydrogen sulphide gas **Now answer** 1). Did the material obtained by the two groups look the same? 2). Which group has obtained a material with magnetic properties? 3). Can we separate the components of the material obtained? 4). On adding dilute sulphuric acid or dilute hydrochloric acid, did both the groups obtain a gas? 5). Did the gas in both the cases smell the same or different? **Conclusion** Based on the varied behavior of the materials obtained by the two group students, they conclude that "physical change leads to the formation of mixture whereas the chemical change leads to the formation of compound.	

Summary	Factual Questions	CFU	Home Work
Students will be able to understand the properties of compounds	Is copper sulphate is said to be a compound?	Define the term Compound?	Classify the following into elements, compounds and mixtures. (a) Sodium (b) Soil (c) Sugar solution (d) Silver (e) Calcium carbonate (f) Tin (g) Silicon (h) Coal (i) Air (j) Soap (k) Methane (l) Carbon dioxide (m) Blood

PERIOD Number: 4.10

TOPIC: Difference between mixtures and compounds

Learning Outcome: Students will be able to differentiate mixtures and compounds

Teaching Learning Process

I DO	WE DO	YOU DO
Asking previous knowledge questions. Mixtures Compounds	What is mixture? What is compound? How many types of mixtures? What are they? **Differences between mixtures and Compounds** **Mixtures** 1. Elements or compounds just mix together to form a mixture and no new compound is formed 2. A mixture has a variable composition 3. A mixture shows the properties of the constituent substances 4. The constituents can be separated fairly easily by physical methods **Compounds** 1. Elements react to form new compounds	

2. The composition of each new substance is always fixed.
3. The new substance has totally different constituent substances. properties.
4. The constituents can be separated only fairly easily by physical methods. by chemical or electrochemical reactions.

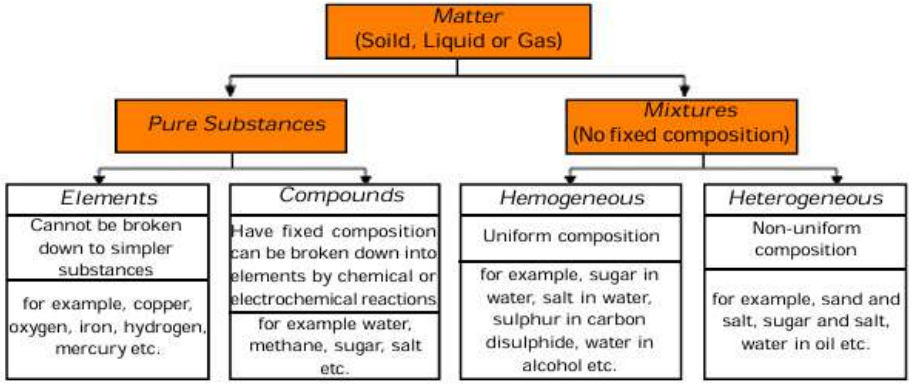
✔ The material obtained by group II is a compound.

✔ On heating the two elements strongly we get a compound, which has totally different properties compared to the combining elements.

✔ The composition of a compound is the same throughout.

✔ We can also observe that the texture and the colour of the compound are the same throughout.

Mixtures	Compounds
1. Elements or compounds just mix together to form a mixture and no new compound is formed.	1. Elements react to form new compounds.
2. A mixture has a variable composition.	2. The composition of each new substance is always fixed.
3. A mixture shows the properties of the constituent substances.	3. The new substance has totally different properties.
4. The constituents can be seperated fairly easily by physical methods.	4. The constituents can be separated only by chemical or electrochemical reactions.



Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate mixtures and compounds	How do you establish that air is a mixture and not a pure substance?	Write the characteristics of compounds.	Write the differences between Mixtures and Compounds.

Teacher's Diary

Name of the Teacher:		**Name of the Month:**	
Name of the Lesson: 4. Is Matter Around Us Pure?		Class: 9	

Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	**Remarks**
4.1	Introduction 4.1 What is a mixture 4.1.1 types of mixtures				
4.2	Activity 4.1 , Activity 4.2				
4.3	4.2 what is a solution? Properties of a solution				
4.4	4.2.1 Concentration of a solution Activity 4.3				
4.5	4.2.2 What is a suspension? Properties of suspensions				
4.6	4.2.3 What is a colloidal solution? Properties of colloid				
4.7	4.3 Physical and chemical changes				
4.8	4.4 What are the types of pure substances? 4.4.1 Elements				
4.9	4.4.2 Compounds Activity 4.4				
4.10	Differences between mixtures and compounds				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature **Teacher's Signature**

Head Teacher's Suggestions:

Teacher Notes:

October 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
 No bag Day	 Cluster Complex	 Teacher Resources	1 Dussera Holiday	2 Mahatma Gandhi Jayanti Vijayadashami	3 5.1	4 No Bag Day Sakunthala Devi Jayanti
5 Sunday International Teacher's Day	6 5.2	7 5.3	8 5.3	9 NI	10 NI	11 Second Saturday
12 Sunday	13 FA - 2	14	15	16	17 5.4	18 No Bag Day Cluster meeting
19 Sunday	20 Deepavali	21 5.5	22 5.6	23 5.7	24 NI	25 No Bag Day SMC Meeting
26 Sunday	27 5.8	28 5.9	29 5.10	30 5.11	31 National Unity Day NI	

Week1: **5.1**

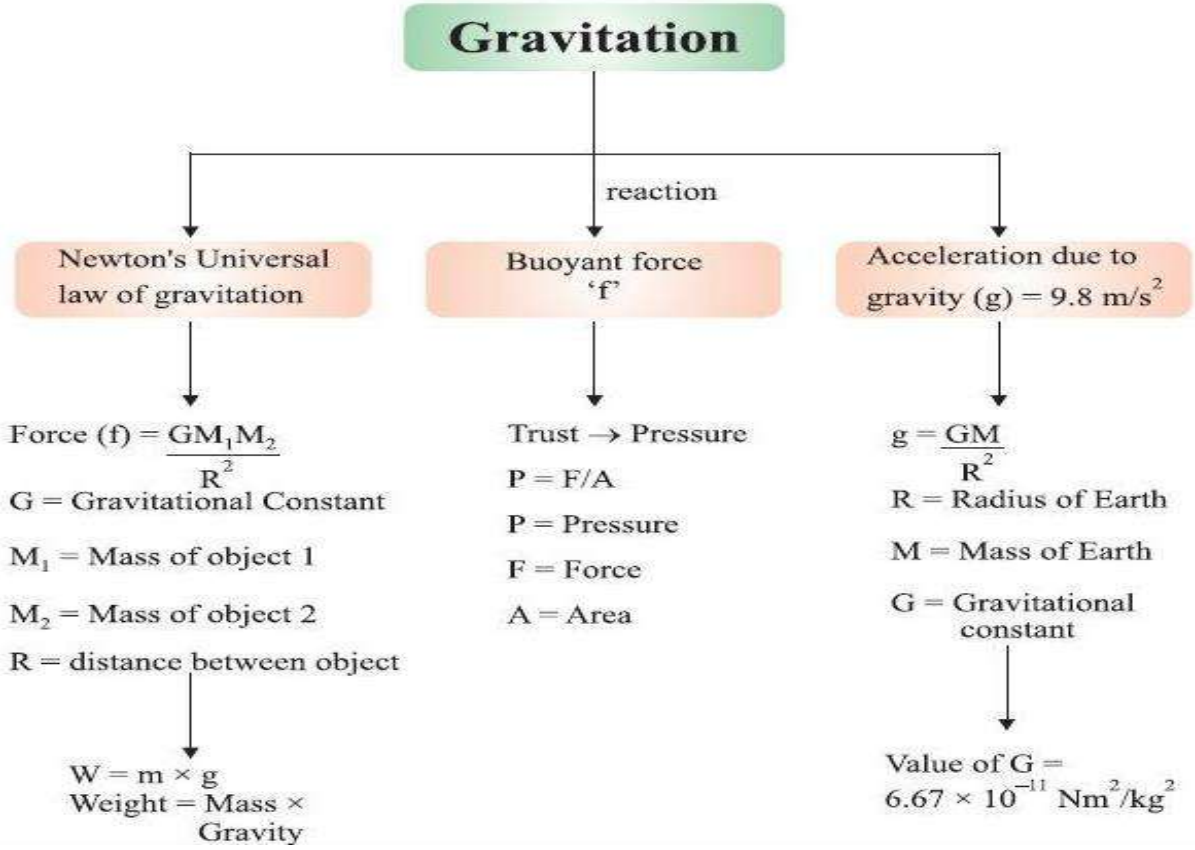
Week 2: **5.2 – 5.3**

Week 3: **5.4**

Week 4: **5.5 – 5.7**

Week 5: **5.8 – 5.11**

↓ TEACHER'S NOTES ↓



Synopsis

- **Gravitation:** It is the force of attraction between any two bodies in the universe.
- **Universal law of Gravitation:** The law of gravitation states that the force of attraction between any two objects is proportional to the product of their masses and inversely proportional to the square of the distance between them. The law applies to objects anywhere in the universe.
- **Free Fall:** The motion of a body under the influence of gravity alone is called a free fall
- **Mass:** The mass of a body is the quantity of matter contained in it.
- **Weight:** The weight of a body is the force with which the earth attracts it. The weight is equal to the product of mass and acceleration due to gravity

- **Thrust:** The total force exerted by a body on any surface in contact with it is called thrust.
- **Pressure:** The thrust acting per unit area of the surface of a body is called pressure
- **Buoyancy:** The upward force acting on a body immersed in a fluid is called upthrust or force of buoyancy.
- **Float or Sink concepts:** Objects having density less than that of the liquid in which they are immersed, float on the surface of the liquid. If the density of the object is more than the density of the liquid in which it is immersed then it sinks in the liquid.
- **Archimedes' Principle:** When a body is immersed fully or partially in a fluid, it experiences an upward thrust equal to the weight of the fluid displaced by it.

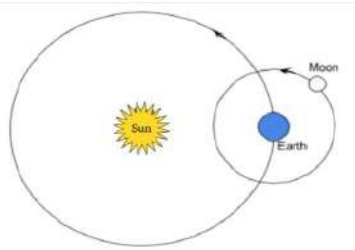
Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes
5.1	5.1 Gravitation A ctivity 5.1	derive the formula of universal law of gravitation
5.2	5.1.1 Universal law of gravitation , example 5.1	understand the concept of gravitation
5.3	5.1.2 Impotance of universal law of gravitation 5.2 Free fall Activity 5.2	explain the importance of gravitation
5.4	.5.2.1 To calculate the value of g 5.2.2 Motion of objects under the influence of gravitational force of the earth	calculates the value of g by using the given values
5.5	Example 5.2 and Example 5.3	Do calculations related to g and G
5.6	5.3 Mass 5.4 weight	differentiate mass and weight
5.7	5.4.1 weight of an object on the moon Ex.5.4 , 5.5	Calculate the g value on moon
5.8	5.5 Thrust and pressure Example 5.6	relates thrust and pressure
5.9	5.5.1 Pressure in fluids 5.5.2 Buoyancy Activity 5.4 5.5.3 Why objects float or sink when placed on the surface of water	explains the concept of buoyancy
5.10	5.5.3 Why objects float or sink when placed on the surface of water Activity 5.5 and Activity 5.6	give reasons for sinking and floating of bodies
5.11	5.6 Archimedes principle Activity 5.7	explain archimedes principle in his own words

PERIOD Number: 5.1 **TOPIC: 5.1 Gravitation** **Activity 5.1**

Learning Outcome: Students will be able to derive the formula of universal law of gravitation

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory activity:  1.what is position of the bicycle? 2.is it at rest or in motion? 3.Is there any change in its position? 4.what causes the change of position of the bicycle? 1.Teacher shows above picture and video and discuss (teacher splits students into group and ask them to discuss on the following questions) **(Teacher gives more example.)** 1. What are these objects? 2. Where are they seen? 3. Do tese objects move?	 **Activity 5.1:** (Teacher make groups and ask students to do this activity) 1.What will happen if we tie this stone and revolve? 2.Why it revolves? 3.Who was pushing ? 4.What is the shape of the path when it is moving? 5.What is the reason for revolving? 6.What happens if you leave the thread? (Discuss about the direction of the path of the stone) ● What do we need to change the direction of an object? (student's response: force) ● In this case tensen force changing the	What is the cause of motion? What is the name of the force which attracts the objects towards earth or sun? What is the name of the force which is generated in the thread?

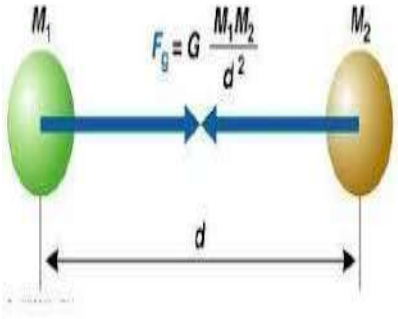
4. How do they move? Why? 5. Why it is not fallen?	direction stone. • This force which changes direction of object continuously is called centripetal force. • What if we cut the thread ?/ what if if this force is not acting?(stone will travel in straight path) • This straight path will be a tangent to the circle. • **Let's check earth –moon system once more**: • Now, do you find any similarity? (response: moon is revolving around the earth just like the stone in the activity) • This force which is holding the moon is centripetal force generated by the attraction between earth and moon. • Let's seen the solar system also. • What is the center of the solar system, and what revolves around it? (student's response: sun is the center and palnets revolve around the sun) • Can you guess why planets revolve around the sun? Just like earth –moon system there exist an attraction force which is nothing but gravitational force between sun and the planes.	What is centripetal force? What is tangent drawn to circle? Prepare a chart having examples from real life or systems around you. Incidents we can see gravitational force and centripetal force. Do you see any connection between movement of stone and the movement of earth/moon around sun/earth?

Summary	Factual Questions	CFU	Home Work
Students are able to derive the formula of universal law of gravitation and solve the problems on Universal law of gravitation.	Is gravitation is available in the universe?	1. Define the term gravitation 2. Define the term gravity	What is centripetal force?

PERIOD Number: 5.2 TOPIC: 5.1.1 Universal law of gravitation Example 5.1

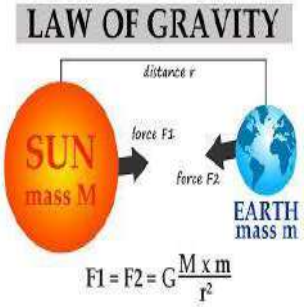
Learning Outcome: Students will be able to understand the concept of gravitation

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory activity: Previous knowledge questions: 1.How can we walk in the ground? 2. What force acting to makes us to walk on the ground? 3. What force is caused to occur tides in the ocean? 4. What is the force responsible for the revolution of the moon? **(teacher activity)** Showing the picture on IFP: - 1. Between twoobjects can gravitational force be changed?	 The force between two objects is directly proportional to the product of their masses. $f \propto m_1 * m_2$	What is the relationship between force and masses? What the relationship between force and distance between the objects? What is the units of force in SI system? (newton)

2. If we go on the moon,what changes we see?
3. Why, force between you and moon is less?

Why,force between you and earth is more?



Is there any other way to decrease the force

Q.What happens when we move far away from the earth?

The force between two objects is inversely proportional to the square of the distance between them.

$$f \propto \frac{1}{d^2}$$

Combining both equations , we get

$$f \propto \frac{m_1 * m_2}{d^2}$$

Where G is the constant of proportionality and is called the universal gravitation constant.

Let's write units for G in SI system

How can you re arrange the formula to calculate G?

Expected answer:

$$G = \frac{f d^2}{m_1 m_2}$$

So, what is the units of G?

$N\ m^2\ kg^{-2}$.

Can anyone guess why it is called as universal law?

(Applicable to all bodies, big or small, whether they are celestial or terrestrial.)

What is the units of force in SI system? (newton)

What are units of mass in SI system? (kg)

What are the units of distance in SI system? (m)

What is the formula for force?

The mass of the earth is $6\ X10^{24}$ kg and that of the moon is $7.4X10^{22}$ kg. If the distance between the earth and the moon is $3.84X10^5$ km, calculate the force exerted by the earth on the moon.

(Take G = $6.7X10^{-11}\ Nm^2\ kg^{-2}$)

$2.02X10^{20}$ N

	Example 5.1: As a student to read the problem which is displayed on IFP. What is the mass of the earth given in problem? ($M=6X10^{24}kg$) What is the mass of the moon? ($m=7.4X10^{22}kg$) What is the distance between moon and earth given? ($3.84X10^{5}km=3.84X10^{8}m$) **Let's substitute:** In the formula $$f = G\frac{m_1 * m_2}{d^2}$$ What will be the answer?	

Summary	Factual Questions	CFU	Home Work
Students are able to understand the concept of gravitatio	1. Why is Newton's law of gravitation called the universal law? 2. Do the gravitational forces obey Newton's third law of motion?	What happens to the force between two objects, if (i) the mass of one object is doubled? (ii) the distance between the objects is doubled and tripled? doubled?	How does the force of gravitation between two objects change when the distance between them is reduced to half ?

PERIOD Number: 5.3 **TOPIC: 5.1.2 Importance of the universal law of gravitation**

5.2 Free fall **Activity 5.2**

Learning Outcome: Students will be able to explain the importance of gravitation

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory activity: To recall the concept of gravitation,the following questions may be probed. 1. What is the force that binds us to the earth? 2. Why planets revolving around the Sun? **3.** why tides are forming? **(Teacher will demonstrate the activity)**	Today we will try to get an idea on **free fall**. 1. what happens when you through a stone in the upward? ✔ Why stone is falling down? ✔ What are the forces acting on the stone while it is falling down? ✔ The motion of an object only under the influence of gravitational force itself is called free fall. ✔ While it is falling down is there any change in its direction? ✔ Is there any change in its magnitude? ✔ (Teacher introduces acceleration) ✔ This acceleration is called as acceleration due to gravity – reason is very clear because of gravity this acceleration generated. ✔ Do any one know—what is the notation of acceleration due to gravity is? **Relationship between G and g**	What is free fall? How do we to change the velocity of an object? what is newton's second law mathematically? In free fall what is the acceleration? Q. The gravitational force between two objects is F. If the masses of both objects are halved without changing the distance between them, then the gravitational force would become a. F/4 b. F/2 c. F d. 2F

	Now in free fall- how to express force according to newton's second law? F=mg (a=g) We know formula for universal gravitation. $F = G\frac{m_1 * m_2}{d^2}$ By equation above equations we will get $mg = G\frac{M * m}{d^2}$ M gets cancelled gives us $g = G\frac{M}{d^2}$ If we are near to earth we can replace d with R Leads us to $g = G\frac{M}{R^2}$	what is newton's second law mathematically? In free fall what is the acceleration? How do we express universal law of gravitation mathematically?

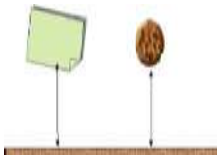
Summary	Factual Questions	CFU	Home Work
Students are able to explain the importance of gravitation	1.Does the value of G depend on the medium present between the two bodies? 2.Is it possible to shield a body from gravitational field?	1.Gravitational force acts on all objects in proportion to their masses. Why then, a heavy object does not fall faster than a light object	1.What is the importance of universal law of gravitation? 2. What is the acceleration of free fall?

PERIOD Number: 5.4 **TOPIC: 5.2.1 To calculate the value of g**

5.2.2 Motion of objects under the influence of gravitational force of the earth

Learning Outcome: Students will be able to calculates the value of g by using the given values.

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory activity: 1. What is themathematical expression newton's second law ? 2. Based on 2nd law of motion,can you Write equation for weight? 3. What is the relation between G and g?	$g = G\frac{M}{R^2}$ Now let's try to calculate the value of 'g': What are the terms in the above equation: 1. G= universal gravitation constant 2. M=mass of earth 3. R= radius of earth. On solving we can get the value of acceleration due to gravity g=9.8 m/sec^2 1. Acceleration due to gravity (5.2.2) 2. 'g' doesn't depend on mass of object. Now let's check this with an activity  Take a sheet of paper and a stone, drop them simultaneously from some height. 1. What do you observe? (stone reached first)	Does the value of g depends on mass of object? How can you say that? What is the value of g?

	2. But according to the theory who should reach first ?(both should reach at one time) 3. What may be the reason for paper to reach lately/who is stopping the paper?(air resistance) What if we do the same activity without the presence of air? Both paper and stone reach the ground at the same time. Let's watch this experiment in the absence of air.	

Summary	Factual Questions	CFU	Home Work
Students are able to calculates the value of g by using the given values. And also observe the variation of g	Is there any change in the velocity of a freely falling body?	Differentiate between g and G in a tabular form	Calculate the g value on the surface of earth

PERIOD Number: 5.5 TOPIC: Example 5.2 and Example 5.3

Learning Outcome: Students will be able to Do calculations related to g and G

Teaching Learning Process

I DO	WE DO	YOU DO
Let's solve a problem: **Example 9.2** 1. A car falls off a ledge and Drops to the ground in 0.5 s. Let g = 10 m s–2 (for simplifying the calculations). I. What is its speed on striking the ground?	Time, t = ½ second Initial velocity, u = 0 m s^{-1} Acceleration due to gravity, g = 10ms^{-2} Acceleration of the car, a = + 10 m s^{-2} Do you remember the avg. speed	

II. ii) What is its average speed during the 0.5 s? III. How high is the ledge from the ground? **Example 5.3** An object is thrown vertically Upwards and rises to a height of 10 m. Calculate (i) the velocity with which the object was thrown upwards and (ii) the time taken by the object to reach the Highest point.	formula: Yes- $v_{avg}=u+v/2$ Similarly find out displacement – identify which equation is suitable for finding displacement. Yes, you can use second or third as well. Distance travelled, s = 10 m Final velocity, v = 0 m s^{-1} Acceleration due to gravity, g = 9.8 m s^{-2} Acceleration of the object, a = –9.8 m s^{-2}	s = ½ a t2 = ½ * 10 m s–2 *(0.5 s)2 = ½ *10 m s–2 *0.25 s2 = 1.25 m (i) $v^2 = u^2 + 2as$ $0 = u^2 + 2 \times (-9.8\ m\ s^{-2}) \times 10\ m$ $-u^2 = -2 \times 9.8 \times 10\ m^2 s^{-2}$ $u = \sqrt{196}\ m\ s^{-1}$ $u = 14\ m\ s^{-1}$ (ii) $v = u + at$ $0 = 14\ m\ s^{-1} - 9.8\ m\ s^{-2} \times t$ $t = 1.43\ s.$ Thus, (i) Initial velocity, $u = 14\ m\ s^{-1}$, and (ii) Time taken, $t = 1.43\ s.$

Summary	Factual Questions	CFU	Home Work
Students know the relation between g and G and do the problems.	A ball is thrown vertically upwards with a velocity of 49 m/s. Calculate (i) the maximum height to which it rises, (ii) the total time it takes to return to the surface of the earth.	1. Write the equations of motion. 2. What is mass of the earth 3. What is the SI unit of g	What is the magnitude of the gravitational force between the earth and a 1 kg object on its surface? (Mass of the earth is 6 10^{24} kg and radius of the earth is 6.4 x10^6 m.)

PERIOD Number: 5.6 TOPIC: 5.3 Mass 5.4 Weight

Learning Outcome: Students will be able to differentiate mass and weight

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory Activity	❖ What are objects in our class room at rest? (Chair, Table, Desk, IFP Board) ❖ What are the objects moving in the class room ? (By showing moving fan) ❖ Do they change their state of rest or motion ❖ Why they resist change in its state of motion or rest ❖ What we call the natural tendency of the object to resist the change in its state of motion or rest. ❖ How the inertia related to mass Let us explore mass and weight in this chapter Why an elephant is heavier that leaf (The amount of matter in elephant is more than leaf) ● The measure of amount of matter in a body is called (Mass) ● Does the mass of the object change place to place? (No) ● Mass remains same when the object is on the earth, the moon or in orator space. ● What does not change from place to place. ● What is the difference between mass and weight? ● On what factors the gravitational force depends ?	What is the reason for state of rest? What is the reason for state of the motion? What is inertia of motion or rest? What is the measure of inertia (mass of an object)? What is mass ?

Teachers statement:

- The mass of an object is the measure of inertia

ACTIVITY:

- What is the equation newton second law of motion?

  F= ma

- If you replace a(acceleration) with g(=acceleration due to gravity) it will become earth's gravitational force.

  F=mg

- The force of attraction of earth on an object is called as (Weight)

  F = W

  Therefore W = mg

- In which direction the force is acting.
- Does it has magnitude
- What is the SI unit of Weight?
- In W=mg what is constant an given place?

  Therefore $W \propto m$

- At a given place what is measure of its mass

Mass is the amount of matter in a body	The force of attraction on an object
It is denoted by M	It is denoted by W
It is a scalar quantity (it has no directions	It is a vector quantity (it has both magnitude and direction)
SI Unit of mass is Kg	SI unit of Weight is newton (N)

SI unit of Mass is _____

What remains the same whether the object on earth, the moon, or even in outer space?

Does the mass of object change from place to place?

What is gravitational force?

What is Weight?

What is the formula or equation used to find the weight of the object?

SI Unit of weight is ____?

is weight a vector quantity?

If a physical quantity is having both magnitude and direction what we call it as ?

		What is dependent on its location (weight) What are the differences between mass and Weight? What are the differences between mass and Weight?

Summary	Factual Questions	CFU	Home Work
Students are able to differentiate mass and weight	1. Can the mass of a body be zero? 2. How does the mass of a body change when it is taken into deep space?	Why will a sheet of paper fall slower than one that is crumpled into a ball?	1. Define Mass of a body 2. Define weight of a body 3. What is the formula of weight 4. Write the differences between mass and weight.

PERIOD Number: 5.7 TOPIC: 5.4.1 weight of an object on the moon Example.5.4 , 5.5

Learning Outcome: Students will be able to Calculate the g value on moon

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory Activity	❖ Can we walk on earth? ❖ Can we walk on the moon as we walk on the earth? ❖ Why can't we walk normally on the moon? ✔ Gravity on any celestial body(planet,moon,stars)depends up	On what factors the gravity of a celestial body depends?

	on the mass (m) and size of the celestial body. ✔ Since the mass and size of the moon is very less compared to earth we can say that gravity on moon is very less compared to earth. ✔ Due to this the moon exerts lesser force of attraction on objects. ✔ Let the mass of the object be --- m ✔ Let the weight on the moon be --- W_m ✔ Let the mass of the moon be --- M_m ✔ It radius be --- R_m ❖ What is the universal law of gravitation $(F= G \frac{m_1 m_2}{r^2})$ ✔ By applying universal law of gravitation. ✔ The weight of the object on the moon will be $W_m= G.\frac{M_m.m}{R^2m}$ ---① ❖ Let the weight of the same object on earth be --- W_e ✔ The mass of the earth is ----M ✔ Radius of earth is --- --------R ❖ Weight of the object on the earth $W_e =mg$ ------② We know that $g= G.\frac{M}{R^2}$ -----③	Which formula is used to find the weight of object on moon? where our weight will be more on earth or on moon? Where we can jump high? Why can't we walk as smooth as on earth? **An object weighs 10 N when measured on the surface of the earth** Lets estimate/guess will it be higher or lesser than earth? What would be its weight when measured on the surface of the moon?

❖ Let's check consept of weight of an object on the moon.

Example 9.4

Mass of an object is 10 kg. What is its weight on the earth?

from equations ② & ③

$W_e = mg$

$= m. G. \frac{M}{R^2}$

$W_e = G. \frac{M.m}{R^2}$ ----- (4)

❖ On substituting the values from the table in equation ① & (4)

$W_m = G. \frac{7.36 \times 10^{22} \text{ kg} \times m}{(1.74 \times 10^6 \text{ m})^2}$

$= 2.431 \times 10^{10} \text{ G} \times m$ ----(5)

And $W_e = 1.474 \times 10^{11} \text{ G} \times m$ -----(6)

On dividing equation --(5) by (6)

We get

$\frac{W_m}{W_e} = \frac{2.431 \times 10^{10}}{1.474 \times 10^{11}}$

$= 0.165 = \frac{1}{6}$ ---------(7)

$= \frac{\text{weight of the object on the moon}}{\text{Weight of the object on the earth}} = \frac{1}{6}$

Actually gravity on moon is 6 times lesser than the gravity on earth and we also fell 6 time lighter on the moon's surface. Hence we are not pulled towards the surface of the moon as effectly as we are pulled towards surface of the earth that's why it's very hard to walk on the moon.

We know,
Weight of object on the moon
= (1/6) × its weight on the earth.
That is,

$W_m = \frac{W_e}{6} = \frac{10}{6}$ N.

= 1.67 N.

Example 9.5 An object weighs 10 N when measured on the surface of the earth. What would be its weight when measured on the surface of the moon?	Mass, m = 10 kg Acceleration due to gravity, g = 9.8 m s^{-2} W = m g W = 10 kg x 9.8 m s^{-2} = 98 N Thus, the weight of the object is 98 N.	

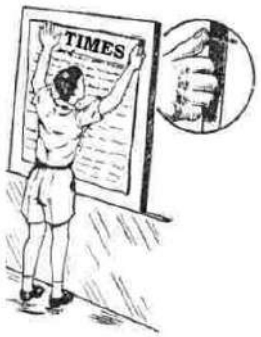
Summary	Factual Questions	CFU	Home Work
Students are able to Calculate the g value on moon.	Can a body have mass, but no weight? Give reason	Gravitational force on the surface of the moon is only 1 /6 as strong as gravitational force on the earth. What is the weight in newtons of a 10 kg object on the moon and on the earth?	If the moon attracts the earth, why does the earth not move towards the moon?

PERIOD Number: 8 **TOPIC: 5.5 Thrust and pressure Example 5.6**

Learning Outcome: Students will be able to relates thrust and pressure

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory Activity	❖ What do you observe from the picture? ❖ What is acting on nail? ❖ In which direction the force is acting? ❖ What we call the force acting on an object perpendicular to the surface (thrust) **Situation 1** ✔ What is fixed on the bulletin board (poster) ✔ To fix poster what are used?	

Let us explore thrust and pressure the pin.  	✔ What is applied on the surface area of the head of pins? ✔ In which direction the force is applied? ✔ What we call this force? ✔ Is thrust vector quantity? ✔ Why it is called vector? ✔ the S.I.unit of thrust (Newton) ✔ A thrust is a force but not every force can be counted as thrust ❖ **situtaion 2** ✔ when you stand loss sand what happens(feet go deep into the sand) ✔ if you lie down on the same sand dose your body go into the sand deep? (will not go that in the sand) ✔ In both cases the force exerted on the sand is weight of the body ✔ Here what is acting perpendicular to the sand (a force) ✔ The force acting on the object perpendicular to surface is called thrust ✔ In the above what is same (thrust) ✔ What are different effects are different ✔ the effect of thrust on sand is larger while standing than while lying **Pressure** ✔ Why the JVK bags have wide straps so? ✔ Why does a sharp knife cut object more effectively than a blunt knife ? ✔ Why dose the tip of sewing needle is sharp? ✔ Why it is easier to walk on soft sand if we have flat shoes rather than shoes with sharp heels? ✔ Why the tractor have broad tyres ?	1.force – more force more pressure. 2.Area – more area less pressure.

The following diagram gives the formula for pressure: pressure = force ÷ area.  	✔ Why the foundations of buildings and dams are laid on larger area ? ✔ To answer all these questions let us explore about pressure. ❖ The Thrust on unit area is called pressure. ✔ What is the formula for pressure Pressure(P)=force / Area S.I.unit of pressure is N/m^2 or Nm^{-2} ✔ In honour of scientist Blaise Pascal The S.I.unit of pressure is pascal ✔ On what factors the pressure depends(2 factors) **Conclusion :-** The same force acting on a smaller area exerts a larger pressure. ✔ A smaller pressure on a larger area. This is the reason why a nail has pointed tip knives have sharp edges buildings have wide foundations Solve this problem	

Summary	Factual Questions	CFU	Home Work
Students are able to relates thrust and pressure. And their differences	Why are railways tracks laid on large sized concreate sleepers?	1. Define the term thrust. Give its SI unit 2. What is pressure? Write its mathematical expression and SI unit.	Why do buildings have wide foundations?

PERIOD Number: 5.9 **TOPIC: 5.5.1 Pressure in fluids 5.5.2 Buoyancy** **Activity 5.4**

Learning Outcome: Students will be able to explains the concept of buoyancy

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory Activity **All liquids and gases are fluids.** **The upward force exerted by the water on the bottle is known as buoyancy.** **Teacher activity**	**Introductory Activity** ❖ Why we feel lighter when we swim in a pool? ❖ When you draw water fom a well we feel that the bucket of water is heavier when it is out of the water why? ❖ Why a ship made of iron and steel dose not sink in sea water? ❖ Why the same amount of iron and steel is in the form of a sheet would sink? ❖ Why an helium filled balloon goes up ? ❖ Let us explore pressure in liquids and buoyancy. **pressure in fluids** ❖ The substances which can flow easily are called fluids. ❖ All the liquids and gases are fluids . ❖ Like solids exert pressure due to their weight, liquids also have weight and exert pressure. ❖ But a fluid (liquid or gas)exert pressure in all directions – sideward, downward and even upward. Activity (buoyancy): ✔ Take an empty plastic bottle.Close the mouth of the bottle with an	Previous knowledge question What are fluids? In which direction fluid exert force? Is a gas fluid Do the liquids exert pressure? In which direction fluid exert pressure?(all directions Which floats on water? Which exert a force in upward

Activity 5.4	air tight stopper. put it in bucket filled with water. the bottle floats. ✔ Push the bottle in to the water we feel an upward push.try to push in further down .we will find it difficult to push deeper and deeper .this indicates that water exerts a force in the upward direction. ✔ The upward force exerted by the water goes on increasing as the bottle is pushed deeper till it is completely immersed. ✔ Now release the bottle. It bounces back to the surface. ✔ The force due to the gravitational attraction of the earth acts on the bottle in the downward direction ,so the bottle is pulled downward direction. ✔ But the water exerts an upward force on the bottle. Thus the bottle is pushed upward . ✔ Weight of an object is force due to gravitational attraction of the earth. ✔ The upward force exerted by the water on the bottle is greater than its weight. So the bottle raises up when it is released. ✔ The upward force exerted by the water on the bottle is known as up thrust or buoyant force.	direction? Dose the upward force increase as the depth of water increase? Which force acts on the bottle to bounces back to the surface ? Which force acts on the bottle in downward direction ? In which direction the force is more (upward or downward direction)? Why it is more?

Summary	Factual Questions	CFU	Home Work
Students are able to explains the concept of buoyancy in their own words.	Why a bucket of water is lighter when in water than when it is taken out of water?	Draw a diagram to show two forces acting on an object floating half immersed in a liquid.	1. What is buoyant force? 2. What is the other name of the buoyant force? 3. On what factors the buoyant force depends?

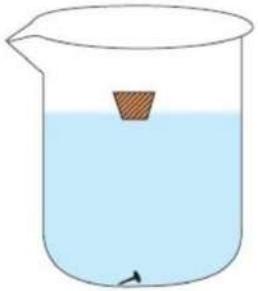
PERIOD Number: 5.10

TOPIC: 5.5.3 Why objects float or sink when placed on the surface of water Activity 5.5 and Activity 5.6

Learning Outcome: Students will be able to give reasons for sinking and floating of bodies

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory Activity ✔ observe the figure can you identify what are sinking objects.	✔ What are floating object? ✔ From the **second figure** identify what are sinking objects? What are floating objects? Which objects sink or float in liquids, can you guess the reason? ✔ To answer all these questions let us explore ‘’why objects float or sink when placed on the surface of water’’. **Activity: 5.4** Take a beaker filled with water. ✔ Take an iron nail and place it on the surface of water. ✔ Observe what happens. (The nail sinks). ✔ The force due to the gravitational attraction of the earth on the iron nail pulls it downwards. ✔ There is up thrust of water on the nail which pushes it up wards. ✔ But the downward force is acting on the nail is greater than the up thrust of water on the nail. **Activity: 5.5**	✔ Which objects sink in the water ? ✔ Which objects float in water? ✔ Identify the sinking object from the figure. ✔ Identify the floating objects from the figure. **Activity 5.4** ✔ Dos the iron nail sink in the water? ✔ Which force pulls down the iron nail? ✔ Which force acts on the nail in the upward direction? ✔ Which force is greater on the nail? ✔ What were placed on the surface of water?

Activity 5.5 	✔ Take a beaker filled with water take a piece of cork and iron nail of equal masses. ✔ Place them on the surface of water. ✔ Observe what happens? ✔ The cork floats while the nail sinks. ✔ This happens because of the difference in their **densities.** The density of a substance is defined as ‘’The mass per unit volume’’. ✔ Density (D)= Mass(M) / Volume (V) ✔ What are equal here? What are different? ✔ If volume increases does the density decrease or increase? ✔ Why the density decreases? The cork is less than that of density of water. ✔ The up thrust or buoyant force on cork is greater than the weight cork.	✔ Which one floats on the water? ✔ Which one sinks in the water? ✔ What is the reason behind sink or float of objects? ✔ What is density? ✔ Which object density is less than that of water? ✔ Which of the objects density is more than the density of water? ✔ Which objet float on the water? ✔ Which objects sink in the object sink in the water ? On which factor of an object depends sinking or floating ?

	✔ The density of iron nail is more than the density of water. The sinking or floating of an object is determined by its density.	

Summary	Factual Questions	CFU	Home Work
Students are able to give reasons for sinking and floating of bodies with their own examples.	The volume of 50 g of a substance is 20 cm^3. If the density of water is 1 g cm^{-3}, will the substance float or sink?	1.In what direction does the buoyant force on an object immersed in a liquid act?	1.Why does a block of plastic Kept under water come up to the surface of water?

PERIOD Number: 5.11 TOPIC: 5.6 Archimedes principle Activity 5.7

Learning Outcome: Students will be able to explain Archimedes' principle in his own word

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory Activity observe the Youtube link and find which objects sink are float in the water . Explain the procedure	✔ Why huged bodies like ships and submarians float on the water? ✔ To answer this question let us explore Archimedes principle. **Activity 5.7** Take a piece of stone and tie it to on end of rubber string or a spring balance. ✔ Suspend the stone by holding the balance or the string. Note the elongation of the string or the reading on spring balance due to the weight of the stone. ✔ Now slowly dip the stone in the water in a container . ✔ We find the elongation of the string or reading of the balance	✔ Which objects sink in the water ? ✔ Which objects float on the water? ✔ What is the reason for sink or float in the water? ✔ What is suspended to the string? ✔ Why elongation in the string or spring balance takes place ? ✔ What do you observe in the elongation of the string or the reading on the balance? ✔ Why the extension in the string decreases when the

	decreases as the stone is gradually lowered in the water. ✔ No further change is observe once the stone gets fully immersed in the water. ✔ We know that the elongation produced in the string or spring balance is due to the weight of the stone. ✔ Since the extension decreases once the stone is lowered in the water it means that some force acts on the stone in upward direction . **Archimedes principle**:- ✔ When a body is immerse fully or partially in fluid it experiences an upward force that is equal to the weight of the fluid displaced by it. ✔ Buoyancy describes the force that a fluid exerts on an object of different/ lesser density than itself. ✔ Archimedes principle describes one aspect of buoyancy. That the buoyant force exerted on an object under a fluid is equal to the weight of the fluid it displaces. Applications of Archimedes principle:- ✔ 1. Used in designing ships and submarines. 2.lactometers (used to determine the purity of a sample milk) 3.Hydrometers(used for determining density of a liquid)	stone is lowered in the water? ✔ In which direction the force acts on the stone ? ✔ What we call this upward force? ✔ On what factors the buoyant force acting on the object depends? ✔ What is the magnitude of the buoyant force experienced by a body? ✔ Do all bodies in a given fluid experience the same buoyant force? ✔ What is Archimedes principle? ✔ What is the difference between buoyancy and Archimedes principle ? ✔ Which principle is used in designing ships and submarines? ✔ On which principle the lactometer is made? ✔ What is the purpose of hydrometer and under which principle it is made? ✔ What are the applications of Archimedes principle in our daily life?

Summary	Factual Questions	CFU	Home Work
Students applies the Archimedes' principle in daily life situations.	Do all bodies immersed in a given fluid experience the same buoyant force?	1.State Archimedes' principle 2.Write two applications of Archimedes' principle	The volume of a 500 g sealed packet is 350 cm^3. Will the packet float or sink in water if the density of water is 1 g cm^{-3}? What will be the mass of the water displaced by this packet?

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson: 5. Gravitation		Class: 9	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
5.1	5.1 Gravitation Activity 5.1				
5.2	5.1.1 Universal law of gravitation, example 5.1				
5.3	5.1.2 Importance of the universal law of gravitation 5.2 Free fall Activity 5.2				
5.4	5.2.1 To calculate the value of g 5.2.2 Motion of objects under the influence of gthe ravitational force of the Earth				
5.5	Example 5.2 and Example 5.3				
5.6	5.3 Mass 5.4 Weight				
5.7	5.4.1 Weight of an object on the moon Ex . 5.4, 5.5				
5.8	5.5 Thrust and pressure Example 5.6				

5.9	5.5.1 Pressure in fluids 5.5.2 Buoyancy Activity 5.4 5.5.3 Why objects float or sink when placed on the surface of water				
5.10	5.5.3 Why objects float or sink when placed on the surface of water Activity 5.5 and Activity 5.6				
5.11	5.6 Archimedes' principle Activity 5.7				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?

4	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

November 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						1 No Bag Day Andhra Pradesh Avatarana
2 Sunday	3 6.1	4 6.2	5 6.2	6 NI	7 NI	8 Second Saturday
9 Sunday	10 SA - 1	11 Day	12	13	14 n's Day	15 No Bag Day
16 Sunday	17 SA - 1	18	19	20 6.3	21 6.4	22 No Bag Day Cluster meeting Fibonacci Day
23 Sunday	24 6.5	25 6.6	26 National Constituti 6.7	27 Sanjeevaraya 6.8	28 NI	29 No Bag Day SMC Meeting
30 Sunday						

14 - 20 National Library Week

TEACHER'S NOTES

Week1: **6.1 – 6.2**

Week 2:

Week 3: **6.3 – 6.4**

Week 4: **6.5 – 6.8**

Chapter: 6. Atoms and Molecules

K8Q8S1

Learning Outcomes:

1. 1.Explain the law of chemical combination
2. Conduct experiments to prove law of constant proportion
3. 3.Write modern symbols of atoms
4. 4.Calculate the atomic mass of some elements
5. 5.Differentiate molecule of elements and molecules of compunds
6. 6.Write the chemical formula for compounds using criss cross method
7. 7.Write chemical formulas of some compounds
8. 8.Calculates molecular mass from atomic mass

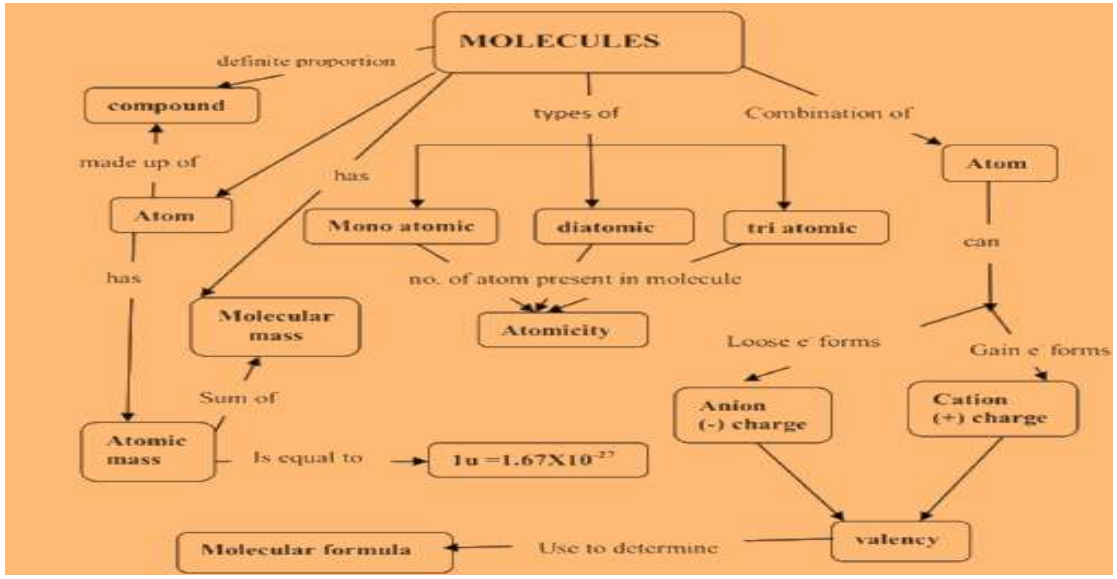
Prior Concepts/Skills: (Essential concepts and skills to be checked/bridged before teaching the current concept)

1. What is the formula for water molecule?

2. Does the weight of an iron rod increase or decrease, on rusting?

3. What is the symbol of Hydrogen?

Concept Map



Synopsis

- **Laws of conservation of mass:** During a chemical reaction, the sum of the masses of the reactants and products remains unchanged. This is known as the Law of Conservation of Mass.
- **Law of constant propotions:** In a pure chemical compound, elements are always present in a definite proportion by mass. This is known as the Law of Definite Proportions.
- **Atom:** An atom is the smallest particle of the element that cannot usually exist independently and retain all its chemical properties.
- **Valency:** The combining capacity of an element is known as its valency.
- **Molecule:** A molecule is the smallest particle of an element or a compound capable of independent existence under ordinary conditions. It shows all the properties of the substance.
- **Ion:** Ion is a charged particle and can be positively or negatively charged.
- **Chemical formulae:** A chemical formula of a compound shows its constituent elements and the number of atoms of each combining element.
- **Molecular mass:** The molecular mass of a substance is the sum of the atomic masses of all the atoms in a molecule of the substance.
- **Formula of unit mass:** The formula unit mass of a substance is a sum of the atomic masses of all atoms in a formula unit of a compound.

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
6.1	6.1 Law of chemical combination 6.1.1 Law of conservation of mass Activity 6.1	explain the law of chemical combination	
6.2	6.1.2 Law of constant proportions	conduct experiments to prove law of constant proportion	
6.3	6.2 What is an atom? 6.2.1 What are the modern day symbols of atoms of different elements?	write modern symbols of atoms	
6.4	6.2.2 Atomic mass 6.2.3 how do atoms exist?	Calculate the atomic mass of some elements	
6.5	6.3 What is a molecule? 6.3.1 Molecules of elements 6.3.2	differentiate molecule of elements	

	Molecules of compounds	and molecules of compunds	
6.6	6.3.3 What is an ion? 6.4 Writing chemical formula	write the chemical formula for compounds using criss cross method	
6.7	6.4.1 Formulae of simple compounds	write chemical formulas of some compounds	
6.8	6.5,1 Molecular mass 6.5.2 formula unit mass	calculates molecular mass from atomic mass	

PERIOD Number: 6.1

TOPIC: 6.1 Law of chemical combination , 6.1.1 Law of conservation of mass , Activity 6.1

Learning Outcome: Students will be able to explain the law of chemical combination

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. Students were asked to break a chalk piece or a paper into further smaller pieces until it cannot be	What is the difference between element and compounds? Do you know how compounds are formed? How elements combine to form compounds? What happens when they combine? Antoine Lavoisier established two important laws of chemical combination. They are	1. What is the difference between atom and element? 2. Which of the following statements is not true about an atom? a. Atoms are not able to exist independently b. Atoms are the basic units from which molecules and ions are formed c. Atoms are always neutral in nature d. Atoms aggregate in large numbers to form the matter that we can see, feel or touch

broken further.

Do you know the name of the smallest particle beyond which further division will not be possible?

Ancient Greek Philosophers.

Democritus and **Leucippus** suggested that the smallest indivisible particles are called 'Atoms'

1. Law of Conservation of Mass
2. Law of Constant Proportions

Activity 6.1

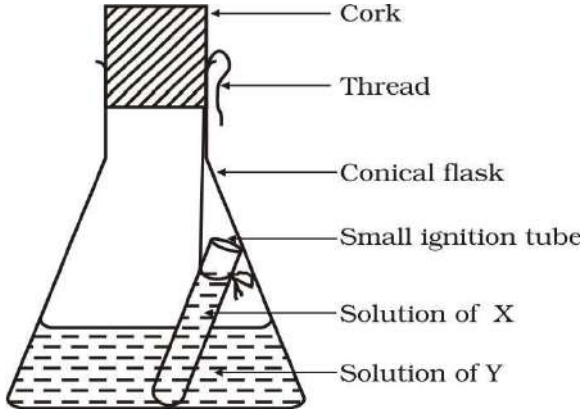
Take a conical flask and take 10ml of 5% Na_2SO_4 .

Take 10ml of $BaCl_2$ in the ignition tube. Hang the ignition tube in the conical flask carefully without mixing the two solutions.

What happens if they are mixed?

While doing this experiment put the cork on the mouth of the flask.

Weigh the flask with contents carefully and record as W_1



Now tilt and swirl the flask, so that two solutions are mixed and weigh again after tilting.

3. Repeat activity 3.1 with copper sulphate solution in the conical flask and sodium carbonate solution in the ignition tube.

4. Does the same experimental results observed?

5. Complete the following reaction

6. i)Copper sulphate + Sodium carbonate---□ ____ + _____

7. Which of the following statements depict the law of conservation of mass?

   a. In a chemical reaction, the sum of masses of the reactants in always equal to the sum of masses of the products.

   b. Physical states of substances taking part in a reaction change during the reaction.

   c. Mass is created in a chemical reaction.

8. Energy of reactants changes during a chemical reaction.

9. Choose the products when lead Nitrate reacts with Sodium Chloride

   a. Lead sulphate

   b. sodium Nitrate

   c. lead chloride

   d. sodium carbonate

	Record the weight as W_2. Do you observe what happened to the contents in the conical flask? Do you know the name of the white precipitate formed in the flask? The white precipitate formed in the flask after mixing the solution is Barium Sulphate ($BaSO_4$) What is the weight of the contents before and after mixing? Is it same? What are the products formed if contents in the flasks are reversed? Yes, the weight of the reaction before mixing (W_1) is equal to Weight of the reaction after mixing(W_2) So $\mathbf{W_1=W_2}$ **Law of conservation of mass** states that "**Mass can neither be created nor destroyed in a chemical reaction"**	**10.** a&b 2) b&c 3) c&d 4) d&a

Summary	Factual Questions	CFU	Home Work
Students will be able to explain the law of chemical combination	Is laws of conservation of mass holds good in everywhere?	A 0.24 g sample of compound of oxygen and boron was found by analysis to contain 0.096 g of boron and 0.144 g of oxygen. Calculate the percentage composition of the compound by weight.	State the law of conservation of mass in chemical reaction. Describe an activity that can be used to verify this law

PERIOD Number: 6.2

TOPIC: 6.1.2 Law of constant proportions

Learning Outcome:

Students will be able to conduct experiments to prove law of constant proportion

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. Who proposed law of conservation of mass? What is law of conservation of mass? What is a compound? What is the chemical formula of water?	The ratio of atoms in water molecule is same everywhere is 2:1(H:O) According to Lavoisier "**many compounds were composed of two or more elements and each such compound had the same elements in the same proportions, irrespective of where the compound came from or who prepared it**". In a compound such as water, the ratio of the mass of hydrogen to the mass of oxygen is always 1:8, whatever the source of water. • This led to the **law of constant proportions** which is also known as the **law of definite proportions**. This law was stated by Proust as **"In a chemical substance the elements are always**	1. How many grams of hydrogen and oxygen are obtained when 9 gms of Water is decomposed? 2. 2.How many grams of nitrogen is required to prepare ammonia if 3 grams of Hydrogen is used? 3. What is law of constant proportions? 4. The ratio of the mass of hydrogen to the mass of nitrogen in NH_3. 5. In a reaction 5.3g of sodium carbonate reacted with 6g of Acetic acid. The products were 2.2g of carbon dioxide, 0.9gm of water and 8.2g of sodium acetate. Show that these observations are in agreement with the law of conservation of mass? 6. Hydrogen and oxygen combinein the

How many hydrogen and oxygen atoms present in one water molecule and what is their ratio?	**present in definite proportions by mass"** Dalton's atomic theory provided an explanation for the law of conservation of mass and the law of definite proportions. • According to Dalton's atomic theory, all matter, whether an element, a compound or a mixture is composed of small particles called atoms.	ratio of 1:8 by mass to form water. 7. What mass of oxygen gas would be required to react completely with 3 g of hydrogen gas? 8. What is law of constant proportions? 9. Which postulate of theory is the result of the law of conservation of mass? 10. Which postulate of Dalton's atomic theory can explain law of constant proportions?

Summary	Factual Questions	CFU	Home Work
Students will be able to conduct experiments to prove law of constant proportion	Which postulate of Dalton's atomic theory can explain the law of definite proportions?	Which postulate of Dalton's atomic theory is the result of the law of conservation of mass?	Explain the law of constant proportion

PERIOD Number: 6.3

TOPIC: 6.2 What is an atom? , 6.2.1 What are the modern day symbols of atoms of different elements?

Learning Outcome: Students will be able to write modern symbols of atoms

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. What is law of constant proportions?		

1) Which postulate of Dalton's atomic theory is the result of the law of conservation of mass?

2) Which postulate of Dalton's atomic theory can explain the law of constant proportion?

3) Have you ever observed a mason building Walls?(Yes/No)

4)These walls form a room?(Yes/No)

5)A collection of rooms will form a building? (Yes/No)

6)What is the building block of the huge building?

7)What about the building block of an ant- hill?

8)Similarly what are the building blocks of matter?

9) What are the units of atomic radius?

ATOM is the building block of matter.

Atomic radius is measured in nanometers.Express atomic radius in meters and give the relative sizes?

$1/10^9$ m = 1 nm=10^{-9} m 1 m = 10^9 nm

From the given table find out the particle of small size and large size?

Give their increasing order ?

Radii (in m)	Example
10^{-10}	Atom of hydrogen
10^{-9}	Molecule of water
10^{-8}	Molecule of haemoglobin
10^{-4}	Grain of sand

1)Define an atom?

2)What is the unit for the measurement of atomic radius?

3)Who is the first scientist to use the symbol for elements?

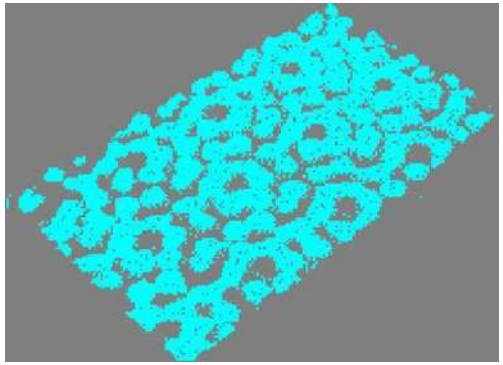
9)How big are atoms?

10)Can we see an atom with our naked eye or by microscope?

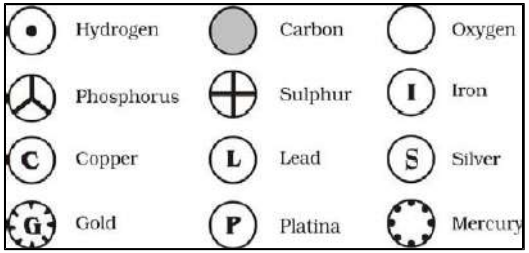
- An image of the surface of silicon

11)Which instruments are used to see an atom?

Using modern techniques,we can produce magnified images of surfaces of elements showing atoms.What are the modern techniques used?



Dalton represented the symbols of the some elements as shown below



What is the difference between these symbols and the modern day symbols?

How the elements symbols are unique worldwide?

What is IUPAC?

4)Which scientist proposed the names of the elements based on their names?

5)What is the symbol of

If atoms are so insignificant in size, why should we care about them?

What are the modern day symbols of atoms of different elements?

Many of the symbols are the first one or two letters of the element's name in English.

For example

(i) Hydrogen	**H**
ii) Boron	**B**
iii) Carbon	**C**
iv) Nitrogen	**N**

The first letter of a symbol is always written as a capital letter (uppercase) and the second letter as a small letter (lowercase).

For example

(i) Lithium	**Li** and not LI
ii) Beryllium	**Be** and not BE
(iii) Aluminium	**Al** and not AL
(iv) Cobalt	**Co** and not CO.

Symbols of some elements are formed from the first letter of the name and a letter,appearing later in the name.

Examples are:

(i)Chlorine, Cl, (ii) Zinc, Zn etc.

What is the symbol of the element Iron?

sodium?

6)

Match the following		
1)	Carbon	a)Ca
2)	Calcium	b)Co
3)	Cobalt	c)Cu
4)	Copper	d)C

7)The chemical symbol for nitrogen gas is

(a) Ni (b) N_2 (c) N^+ (d) N

8)The chemical symbol for sodium is

(a) So (b) Sd (c) NA (d) Na

Can you tell the Modern day symbols of the above elements? What is Berzilius suggestion on symbols of elements?	Why is it written as Fe? Which symbols have been taken from the names of elements in Latin, German or Greek? For example, the symbol of Iron is Fe from its Latin name Ferrum. List out the symbols of other elements from their Latin, German or Greek?	9)Which of the following symbols of elements are incorrect? Give their correct symbols? (a) Cobalt CO (b) Carbon c (c) Aluminium AL (d) Helium He (e) Sodium So

Summary	Factual Questions	CFU	Home Work
Students will be able to write modern symbols of atoms	1. What do you mean by symbols of elements?	2. Write the symbol for the following elements. a) Gold b) Mercy 3. What is an atom?	1. Write symbols of first 10 elements?

PERIOD Number: 6.4

TOPIC: 6.2.2 Atomic mass , 6.2.3 how do atoms exist?

Learning Outcome: Students will be able to Calculate the atomic mass of some elements

Teaching Learning Process

I DO	WE DO	YOU DO
1)What is the unit for the measurement of atomic radius?	The atomic mass unit is equal to the mass of one carbon atom, then we would assign carbon an atomic mass of 1.0 U and oxygen an	1)What do you mean by atomic mass?

2)What is the mass of an atom?

3)If atom cannot be seen with a microscope how it can be weighed?

4) Do you know the mass of one atom of oxygen?

5)What is relative atomic mass?

6)Why is it difficult to determine the exact mass of atom?

7)How Laws of chemical combinations are used to determine the relative atomic mass?

8)

9)What is atomic mass unit?

atomic mass of 1.33 U.

where U is the unit for atomic mass called unified mass.

As it is more convenient to have these numbers as whole numbers or as near to a whole number as possible.

After considering atomic mass unit as the 1/16 of the mass of the oxygen atom ,

Later in 1961 universally accepted atomic mass unit, carbon-12 isotope was chosen as the standard reference for measuring atomic masses.

One atomic mass unit is a mass unit equal to exactly one-twelfth(1/12th) the mass of one atom of carbon-12.

The relative atomic masses of all elements have been found with respect to an atom of carbon-12.

The relative atomic mass of the atom of an element is defined as the

Average mass of the atom, as compared to 1/12th the mass of one carbon-12 atom.

2)What is atomic mass unit and how is it measured?

3)Complete the table

Carbon	12
Nitrogen	14
Oxygen	16
Sodium	?
calcium	?

4)Why it is not possible to see an atom with naked eyes?

5)Match the following

a) Chlorine	i)14
b) Sodium	ii)35.5
c) Nitrogen	iii)16
d) Oxygen	iv)23

6)Carbon monoxide (CO) is formed by carbon and oxygen. It was observed experimentally that 3 g

Element	Atomic Mass (u)
Hydrogen	1
Carbon	12
Nitrogen	14
Oxygen	16
Sodium	23
Magnesium	24
Sulphur	32
Chlorine	35.5
Calcium	40

Existence of an Atom:

Atoms of most elements exist in the form of molecules and ions which aggregate in large numbers to form matter.

of carbon combines with 4 g of oxygen to form CO. In other words, carbon combines with 4/3 times its mass of oxygen.If carbon and oxygen form CO_2 then --.

a) 3g of carbon combines with 6g of oxygen

b) 3g of carbon combines with 8g of oxygen

c) 4g of carbon combines with 8g of oxygen

4g of carbon combines with 6g of oxygen

Summary	Factual Questions	CFU	Home Work
Students will be able to Calculate the atomic mass of some elements	1. Why is it not possible to see an atom with naked eyes?	1. Define the atomic mass unit. 2. How were the names of elements derivd?	1. With the help of an example, state the significance of a symbol of an element. 2. Mention the unit of measuring the atomic mass of an element.

PERIOD Number: 6.5

TOPIC: 6.3 What is a molecule? , 6.3.1 Molecules of elements , 6.3.2 Molecules of compounds

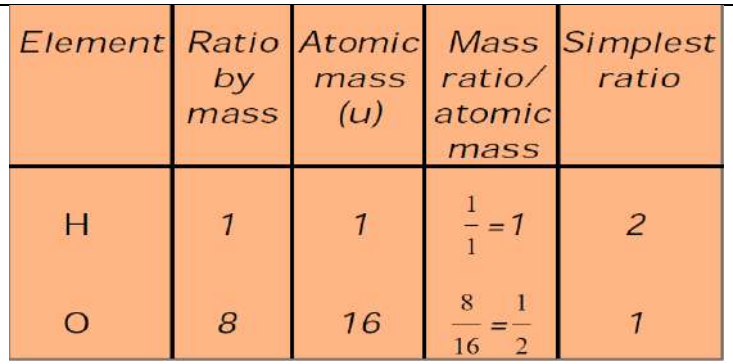
Learning Outcome: Students will be able to differentiate molecule of elements and molecules of compounds

Teaching Learning Process

I DO	WE DO	YOU DO
1. 1.What is the building block of matter? 2. Why do atoms combine to form molecule? 3. Which atoms do not combine or react? 4. What is a molecule? 5. How does a molecule formed from its atoms? 6. Why it is not possible to see molecule with naked eye? 7. Which element is represented with symbol 'N'?	Interaction with students to know their previous knowledge regarding to the Concept of the topic. Molecules of Elements: The molecules of an element are constituted by the same type of atoms. Some molecules are made up of only one atom of that element called **Monoatomic molecule.** Eg:-Gases like Helium(He),Neon(Ne), Argon(Ar), Krypton(Kr). Most of the metals are monoatomic in nature like Sodium(Na), Potassium (K), Gold(Au) etc. What are the poly-atomic elements? List out different poly-atomic elements with examples? If 3 atoms of oxygen unite to form a molecule, we get Ozone with is represented by O_3. What is the name of this molecule? If 4 or more atoms unite to form a molecule then they are called Polyatomic molecules.	1.How do we represent the molecule of element Oxygen and ozone? 2. What is the name of S_8 and P_4? 3.Can you name some other monoatomic molecules? Classify each of the following on the basis of their atomicity. (a) F_2 (b) NO_2 (c) N_2O (d) C_2H_6 (e) P_4 (f) H_2O_2 (g) P_4O_{10} (H) O_3 (i) HCl (j) CH_4 (k) He (l) Ag

8. How many types of molecules are formed? 9. How many types of molecules are formed from atoms? 10. How to find out the mass of a molecule?	Can you give few examples of poly-atomic? What is atomicity? The number of atoms constituting a molecule is known as its **atomicity**. Molecules are formed only by same type of atoms or different atoms? If molecules are formed by different atoms then they are called as ? **Molecules of compounds:** If a molecule is formed from the combination of two or more atoms of different elements in a definite proportion then it is called **Compound. Molecules of some compounds** **Activity: 6.2** With the following procedure we can calculate the ratio of number of atoms in the compound	Can you name some other diatomic molecules? Identify the poly-atomic, mono- atomic and compounds from the above lists? With the given ratio by mass of the compound can you find the ratio of number of atoms ?

Compound	Combining Elements	Ratio by Mass
Water (H_2O)	Hydrogen, Oxygen	1:8
Ammonia (NH_3)	Nitrogen, Hydrogen	14:3
Carbon dioxide (CO_2)	Carbon, Oxygen	3:8

	 Thus, the ratio by number of atoms for water is $H_2 : O = 2 : 1$ Hence we write the formula of water as H_2O.	Calculate the ratio of number of atoms in the compound CO_2 by this method.

Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate molecule of elements and molecules of compounds	1. Name some molecules containing more than four atoms. 2. Can you make a distinction between a molecule and a compound?	1.Write an example of a polyatomic molecule. 2. Define atomicity.	Give the names of the elements present in the following compounds. (a) Quick lime (b) Hydrogen bromide (c) Baking powder (d) Potassium sulphate.

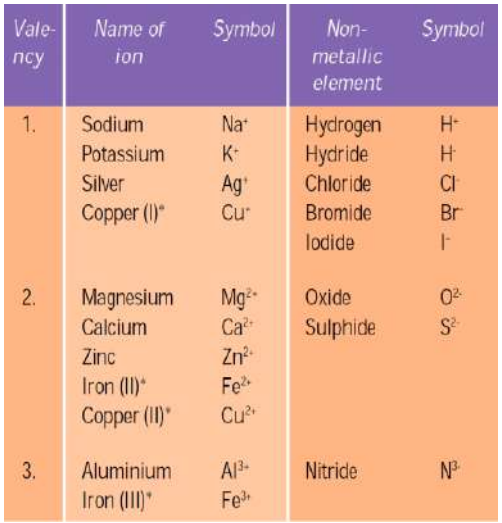
PERIOD Number: 6.6

TOPIC: 6.3.3 What is an ion? , 6.4 Writing chemical formula

Learning Outcome: Students will be able to write the chemical formula for compounds using criss cross method

Teaching Learning Process

I DO	WE DO	YOU DO
Recapitulation 1) What is a molecule?	**What is an Ion:** The substance table salt we use regularly in daily life is	1. What type of batteries used in the mobile phones and rechargable electronic gadgets?

2) What is the formula of hydrogen molecule? 3) What is the atomicity of hydrogen molecule 4) What is atomicity? **5)** What is a compound? 6) What is an ION? 7) What is Cation, and Anion?	chemically called Sodium Chloride The sodium chloride is formed from the charged species of Sodium and chlorine called Ions. The charged species are known as *ions*. Ions may consist of a single charged atom or a group of atoms that have a net charge on them. An ion can be negatively or positively charged. A negatively charged ion is called an 'Anion' and the positively charged ion, a 'Cation'. In sodium chloride (NaCl). Its constituent particles are positively charged sodium ions (Na^+) and negatively charged chloride ions (Cl^-). **Examples of cations and anions: (time10M)**  A group of atoms carrying a charge is known as a	(Lithium ion batteries) 2. What is an Ion? 3. What is the name of the substance which added to the food while cooking gives taste to the food? (Answer is table salt) **4.** What is the chemical name of table salt? 5. What solution is given to the person who is suffering from diarrhea or dehydrated due to hot summer? (Answer is ORS- Oral rehydration salt) 6. Define Ion. 7. Define cation and anion and give examples?

Valency	Name of ion	Symbol	Non-metallic element	Symbol
1.	Sodium	Na^+	Hydrogen	H^+
	Potassium	K^+	Hydride	H^-
	Silver	Ag^+	Chloride	Cl^-
	Copper (I)*	Cu^+	Bromide	Br^-
			Iodide	I^-
2.	Magnesium	Mg^{2+}	Oxide	O^{2-}
	Calcium	Ca^{2+}	Sulphide	S^{2-}
	Zinc	Zn^{2+}		
	Iron (II)*	Fe^{2+}		
	Copper (II)*	Cu^{2+}		
3.	Aluminium	Al^{3+}	Nitride	N^{3-}
	Iron (III)*	Fe^{3+}		

List of CATIONS and ANIONS:

Polyatomic ion (Time : 10Min)

Vale-ncy	Polyatomic ions	Symbol
1.	Ammonium	NH_4^+
	Hydroxide	OH^-
	Nitrate	NO_3^-
	Hydrogen carbonate	HCO_3^-
2.	Carbonate	CO_3^{2-}
	Sulphite	SO_3^{2-}
	Sulphate	SO_4^{2-}
3.	Phosphate	PO_4^{3-}

Some ionic compounds

Ionic Compound	Constituting Elements	Ratio by Mass
Calcium oxide	Calcium and oxygen	5:2
Magnesium sulphide	Magnesium and sulphur	3:4
Sodium chloride	Sodium and chlorine	23:35.5

Writing Chemical Formula:

The chemical formula of a compound is a symbolic representation of its composition.

8. What are polyatomic ions? Give examples.

9. Define Valency?

10. What is the valency of M in the molecule having chemical formula MCl_3?

11. Write the chemical formulae of the following combination of ions.

How the Polyatomic Ions are formed?

What are the Elemental Raios observed in POLYATOMIC Ions formation?

How to write a CHEMICAL FORMULA by using IONS?

RULES to be fallowed while writing a Chemical Formula:

TO Write a chemical formula we must learn the symbols and combining capacity of the elements.

The combining power (or capacity) of an element is known as its valency.

Valency canbe used to find out how the atoms of anelement will combine with the atom(s) of another element to form a chemical compound.

The following rules to be followed while writing a chemical formula.

1) The valencies or charges on the ions must be balanced.
2) If the compound has both metal and nonmetal then the symbol of the metal should be written first and then nonmetal symbol.

   Ex: Calcium oxide: CaO
   Sodium Chloride: NaCl

   Magnesium Bromide: $MgBr_2$
3) In compounds formed with polyatomic ions, the number of ions present in the compound is indicated by enclosing the formula of ion in a bracket and writing the number of ions outside the bracket

   Ex: Magnesium Hydroxide: $Mg(OH)_2$.

   Aluminium carbonate: $Al_2(CO_3)_3$
4) In case the number of polyatomic ion is one, the bracket is not required.

   Ex: Sodium Hydroxide: NaOH

   Barium Sulphate : $BaSO_4$

Ions	Cl^-	OH^-	SO_4^{2-}
Na^+ following combination			
Mg^{+2}			
Ca^{+2}			

Summary	Factual Questions	CFU	Home Work
Students will be able to write the chemical formula for compounds using criss cross meth	1. What is the difference between 2H and H_2 ? 2. Do all elements have a charged valency?	1. What is meant by the term chemical formula? 2. How many atoms are present in a (i) H_2S molecule and (ii) PO_4^{3-} ion	1. What are polyatomic ions? Give examples

PERIOD Number: 6.7

TOPIC: 6.4.1 Formulae of simple compounds

Learning Outcome: Students will be able to write chemical formulas of some compounds

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. 1. What is an ion? 2. What type compound is formed from ions? 3. What type of forces exist among	Let us discuss about binary compound and writing the formulae of simple compounds. The simplest compounds, which are made up of two different elements are called binary compounds. While writing the chemical formulae for compounds, we write the constituent element sand their valencies and then we must crossover the valencies of the combining atoms. This method is called criss- cross method.	1) Write down the formulae of (i) sodium oxide (ii) aluminium chloride (iii) sodium suphide (iv) magnesium hydroxide 2) Write down the names of compounds represented by the following formulae: (i) $Al_2(SO4)_3$ (ii) $CaCl_2$ (iii) K_2SO_4 (iv) KNO_3

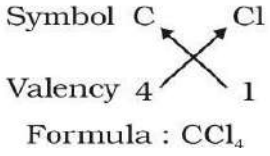
ions?

4. Define valency?

5. What is the formula of sodium chloride?

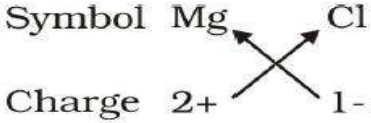
6. What is the valency of Sodium in NaCl?

3. Formula of carbon tetrachloride



Formula : CCl_4

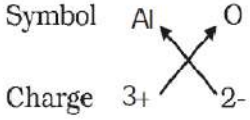
Formula of magnesium chloride



Formula : $MgCl_2$

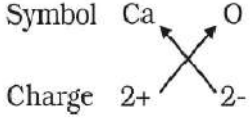
Some more examples

(a) Formula for aluminium oxide:



Formula : Al_2O_3

(b) Formula for calcium oxide:



Here, the valencies of the two elements are the same. You may arrive at the formula Ca_2O_2. But we simplify the formula as CaO.

(v)$CaCO_3$

3)What is meant by the term chemical formula?

4) How many atoms are present in a

(i) H_2S molecule and

(ii) $(PO_4)^{3-}$ ion?

5) Ramu wrote the formula of sodium carbonate as $NaCO_3$. Is he write or wrong? Justify your answer.

6) The formula of calcium hydroxide is $Ca(OH)_2$ and not $CaOH_2$. Justify your answer?

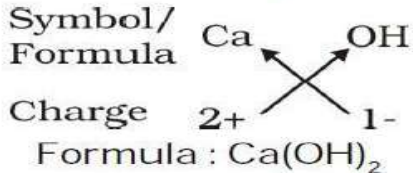
7) Read the following paragraph and answer questions 2(a) -2(d) on the basis of your understanding of the paragraph and the related studied concepts. The compounds containing metal and non-metals consist of charges particles called ions. In ionic compounds the ions are present in a fixed ratio by mass. The formula of an Ionic compound gives the simple whole number

ratio of ions present in it. The valency of an atom may be used to find out how many atoms of one element combine with other to form a chemical formula.

7. What are polyatomic ions? Give example?

8. Do you know what is a binary compound?

Formula of calcium hydroxide:



Three elements A,B and C exhibit the following valencies.

A	A'	B	B'	C	C'
+ 1	+ 3	+ 2	+ 3	+ 2	+ 4

a) Write the chemical formula of B' phosphate.

b) Which out of A or A' has larger atomic mass?

c) Between C sulphate and C' sulphate, larger mass is possessed by__?

In between A nitrate and A' nitrate more number of oxygen atoms are present in_____?

Summary	Factual Questions	CFU	Home Work
Students will be able to write chemical formulas of some compounds	Is Criss Cross method useful for the writing of the chemical formula	1. What is valency? 2. What is use of Valency?	1. An element X has valency 3. Write the formula of its oxides. 2. Write down the names of compounds represented by the following formulae: (i) $Al_2(SO_4)_3$ (ii) KNO_3 (iii)$CaCO_3$

PERIOD Number: 6.8

TOPIC: 6.5.1 Molecular mass 6.5.2 formula unit mass

Learning Outcome: Students will be able to calculates molecular mass from atomic mass

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. 1) What is an atom? 2) Does it have any mass? 3) Can you define atomic mass? 4) What is a molecule? 5) What is the difference between atom and molecule? 6) Does molecule mass calculated from atomic mass? 7) Why is it called relative molecular mass of a molecule? 8) What are the units of relative molecular mass?	The molecular mass of a substance is the sum of the atomic masses of all the atoms in a molecule of the substance. It is expressed in ***atomic mass units (U)***. Ex:1) Water - H_2O Atomic mass of Hydrogen=1 U Atomic mass of Oxygen= 16U Molecular mass of H_2O = 2(Atomic mass of H)+ 1(Atomic Mass of O) =2(1)+1(16) =**18 U** **2) Nitric acid -** HNO_3 Atomic mass of Hydrogen=1 U Atomic mass of Nitrogen= 14 U Atomic mass of Oxygen= 16U Molecular mass of HNO_3 = 1(Atomic mass of H)+ 1(Atomic Mass of N) + 3(Atomic Mass of O)	1.Do molecules experience gravity? 2) Fill the blanks 3) Arrange the molecular masses of H_2, O_2, Cl_2, CO_2, CH_4, C_2H_6,C_2H_4, NH_3, CH_3OH in the increasing order. 4)Calculate the formula unit masses of

SUBSTANCE	ATOMS PRESENT	RELATIVE MOLECULAR MASS
AMMONIA	1XN+3XH	
OXYGEN-O_2		
CARBON DI OXIDE-CO_2	1XC+2XO	1X12+2X32=44
POTASSIUM CARBONATE -K_2CO_3	2XK+1XC +3xO	
SULPHURIC ACID-H_2SO_4		2X1+1X32+4X16=98

9) How to calculate the relative molecular mass of a molecule? 10) What is formula unit mass? 11) How is it different from the relative molecular mass? 12) How to calculate relative formula mass from relative atomic mass?	=1(1)+1(14)+3(16) =1+14+48 = **63U** The formula unit mass of a substance is a sum of the atomic masses of all atoms in a formula unit of a compound. The only difference between Molecular mass and Formula unit mass is that we use the word Formula unit mass for those substances whose constituent particles are ions. Ex:1) Calculate the formula unit mass of NaCl Sol: Atomic mass of Na=23 U Atomic mass of chlorine=35.5 U Molecular mass of NaCl =1(Atomic Mass of Na)+1(Atomic mass of Cl) =1(23)+1(35.5) =58.5 U	ZnO, Na_2O, K_2CO_3. (Given atomic masses of Zn = 65 U,Na = 23 U, K = 39 U, C = 12 U, and O = 16 U). 5) Match the following Molecule / Molecular mass a) Na_2CO_3 — i)098 b) H_2SO_4 — ii)106 c) HNO_3 — iii)040 d) NaOH — iv)063 6) $MgCl_2$ is____U a) 95 b) 97 c) 93 d) 94

Summary	Factual Questions	CFU	Home Work
Students will be able to calculates molecular mass from atomic mass	1. What is Avogadro constant? 2. Define molar mass. 3. What is the mass of 0.5 mole of water molecule?	1. Calculate the molar mass of the following substances. (a) Ethyne, C_2H_2 (b) Sulphur molecule, S_8 (c) Phosphorus molecule, P_4 (Atomic mass of phosphorus = 31) (d) Hydrochloric acid, HCl (e) Nitric acid, HNO	5. Calculate the molecular masses of H_2 , O_2 , Cl_2 , CO_2 , CH_4 , C_2H_6 , C_2H_4 , NH_3 , CH_3OH. 2. Calculate the formula unit masses of ZnO, Na_2O, K_2CO_3 , given atomic masses of Zn = 65 u, Na = 23 u, K = 39 u, C = 12 u, and O = 16 u.

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson: 6. Atoms and Molecules		Class: 9	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
6.1	6.1 Law of chemical combination 6.1.1 Law of conservation of mass Activity 6.1				
6.2	6.1.2 Law of constant proportions				
6.3	6.2 What is an atom? 6.2.1 What are the modern day symbols of atoms of different elements?				
6.4	6.2.2 Atomic mass 6.2.3 how do atoms exist?				
6.5	6.3 What is a molecule? 6.3.1 Molecules of elements 6.3.2 Molecules of compounds				
6.6	6.3.3 What is an ion? 6.4 Writing chemical formula				
6.7	6.4.1 Formulae of simple compounds				
6.8	6.5,1 Molecular mass 6.5.2 formula unit mass				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

December 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 World AIDS 7.1	2 7.2	3 7.3	4 7.4	5 NI	6 No Bag Day Mega PTM/SMC
7 Sunday	8 7.5	9 7.6	10 Human 7.7	11 7.8	12 NI	13 Second Saturday
14 Sunday	15 7.9	16 7.10	17 7.11	18 7.12	19 NI	20 No Bag Day Cluster Meeting
21 Sunday	22 Mathematics Day	23	24 National	25 Christmas	26	27 No Bag
28 Sunday	29 7.13	30 7.14	31 NI			

Christmas Holiday

No bag Day | Cluster Complex | Teacher Resources

Christmas Holidays for Christian Minority Institutions
21.12.25 to 28.12.25

TEACHER'S NOTES

Week1: **7.1 – 7.4**

Week 2: **7.5 – 7.8**

Week 3: **7.9 – 7.12**

Week 4:

Week 5: **7.13 – 7.14**

Chapter: 7. Work and Energy



Learning Outcomes:

1. understand the concept of work
2. explain the scitific concept of work
3. Calculates the workdone by the bodies
4. classify different forms of energy and derive the formula of kinetic energy
5. explain different forms of energy
6. do experiment with trolley and unerstand that objects in motion posses some energy
7. Do the calculations regarding kinetic enegy
8. derive the formula for potential energy
9. understand that object get potential enegy by its position
10. explain that object get potential energy by its height
11. understand that different forms of energies are interconvertible
12. draw the flowchart of conservation of energy
13. calculate the rate of work using the formula
14. calculate the rate of work using the formula

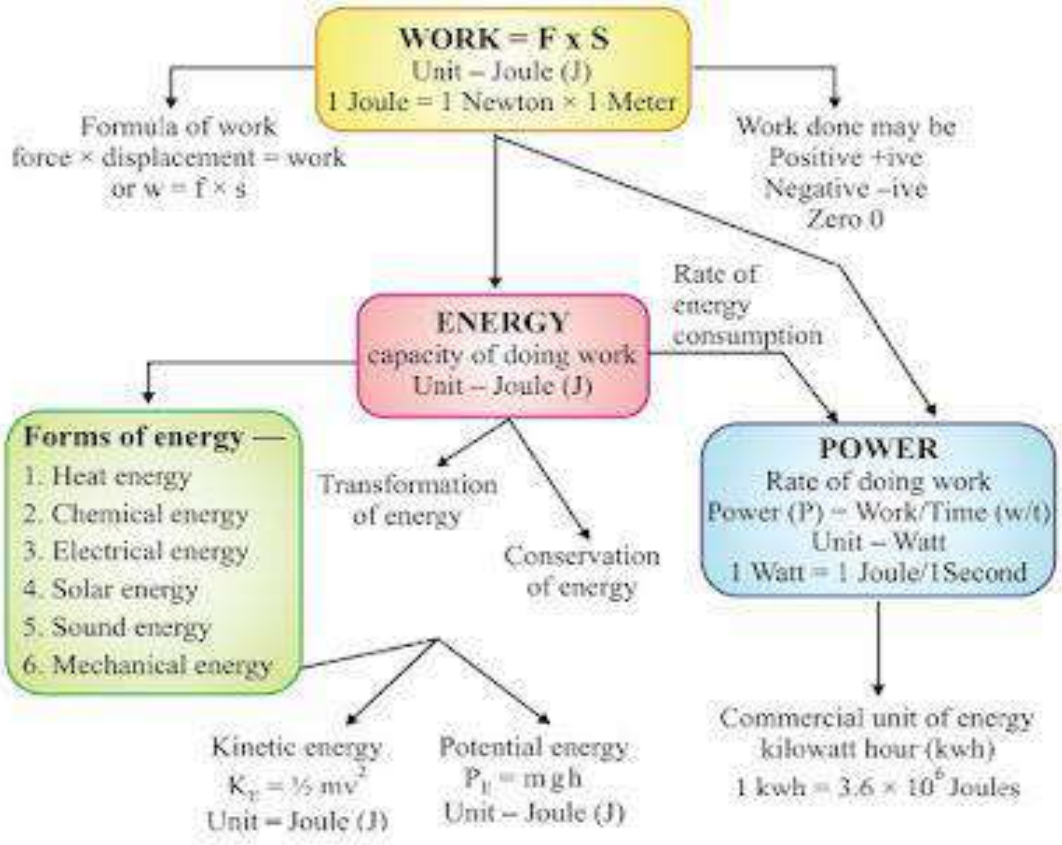
Prior Concepts/Skills: (Essential concepts and skills to be checked/bridged before teaching the current concept)

1. What are S.I units of Force and Displacement?

2. How many forms of energy are there?

3. How do green plants produce food?

Concept Map



Synopsis

- **Work done:** Work done on an object is defined as the magnitude of the force multiplied by the distance moved by the object in the direction of the applied force.
- **Energy:** An object having capability to do work is said to possess energy.

- **Kinetic energy:** An object in motion possesses what is known as the kinetic energy of the object.
- **Potential energy:** The energy possessed by a body due to its change in position or shape is called the potential energy.
- **Law of conservation of energy:** According to the law of conservation of energy, energy can only be transformed from one form to another; it can neither be created nor destroyed. The total energy before and after the transformation always remains constant.
- **Power:** Power is defined as the rate of doing work. The SI unit of power is watt. 1 W = 1 J/s

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
7.1	7.1 Work 7.1.1 Not much work in spite of working hard! Activity 7.1	understand the concept of work	
7.2	7.1.2 Scientific conception of work Activity 7.2 Activity 7.3	explain the scitific concept of work	
7.3	7.1.3 Work done by a constant force Example 7.1 Activity 7.4	Calculates the work done by the bodies	
7.4	7.2.Energy Activity 7.5	classify different forms of energy and derive the formula of kinetic energy	
7.5	7.2.1 Forms of enegy 7.2.2 Kinetic energy Activity 7.6	explain different forms of energy	
7.6	Activity 7.7	do experiment with trolley and unerstand that objects in motion possess some energy	
7.7	Examples 7.3 and 7.4	Do the calculations regarding kinetic enegy	
7.8	7.2.3 Potential enegy Activity 7.8 Activity 7.9	derive the formula for potential energy	
7.9	Activity 7.10 and Activity 7.11 Activity 7.12	understand that object get potential enegy by its position	
7.10	7.2.4 Potential energy of an object at a height example 7.5 example 7.6	explain that object get potential energy by its height	

7.11	7.2.5 Are various energy forms interconvertible? Activity 7.13 Activity 7.14	explain that object get potential energy by its height	
7.12	7.2.6 Law of conservation of energy	draw the flowchart of conservation of energy	
7.13	7.3 Rate of doing work Activity 7.16	calculate the rate of work using the formula	
7.14	Examples 7.7 and Example 7.8 ans Activity 7.17	calculate the rate of work using the formula	

PERIOD Number: 7.1

TOPIC: 7.1 Work 7.1.1 Not much work in spite of working hard!

Activity 7.1

Learning Outcome: Students will be able to understand the concept of work

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons. **Probing questions:** Look at the some pictures listed below to understand the term work. Suma is swimming in a pond? **Probing questions:** Look at the some pictures listed below to understand the term work. Suma is swimming in a pond?	✔ A donkey is carrying a load on its back?. ✔ A wind mill is lifting water from a well?. ✔ A green plant is carrying out photosynthesis?.	✔ What is work? Give some examples of work in everyday life. ✔ Write some situations involving work ✔ Write some situations absence of force

	✔ An engine is pulling a train.? ✔ You are moving huge a rock? ✔ Food grains are getting dried in the sun. ✔ A sail boat is moving due to wind energy. Ask some examples of day to day life on work. ✔ Why not much work in spite of working hard? **Activity.7.1 :** On the basis of this concept, teacher shows some more pictures and activities doing by students and ask the following questions.  Boy is pushing huge rock, but its not moving. ✔ Whether work is done or not scientifically? ✔ Boy is pushing the table & table displaced? ✔ Wheather the work is done or not scientifically?	✔ What is the scientific conception of work, give ? ✔ What is the concept of work in real life ✔ Write yes or no for the following questions that work is being done? ✔ 1.A scientist delivers a speech to an audience of his peers?

	 ✔ You are pushing wall? But wall not moving. .  ✔ Whether work is done or not scientifically?	

Summary	Factual Questions	CFU	Home Work
Students will be able to understand the concept of work	1. Is any work done on a body in uniform circular motion? Justify your answer.	Look at the activities listed below. Reason out whether or not work is done in the light of your understanding of the term 'work'. 1. Suma is swimming in a pond. 2. A donkey is carrying a load on its back. 3. A wind-mill is lifting water from a well. 4. A green plant is carrying out photosynthesis. 5. An engine is pulling a train	1. What is work? Give some examples of work in everyday life.

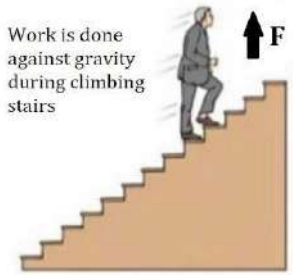
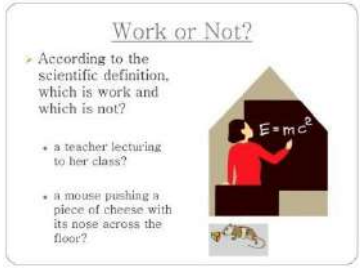
PERIOD Number: 7.2

TOPIC: 7.1.2 Scientific conception of work Activity 7.2 Activity 7.3

Learning Outcome: Students will be able to explain the scitific concept of work

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons.	 ✔ Whether work is done or not scientifically? ✔ Why you are not moving those objects by using your effort completely exhausted. ✔ You are pushing the chair, but it is moving.  ✔ you are pulling the table, but the table is not moving.  ✔ You are holding the book, is the work done or not ?	✔ Students will be participating group discussion ✔ Students will be list out more examples work is done/not.
Teacher will be show video on IFP and discuss with students so many activities in our daily life and asking some questions regarding this ✔ Climbing up the staircase work is or not		✔ Students will be given answer

 ✔ when do work is done ✔ Who is doing the work ✔ What are the conditions to be work is done	**Activity 7.2 :** Teacher will be discuss with students ,list out some situations in your daily life involving work and, find out in each situations work is done or not  **Activity 7.3 :** Teacher will be discuss with students, think of situations when the object is not displaced in spite of a force acting on it list out them	✔ Students solve this problem ✔ Students will be participating group discussion ✔ Students will be list out more examples work is done/not.

Summary	Factual Questions	CFU	Home Work
Students will be able to explain the scitific concept of work	1. A man is holding a suitcase in the state of rest. What is the work done by him? 2. What is the amount of work done by a man in pressing a rigid wall with a force of 500 N?	3. Define the term work. 4. How is work done measured when a body moves in the direction of the applied force?	1. When is the work done by a force said to be (a) positive (b) negative Give examples

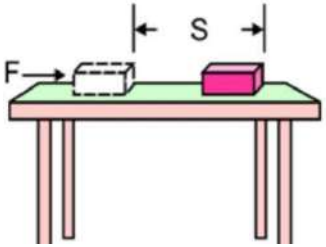
PERIOD Number: 7.3

TOPIC: 7.1.3 Work done by a constant force

Example 7.1 Activity 7.4

Learning Outcome: Students will be able to Calculates the work done by the bodies

Teaching Learning Process

I DO	WE DO	YOU DO
The teacher will start the lesson with a discussion to check for prior knowledge according to previous lessons.	Teacher will ask the some pre test questions of students how is work done in science ✔ Do you know the formula for work done? ✔ What are the units of work done? Let's move to the concepts to learn Conduct an activity with the students, one student moves the brick on the table  ✔ What do you observe, is any displacement in the position of brick happened? ✔ In which direction displacement is happened? ✔ Is any force applied on brick? ✔ In which direction the force is applied? **Example 1:** A force of 7N acts on an object.The displacement is say 8m ,in the direction of the force .let us take it that the force acts on the object through the displacement .what is the work done	✔ When do we say that work is done? ✔ What is the definition of work? ✔ What are the units of force? ✔ What are the units of displacement?

	in this case ✔ What is the magnitude of force? ✔ What is the displacement? ✔ What is the formula for finding work done? **Activity 7.4** Students will do this activity, teacher asking some questions regarding this activity  ✔ What are the forces working here ? ✔ Which one is +ve and -ve force ? ✔ Which one of these forces is doing positive work? ✔ Which force is doing negative work ? ✔ Why the work done is taken as –ve against gravity? ✔ Which force is acting on free fall body? Is it +ve or –ve? ✔ What is the work done when the force on the object is zero? 	✔ What is the formula for work done? ✔ What are the units of work done? ✔ Define 1J of work?

	✔ What is the work done when displacement is zero even though force is applied? W = F X S , S=F=0 ✔ What would be the work done when the displacement (S=0) of the object is zero? ✔ What kind of work done by the moon while revolving around the earth? ✔ Teacher will give the problem and make the students to solve it? ✔ By give problems and asking the students what values are given and find the work done in the given problem?	

Summary	**Factual Questions**	**CFU**	**Home Work**
Students will be able to Calculates the work done by the bodies	What is the amount of work done when an artificial satellite revolves around the earth?	1. When do we say that work is done? 2. Define I J of work	1. A mass of 10 kg is at a point A on a table. It is moved to a point B. If the line joining A and B is horizontal, what is the work done on the object by the gravitational force? Explain your answer.

PERIOD Number: 7.4

TOPIC: 7.2 Energy Activity 7.5

Learning Outcome: Students will be able to classify different forms of energy and derive the formula of kinetic energy

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask questions to children, What they understand by the term energy? ✔ Give various sources of energy? ✔ Is life possible without energy? ✔ Do we get energy from nuclei of atoms? ✔ What is the energy stored in batteries? ✔ Do you get energy from water? ✔ How do you energy for our muscles? ✔ Do we get energy from tides? ✔ Do we get energy from winds? ✔ What will happen without energy? ✔ Where children have seen energy? Tell examples. ✔ How can we say that an object have energy? Think and tell. ✔ Which is the biggest natural source of energy	**Activity 7.5 :** Divide two groups discuss sources of energy and list out many other sources of energy ✔ Do you know how many types of energies in the universe? ✔ The ability to do work is called of energy? Teacher Images show on IFP and ask some questions	✔ What is energy? ✔ List out some sources of energy from our nature? ✔ A larger unit of energy is called------ ✔ I am a capacity of to do work, what is my name? ✔ The S I unit of energy?

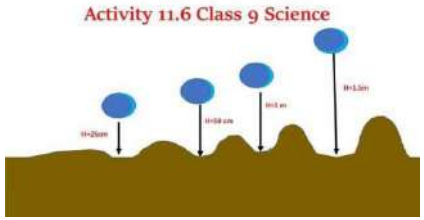
Summary	Factual Questions	CFU	Home Work
Students will be able to classify different forms of energy and derive the formula of kinetic energy	1. Is energy a scalar or vector quantity? 2. Why are the units of energy the same as those of work?	Define the term energy Name the biggest natural sources of energy. Name some other sources of energy also.	Certain force acting on a 20 kg mass changes its velocity from 5 m s^{-1} to 2 m s^{-1}. Calculate the work done by the force.

PERIOD Number: 7.5

TOPIC: 7.2.1 Forms of energy 7.2.2 Kinetic energy Activity 7.6

Learning Outcome: Students will be able to explain different forms of energy

Teaching Learning Process

I DO	WE DO	YOU DO
Divide two groups discuss sources of energy and list out many other sources of energy ✔ Do you know how many types of energies in the universe? ✔ The ability to do work is called of energy? Teacher Images show on IFP and ask some questions	✔ Match the following ✔ How many types of energy you have heard about? ✔ Name device which converts electrical energy into sound energy? ✔ Name device which converts light energy into heat energy? **ACTIVITY 10.6 :** Teacher conducts as activity with students. 	✔ What is the kinetic energy? ✔ What is the expression of kinetic energy? ✔ Give ten examples of kinetic energy?

	Doing this activity by students and ask questions regarding this activity ✔ Which one of them is deepest? ✔ Which one is shallowest? why? ✔ What has caused the ball to make a deeper dent? ✔ In which case work done is more or less? Why? Student observed that more is the displacement of ball more is the work done so the deepest.	✔ Derive the expression of kinetic energy? ✔ observe the below picture and Give answers

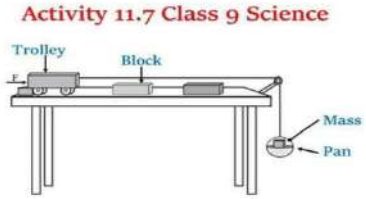
Summary	Factual Questions	CFU	Home Work
Students will be able to explain different forms of energy	1. Write the formula of KE 2. If mass of the body is doubled, then what is KE of the body?	**3.** An object of mass, m is moving with a constant velocity, v. How much work should be done on the object in order to bring the object to rest?	1. A battery lights a bulb. Describe the energy changes involved in the process. 2. Name the different forms of energy

PERIOD Number: 7.6

TOPIC: Activity 7.7

Learning Outcome: Students will be able to do experiment with trolley and unerstand that objects in motion possess some energy

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask questions to children, What they understand by the Kinetic energy?	**ACTIVITY 7.7 :** Teacher conducts this activity and ask questions regarding this activity  ✔ If block is displaced with the gained energy, where does this energy coming from? ✔ We do repeat this activity with different weights, what do you observe? ✔ Write the observations in your note book. ✔ Do you know the moving object can do work ? if yes, what is it? ✔ Which energy is possessed by moving objects? ✔ Give more examples of moving objects?	 ✔ what kind of energies of a flying Aeroplane has?

	✔ Every objects in motion possess energy, what do we call this energy? ✔ If two persons coming by walking in opposite direction strikes each other what happens? ✔ If two persons coming by running in opposite directions striking each other .what happens? ✔ In which case can both of them feel more pain .why? ✔ what do know about kinetic energy? ✔ Give some examples of kinetic energy? ✔ How do you derive the expression of kinetic energy? ✔ What is the symbol using for initial energy? ✔ What is the symbol using for final velocity? ✔ What is the third equation of motion? ✔ What is the formula of force? By giving one problem and ask students what values are given ,what values are find **Example 1 :** ✔ What is the kinetic energy of a 50 kg	 - What kind of energy in windmills? ✔ Define K.E and write formula for K.E

	skateboarder moving at 3 m/sec2?  ✔ What is the mass of skateboarder? ✔ What is the velocity of skateboarder? ✔ What we find this problem? ✔ What is the formula used solve this problem? Teacher will give one more problem to solve students	

Summary	Factual Questions	CFU	Home Work
Students will be able to do experiment with trolley and unerstand that objects in motion possess some energy	1.The speed of a car increases four times. What is the increases in its kinetic energy?	1. What is the SI unit of KE? 2. Calculate the KE of a ball of mass 10g and momentum 100 g cm/s ?	Calculate the work required to be done to stop a car of 1500 kg moving at a velocity of 60 km/h?

PERIOD Number: 7.7

TOPIC: Examples 7.3 and 7.4

Learning Outcome: Students will be able to Do the calculations regarding kinetic energy.

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask some brain storming questions. **Example 7.3** An object of mass 15 kg is moving with a uniform velocity of 4 m s^{-1}. What is the kinetic energy possessed by the object? **Example 7.4** What is the work to be done to increase the velocity of a car from 30 km h^{-1} to 60 km h^{-1} if the mass of the car is 1500 kg	What is the mass of the object? What is the velocity of the object? What is the formula of kinetic energy? What is the Mass of the car? What is the initial velocity of car? What is the final velocity of the car? What is the initial KE of the car? What is the final KE of the car?	m = 15 kg, v = 4 m $s^{-1.}$ $E_k = \frac{1}{2} m v^2$ $= \frac{1}{2} \times 15\ kg \times 4\ m\ s^{-1} \times 4\ m\ s^{-1}$ $= 120\ J$ The kinetic energy of the object is 120 J. m =1500 kg u = 30 km h^{-1} = 30 ×1000m 60×60s = 25/3 m s^{-1}. v = 60 km h–1 = 50/3 m $s^{-1.}$

		Therefore, the initial kinetic energy of the car, $E_{ki} = \frac{1}{2} m u^2$ $= \frac{1}{2} \times 1500\ \text{kg} \times (25/3\ \text{m s}^{-1})^2$ $= 156250/3\ \text{J}.$ The final kinetic energy of the car, $E_{kf} = \frac{1}{2} \times 1500\ \text{kg} \times (50/3\ \text{m s}^{-1})^2$ $= 625000/3\ \text{J}.$ Thus, the work done = Change in kinetic energy $= E_{kf} - E_{ki}$ $= 156250\ \text{J}.$

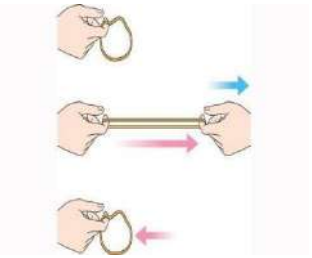
Summary	Factual Questions	CFU	Home Work
Students will be able to Do the calculations regarding kinetic energy.	1. A freely falling object eventually stops on reaching the ground. What happenes to its kinetic energy	1. Give some examples of kinetic energy in our surroundings.	1. A bullet of mass 10g is fired with a velocity of 80 m/s. What is its kinetic energy? 2. Find the momentum of a body of mass 100 g having kinetic energy of 20 J

PERIOD Number: 7.8

TOPIC: 7.2.3 Potential energy , Activity 7.8 , Activity 7.9

Learning Outcome: Students will be able to derive the formula for potential energy

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will do some activities by students and ask some questions on it.	**ACTIVITY 7.8 :** ✔ Doing this activity what do you observe? ✔ The rubber band is released ,what happens? ✔ How did it acquire energy when stretched?  **ACTIVITY 7.9 :**	✔ What is potential energy? ✔ Give ten examples of potential energy?

	 By doing this activity what will be observed. ✔ The slinky is released ,what happens? ✔ How did the slinky acquire energy when stretched? ✔ Would the slinky acquire energy when it is compressed?	✔ Derive the expression of potential energy? ✔ What is the formula of potential energy? ✔ What is S I units of potential energy?

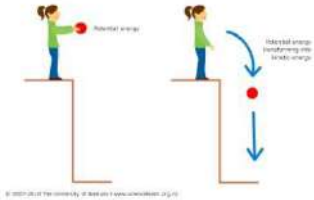
Summary	Factual Questions	CFU	Home Work
Students will be able to derive the formula for potential energy	1. Can the potential energy of an objects be negative?	1. Define Potential energy 2. Write the formula of PE. 3. Write the SI unit of PE.	1. Give some examples of potential energy 2. Give two examples of in which a body possesses both KE and PE at the same time

PERIOD Number: 7.9

TOPIC: Activity 7.10 and Activity 7.11 Activity 7.12

Learning Outcome: Students will be able to understand that object get potential enegy by its position

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will do some activities by students and ask some questions on it.	**ACTIVITY 7.10 :**  ✔ Doing this activity what will be observed? ✔ Why does a wind toy car move on the ground? ✔ Where did it acquire energy from? ✔ Does the energy acquired depend on the number of windings? **ACTIVITY 7.11 :**  ✔ by doing this activity what will be observed?	

	✔ If you lift the object and release it then what will be happened? ✔ does that object get energy? What type of energy is it? ✔ From where did it get that energy? **ACTIVITY 7.12 :**  ✔ After doing this activity what will be observed ✔ What will happen when the arrow is released? ✔ Do you observe any change in the bow? ✔ Which type energy is stored in that arrow? ✔ Give some more examples of this kind that you see in your surroundings?	

Summary	Factual Questions	CFU	Home Work
Students will be able to understand that object get potential enegy by its position	Why does a driver speed up his vehicle when he moves up a hill?	1. How is energy stored in a watch? 2. A bag of wheat weighs 200 kg. To what height should it be raised so that its potential energy may be 9800 joules? Take g = 9.8 ms^{-2}	3. An object of mass 40 kg is raised to a height of 5 m above the ground. What is its potential energy? If the object is allowed to fall, find its kinetic energy when it is half-way down.

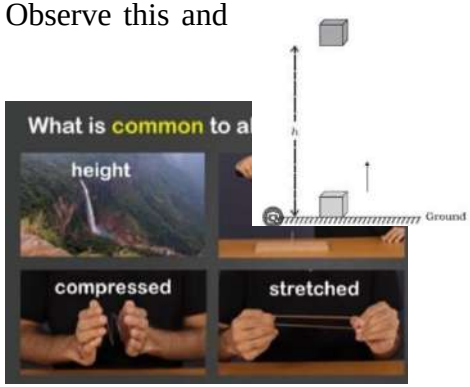
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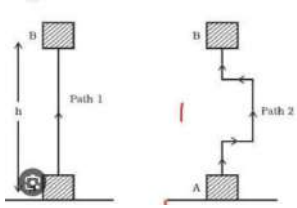
TOPIC: 7.2.4 Potential energy of an object at a height

Example 7.5 Example 7.6

Learning Outcome: Students will be able to explain that object get potential energy by its height

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask some brain storming questions.	Observe this and give answers.  ✔ Observe all this activities use one type of energy, what we call this energy? ✔ What is the formula of potential energy? ✔ How do you derive the formula of potential energy? ✔ What type of energy is with the object if it is at a some height? ✔ What type of forces acting on that object? ✔ Any gravitational potential energy acting on that object? ✔ What is gravitational potential energy?	

	✔ What is the relation between force and displacement? ✔ Do you think that the work done by the gravity depends upon different paths of the object when falling down from same height ?  By giving one problem and ask the students what values are given, what values are find ✔ What is the mass of the object? ✔ What is the P.E of the object? ✔ What will be find this problem?	work done, W = force × displacemen = $mg \times h$ = mgh

Example 10.5 Find the energy possessed by an object of mass 10 kg when it is at a height of 6 m above the ground. Given, g = 9.8 m $s^{-2.}$		Mass of the object, m = 10 displacement (height), h = 6 m, acceleration due to gravity, g = 9.8 m Potential energy = mgh = 10 kg × 9.8 m s^{-2} × 6 m = 588 J. The potential energy is 588 J.
Example 10.6 An object of mass 12 kg is at a certain height above the ground. If the potential energy of the object is 480 J, find the height at which the object is with respect to the ground. Given, g = 10 m s^{-2}		Mass of the object, m = 12 kg, potential energy, E_p = 480 J. $E_p = mgh$ 480 J = 12 kg × 10 m s^{-2} × h $h = \frac{480\ J}{120\ kg\ m s^{-2}} = 4$ m. The object is at the height of 4 m.

Summary	Factual Questions	CFU	Home Work
Students will be able to explain that object get potential energy by its height	1, Soni says that the acceleration in an object could b e zero even when several forces are acting on it. Do you agree with her? Why?	4. What is the potential energy possessed by an object of mass 1 kg kept at a height of I m? Take g = 10 ms^{-2}	**1.** The potential energy of a freely falling object decreases progressively. Does this violate the law of conservation of energy? Why?

PERIOD Number: 7.11

TOPIC: 7.2.5 Are various energy forms interconvertible? Activity 7.13 , Activity 7.14

Learning Outcome: Students will be able to understand that different forms of energies are interconvertible

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask some brain storming questions. ✔ Do you think how does a fan work? ✔ From where did the energy coming for preparing food? ✔ From where did the energy coming to you for running? ✔ How does a motor cycle run ? ✔ How does a generator produce electric energy? ✔ What is happening in the electric heater? ✔ While we are doing so many activities where the energy coming from? ✔ Do you think the energy changes one form to another?	**ACTIVITY 7.13 :** Divide two groups in whole class and discuss the various ways of energy conversion and asking some questions according to this-  ✔ How do green plants produce food? ✔ How are fuels such coal and petroleum were formed? 	✔ What is energy conversion? ✔ Why are doing energy conversion? ✔ Give some examples of energy conversion? ✔ Give some examples of chemical energy? ✔ Give answers the following man made devices have convert which type energies ✔ 1.fan ✔ 2.electric lamp ✔ 3.radio ✔ 4.water pump ✔ 5.cell ✔ 6.car ✔ 7.solar cell ✔ 8.watch ✔ 9.generator ✔ 10.television ✔ Students participate activity

	✔ What kinds of energy conversions sustain the water cycle? ✔ Which one released more energy in nature? ✔ How does wind turbines to produce electricity?  ✔ Give some more examples of energy conversion? **ACTIVITY 7.14 :** ✔ list out the different devices ,gadges which involve conversion of energy? ✔ Make a list of gadgets in your surroundings ? ✔ Identify what kind of energy conversion takes place? ✔ What happens to the total energy of a system during or after the energy conversion is constant? ✔ Is energy lost or gained?	✔ Write law of conservation of energy? ✔ Explain law of conservation of energy ? ✔ What is the formula of P.E ? ✔ What is the formula of K.E ? ✔

Summary	Factual Questions	CFU	Home Work
Students will be able to understand that different forms of energies are interconvertible	1. Are various forms of energy interconvertible? 2. Does the transfer of energy take place when you push a huge rock with all your might and fail to move it? Where is the energy you spend going?	1. What kind of energy transformation takes place at thermal power station?	1. What are the various energy transformations that occur when you are riding a bicycle?

PERIOD Number: 7.12

TOPIC: 7.2.6 Law of conservation of energy

Learning Outcome: Students will be able to draw the flowchart of conservation of energy

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask some brain storming questions. Consider a simple example. Let an object of mass, m be made to fall freely from a height, h. At the start, the potential energy is mgh and kinetic energy is zero. Why is the kinetic energy zero? It is zero because its velocity is zero. The total energy of the object is thus mgh. As it falls, its potential energy will change into kinetic energy. If v is the velocity of the object at a given instant, the kinetic energy would be $\frac{1}{2}mv^2$. As the fall of the object continues, the potential energy would decrease while the kinetic energy would increase. When the object is about to reach the ground, h = 0 and v will be the highest. Therefore, the kinetic energy would be the largest and potential energy the least. However, the sum of the potential energy and kinetic energy of the object would be the same at all points. That is, potential energy + kinetic energy = constant or	Observe this figure and answer the following  ✔ When the K.E is zero? ✔ When the K.E and P.E is constant? ✔ What is called the sum of P.E and K.E? ✔ When the P.E is decrease? ✔ When the sum of K.E and P.E same at all	✔ What is the formula of P.E ? ✔ What is the formula of K.E ? ✔ Complete the following

$mgh + \frac{1}{2} mv^2 =$ constant.	points? **ACTIVITY 7.15** Doing this activity by students at each case and ask some questions An object of mass 20 kg is dropped from a height of 4 m. Fill in the blanks in the following table by computing the potential energy and kinetic energy in each case. 	Height at which object is located (m)	Potential energy ($E_p = mgh$) (J)	Kinetic energy ($E_k = mv^2/2$) (J)	$E_p + E_k$ (J)
---	---	---	---
4			
3			
2			
1			
Just above the ground			| ✔ What is the mass we taken in this activity? ✔ What is the value of g? ✔ What would have happened it nature had not allowed the transformation of energy?	table An object of mass 20 kg is dropped from a height of 4 m. Fill in the blanks in the following table by computing the potential energy and kinetic energy in each case. | Height at which object is located (m)	Potential energy ($E_p = mgh$) (J)	Kinetic energy ($E_k = mv^2/2$) (J)	$E_p + E_k$ (J)
---	---	---	---
4			
3			
2			
1			
Just above the ground				 According to the law of conservation of energy, energy can only be transformed from one form to another; it can neither be created nor destroyed. The total energy before and after the transformation always remains constant. If energy is neither created nor destroyed them, where do we get energy from?

Summary	Factual Questions	CFU	Home Work
Students will be able to draw the flowchart of conservation of energy	**1.** Water stored in a dam is used to produce electric power. Explain various types of energy transformation taking place in the process.	1. State the law of conservation of energy	1. Illustrate the law of conservation of energy by discussing the energy changes which occur when we draw a pendulum bob to one side and allow it to oscillate. Why does the bob eventually come to rest? What happens to its energy eventually? Is it a violation of the law of conservation of energy?

PERIOD Number: 7.13

TOPIC: 7.3 Rate of doing work Activity 7.16

Learning Outcome: Students will be able to calculate the rate of work using the formula

Teaching Learning Process

I DO	WE DO	YOU DO
Teacher will ask some questions on the concept of rate doing work.	✔ Do all of us work at the same rate? ✔ Do all of us work at the same rate? ✔ Do machines consume or transfer energy at the same rate ? ✔ Do all agents work at same rates ? ✔ In your family who is stronger and why? ✔ Which machines consume more energy? **ACTIVITY 7.16 :** ✔ Doing this activity by students and ask some questions according this. ✔ Who accomplish the task?	✔ Which is more powerful bikes or motor cars? ✔ What is power? ✔ Which law states that all energy forms are convertible? ✔ What is the formula of power?

	 ✔ What is the work done by each? ✔ Who has done more work in a given time? ✔ which measures the speed of the work? ✔ What do we call the speed of work done? ✔ What is the formula of power? ✔ What is the unit of power? ✔ What do you know about watt? ✔ Teacher ask some questions by giving the following problem to solve.	

Summary	Factual Questions	CFU	Home Work
Students will be able to calculate the rate of work using the formula	1. Is power is scalar or vector quantity? 2. Define 1 watt of power	3. Explain the term power. 4. Write the SI unit of power	**1.** A certain household has consumed 250 units of energy during a month. How much energy is this in joules?

PERIOD Number: 7.14

TOPIC: Examples 7.7 and Example 7.8 and Activity 7.17

Learning Outcome: Students will be able to calculate the rate of work using the formula

Teaching Learning Process

I DO	WE DO	YOU DO
Example 7.7: ✔ Two girls, each of weight 400N climb up a rope through a height of 8 m. We name one of the girls A and the other B. Girls A takes 20s while B takes 50s to accomplish this task. What is the power expended by each girl?	✔ What is weight of the girl A? ✔ Displacement of A? ✔ Time Taken by A? ✔ What is power by girl A? ✔ What is weight of the girl B? ✔ Displacement of B? ✔ Time Taken by B? ✔ What is power by girl B?	Write the power formula? Power, P = Work done/time taken $= \frac{mgh}{t}$ $= \frac{400\ \text{N} \times 8\ \text{m}}{20\ \text{s}}$ $= 160\ \text{W}$
Exampe 7.8: ✔ A boy of mass 50 kg runs up a staircase of 45 steps in 9 sec .if the height of each step is 15 cm ,find his power?	✔ What is the mass of boy? ✔ How many staircase in this problem? ✔ What is the value of g? ✔ What will be we find? **ACTIVITY 7.17 :**	Power, $P = \frac{mgh}{t}$ $= \frac{400\ \text{N} \times 8\ \text{m}}{50\ \text{s}}$ $= 64\ \text{W}$

	✔ By doing this activity by students with guidance of teacher in our school ,and ask some questions to know the calculation of consumed eclectic power ✔ Do you observe the electric meter in our school? ✔ How many units are consumed during day time? ✔ How much money will we pay in our school electric bill per month and how many units consumed?	power, P = Work done/time taken $= \frac{mgh}{t}$ $= \frac{500\ \text{N} \times 6.75\ \text{m}}{9\ \text{s}}$ $= 375\ \text{W}$ Power is 375 W.

Summary	Factual Questions	CFU	Home Work
Students will be able to calculate the rate of work using the formula	1. A man does 50 J of work in 5 seconds. What is his power? 2. Write the formula of power.	1.How much power consumption in our school every month? 2.How much cost is 1 unit of electricity? 3. How is your electricity bill calculated?	1. An electric heater is rated 1500 W. How much energy does it use in 10 hours? 2. Find the energy in joules consumed in 10 hours by four devices of power 500 W each

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson: 7. Work and Enery		Class: 9	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
7.1	7.1 Work 7.1.1 Not much work in spite of working hard! Activity 7.1				
7.2	7.1.2 Scientific conception of work Activity 7.2 Activity 7.3				
7.3	7.1.3 Work done by a constant force Example 7.1 Activity 7.4				
7.4	7.2.Energy Activity 7.5				
7.5	7.2.1 Forms of enegy 7.2.2 Kinetic energy Activity 7.6				
7.6	Activity 7.7				
7.7	Examples 7.3 and 7.4				
7.8	7.2.3 Potential energy Activity 7.8 Activity 7.9				
7.9	Activity 7.10 and Activity 7.11 Activity 7.12				

7.10	7.2.4 Potential energy of an object at a height example 7.5 example 7.6				
7.11	7.2.5 Are various energy forms interconvertible? Activity 7.13 Activity 7.14				
7.12	7.2.6 Law of conservation of energy				
7.13	7.3 Rate of doing work Activity 7.16				
7.14	**Examples 7.7 and Example 7.8 ans Activity 7.17**				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?

4	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

January 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1 New year 8.1	2 NI	3 No Bag Day
4 Sunday	5	6	7	8	9 8.2	10 Second Saturday
11 Sunday National Youth Day	12	13	14	15	16	17 No Bag Day
18 Sunday D.R.Kaprekar Jayamti	19 8.2	20 8.3	21 8.4	22 8.4	23 NI	24 No Bag Day Cluster,PTM, SMCmeeting
25 Sunday Republic day	26 8.5	27 8.6 Annual Day	28 8.7	29 8.8	30 NI Gandhi Vardhanti	31

No bag Day | Cluster Complex | Additional Resources

FA - 3

Pongal Holidays

Pongal Holidays 10-01-2026 TO 18-01-2026,
Pongal Holidays for Christian Minority Schools 10-01-2026 TO 15-01-2026

Week1: 7.1

Week 2: 8.2

Week 3:

Week 4: 8.2 – 8.4

Week 5: 8.5 – 8.8

Friday

Chapter: 8. Structure of the atom

I3K7H4

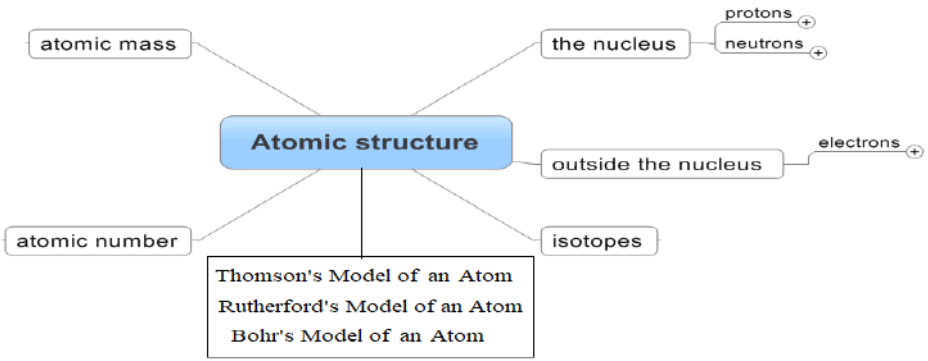
Learning Outcomes:

- Draw the flow chart of subatomic particles of atom
- Explain the Thomson model of atom
- Explain the Rutherford's model of atom and its drawbacks
- Explain about Bohr's model of atom
- Explain the process of distribution of electrons in orbitals
- Understand the concept of valency,
- Differentiate atomic number and mass number
- Differentiate isotopes and isobars

Prior Concepts/Skills: (Essential concepts and skills to be checked/bridged before teaching the current concept)

1. What are atom made of?
2. What is shape of the atom?
3. What is the mass of an atom?

Concept Map



Synopsis

- **Charged particles:** Credit for the discovery of electron and proton goes to J.J. Thomson and E.Goldstein, respectively.
- **Thomson's model:** J.J. Thomson proposed that electrons are embedded in a positive sphere.

- **Rutherford model:** Rutherford's model of the atom proposed that a very tiny nucleus is present inside the atom and electrons revolve around this nucleus. The stability of the atom could not be explained by this model.
- **Bohr's model:** Neils Bohr's model of the atom was more successful. He proposed that electrons are distributed in different shells with discrete energy around the nucleus.
- **Distribution of electrons:** Shells of an atom are designated as K,L,M,N,…
- **Valency:** Valency is the combining capacity of an atom.
- **Valency electrons:** The electrons present in the outermost shell of an atom are known as the valence electrons.
- **Atomic number:** The atomic number of an element is the same as the number of protons in the nucleus of its atom.
- **Mass number:** The mass number of an atom is equal to the number of nucleons in its nucleus.
- **Isotopes:** Isotopes are atoms of the same element, which have different mass numbers.
- **Isobars:** Isobars are atoms having the same mass number but different atomic numbers.

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	QR Code	Any Specific Information
8.1	8.1 Charged particles in matter Activity 8.1	draw the flow chart of subatomic particles of atom		
8.2	8.2 The structure of an atom 8.2.1 Thomson model of atom	explain the Thomson model of atom		
8.3	8.2.2 Rutherford's model of an atom , and drawbacks of model	explain the Rutherford's model of atom and its drawbacks		
8.4	8.2.3 Bohr's model of an atom 8.2.4 Neutrons	explain about Bohr's model of atom		
8.5	8.3 How electrons are distributed in different orbits?	explain the process of distribution of electrons in orbitals		
8.6	8.4 Valency	understand the concept of valency,		
8.7	8.5 Atomic number and mass number	differentiate atomic number and mass		

		number		
8.8	8.6 Isotopes 8.6.1 Isobars applications	differentiate isotopes and isobars		

PERIOD Number: 8.1

TOPIC: 8.1 Charged particles in matter Activity 8.1

Learning Outcome: Students will be able to draw the flow chart of subatomic particles of atom

Teaching Learning Process

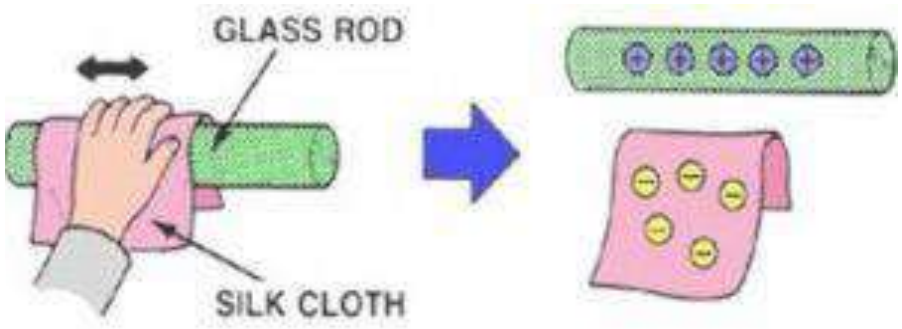
I DO	WE DO	YOU DO
The teacher starts the lesson by probing questions to check for prior knowledge: Today let us learn about sub-atomic particles and the various models that have been proposed to explain how these sub-atomic particles are arranged inside the atom. What are canal rays? These rays were positively charged radiations which ultimately led to the discovery of another sub-atomic particle.	- What are the fundamental building blocks of matter? - What makes the atom of one element different from the atom of another element? - Are atoms really indivisible as proposed by Dalton?(or) Are there smaller constituents inside the atom? Based on the series of experimentations, and from studying the static electricity and the conditions under which electricity is conducted by different substances, it was found that an atom is divisible. **Activity - 4.1** A. Does the comb attract small pieces of paper after combing dry hair? 	1) If an atom contains one electron and one proton, will it carry any charge or not? 2) What are canal rays? 3) What charges are developed on comb and hair after we comb? 4) How electron and proton have equal charge (magnitude) when their masses are different? 5) Dalton's atomic theory successfully explained (i) Law of conservation of mass (ii) Law of constant composition (iii) Law of radioactivity

- This sub-atomic particle had a charge, equal in magnitude but opposite in sign to that of the electron.
- Its mass was approximately 2000 times as that of the electron.
- It was named as proton.

  In general, an electron is represented as ‘e–’ and a proton as ‘p+’.
- The mass of a proton is taken as one unit and its charge as plus one.
- The mass of an electron is considered to be negligible and its charge is minus one.
- An atom was composed of protons and electrons, mutually balancing their charges.
- It also appeared that the protons were in the interior of the atom.

  Electrons could easily be removed off but not protons.

B. Rub the glass rod with a silk cloth and bring the rod near to an inflated balloon.



Observe what happens.

Can we confirm that the rubbing of two objects together, they become electrically charged?

Where does this charge come from?

Many scientists contributed in revealing the presence of charged particles in an atom. An atom is divisible and consists of charged particles.

- The electron identified by J.J. Thomson,
- E. Goldstein discovered the presence of new radiations in gas discharge tubes

(iv) Law of multiple proportion

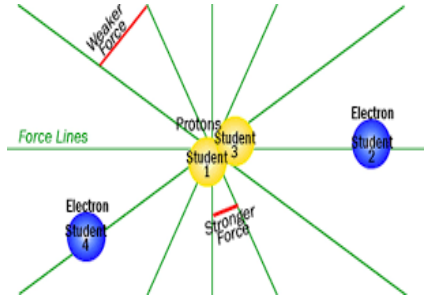
(a) (i), (ii) and (iii)

(b) (i), (iii) and (iv)

(c) (ii), (iii) and (iv)

(d) (i), (ii) and (iv)

6) What are the forces operate between electron and proton?



7) Define a) Weaker forces
(b) Stronger forces.

Collect the information about scientists who discovered fundamental particles and prepare charts with all relevant pictures?

Summary	Factual Questions	CFU	Home Work
Students will be able to draw the flow chart of subatomic particles of atom	1. If an atom contains one electron and one proton, will it carry any charge or not?	1.What are canal rays? 2. Who discovered Electron? 3. Who discovered Proton?	Compare the properties of electrons, protons and neutrons

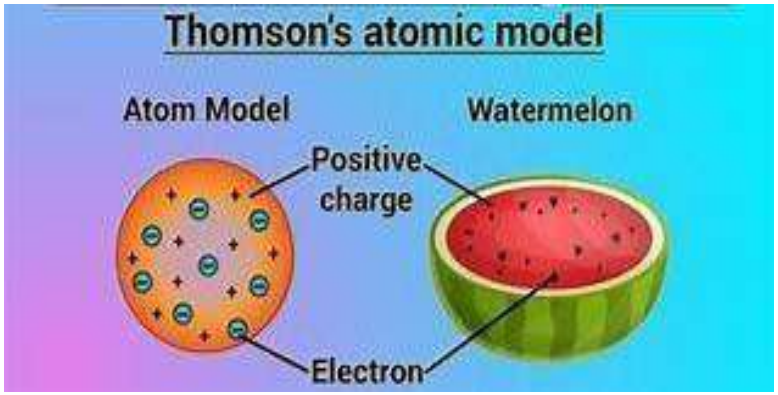
PERIOD Number: 8.2

TOPIC: 8.2 The structure of an atom , 8.2.1 Thomson's model of an atom

Learning Outcome: Students will be able to explain the Thomson's model of atom

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with the students to know their precious knowledge about the Content of the topic.	1. Does the comb attract the small pieces of paper after combing dry hair? 2. Rub the glass rod with a silk cloth and bring the rod near to an inflated balloon, observe what happens. 3. Name the three sub atomic particles of an atom? 4. Write the charge of electron, proton and neutron. **J.J.Thomson's atomic Model:**	1. Thomson atomic model is also known as…… 2. What are embedded in the positive sphere of an atom? 3. Who is the father of Nuclear Physics? 4. In the Thomson's model of atom, which of the following statements are correct? • the mass of the atom is assumed to be uniformly distributed over the

Let us discuss the arrangement of these sub- atomic particles in an atom by using various atomic models	Thomson proposed that: 1. An atom consists of positively charged sphere and the electrons are embedded in it The negative and positive charges are equal in magnitude, So the atom is electrically neutral  The result of this experiment could not explain the proper arrangement of sub-atomic particles; hence other atomic models were deviled. So, the atom as a whole is electrically neutral. Although Thomson's model explained that atoms are electrically neutral, the results of experiments carried out by other scientists could not be explained by this model.	atom • the positive charge is assumed to be uniformly distributed over the atom • the electrons are uniformly distributed in the positively charged sphere 5. Which of the following statements about Rutherford's model of atom are correct? • considered the nucleus as positively charged • established that the α–particles are four times as heavy as a hydrogen atom • can be compared to solar system • agreed with Thomson's model • (i) and (iii) • (ii) and (iii) • (i) and (iv) • only (i)

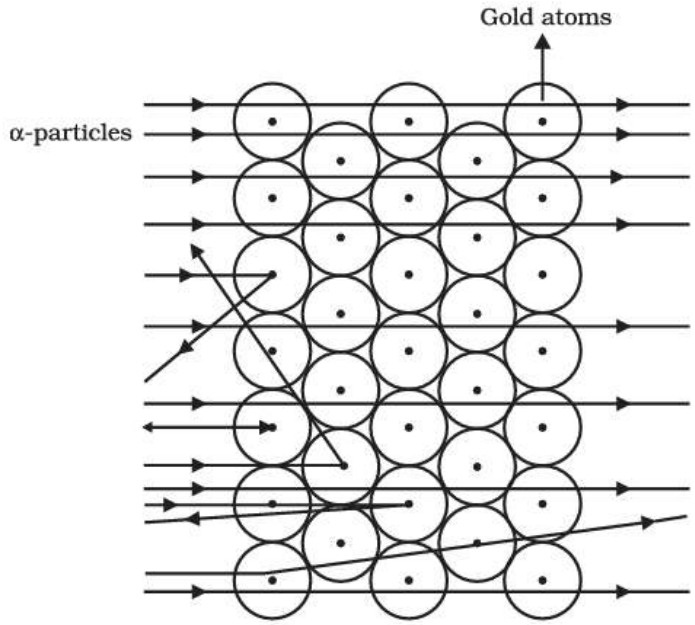
Summary	Factual Questions	CFU	Home Work
Students will be able to explain the Thomson's model of atom	What is the significant of J.J.Thomson's model of an atom	Draw the label diagram of Thomson's model of an atom.	What are the limitations of J.J. Thomson's model of the atom?

PERIOD Number: 8.3

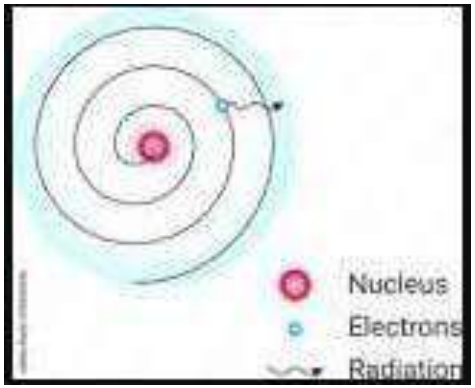
TOPIC: 8.2.2 Rutherford's model of an atom, and drawbacks of model

Learning Outcome: Students will be able to explain the Rutherford's model of atom and its drawbacks

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with the students to know their precious knowledge about the Content of the topic. **Rutherford's Atomic Model:**		1.Which of the following statements about Rutherford's model of atom are correct? (i) considered the nucleus as positively charged (ii) established that the α–particles are four times as heavy as a hydrogen atom (iii) can be compared to solar system (iv) agreed with Thomson's model (a) (i) and (iii) (b) (ii) and (iii) (c) (i) and (iv) (d) only (i)

	Observations: (i) Most of the α-particles penetrate through the gold foil and travel in a straight line. (ii) Some of the α-particles were deflected with some angles. (iii) Surprisingly one out of every 12000 particles appeared to rebound (bounced back). In the words of Rutherford, "This result was almost as incredible as if you fire a 15- inch shell at a piece of tissue paper and it comes back and hits you". (i) Most of the space inside the atom is empty because most of the α-particles passed through the gold foil without deviation. (ii) Very few particles were deflected from their path, indicating that the positive charge of the atom occupies very small space. (iii) A very small fraction of α-particles was deflected, indicating that all the positive charge and mass of the gold atom were concentrated in a very small volume within the atom. ❖ **On the basis of his experiment, Rutherford put forward the nuclear model of an atom, which has the following features:** (i) There is positively charged centre in an atom called as a nucleus. Nearly all the mass of an atom resides in the nucleus.	2. Rutherford's α–particle scattering experiment showed that (i) electrons have negative charge (ii) the mass and positive charge of the atom is concentrated in the nucleus (iii) neutron exists in the nucleus (iv) most of the space in atom is empty 3.Which of the above statements are correct? (a) (i) and (iii) (b) (ii) and (iv) (c) (i) and (iv) (d) (iii) and (iv) 4. Rutherford's α–particle scattering experiment showed that (i) electrons have negative charge (ii) the mass and positive charge of the atom is concentrated in the nucleus (iii) neutron exists in the nucleus (iv) most of the space in atom is

Rutherford concluded from the α-particle scattering experiment that	(ii) The electrons revolve around the nucleus in circular paths. (iii) The size of nucleus is very small as compared to the size of the atom Drawbacks of Rutherford's model of an atom  The revolution of the electron in a circular path is not expected to be stable. Any particle in a circular path would undergo acceleration. During the acceleration, charged particles would radiate energy. Thus, the revolving electron would lose energy and finally fall into the nucleus, if this was so, then the atom should be highly unstable and hence matter would not exist in the form that we know. We know that atoms are quite stable.	empty 5. Which of the above statements are correct? (a) (i) and (iii) (b) (ii) and (iv) (c) (i) and (iv) (d) (iii) and (iv) 6. What are the observations made by Rutherford during the α-particle scattering experiment? 7. What are the postulates proposed by Rutherford for atomic model of an atom? 8. Rutherford's 'alpha (α) particles scattering experiment' resulted in to discovery of (a) Electron (b) Proton (c) Nucleus in the atom (d) Atomic mass 9. What is drawback of Rutherford atomic model?

Summary	Factual Questions	CFU	Home Work
Students will be able to explain the Rutherford's model of atom and its drawbacks	Why did Rutherford's select the gold-foil in his alpha particle scattering experiment?	What are the limitations of Rutherford's model of the atom?	Describe Bohr's model of the atom.

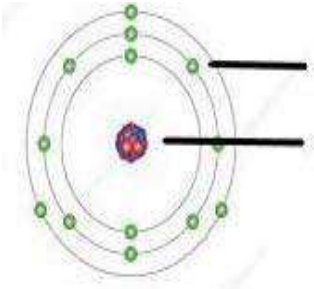
PERIOD Number: 8.4

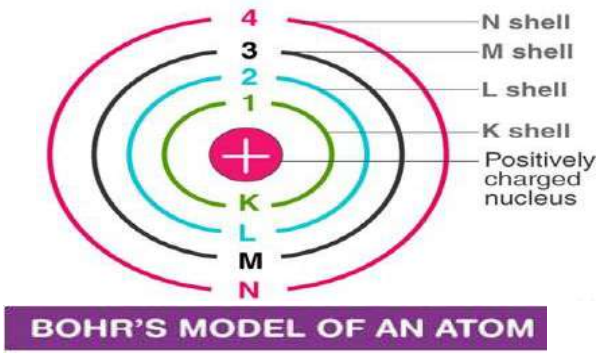
TOPIC: 8.2.3 Bohr's model of an atom

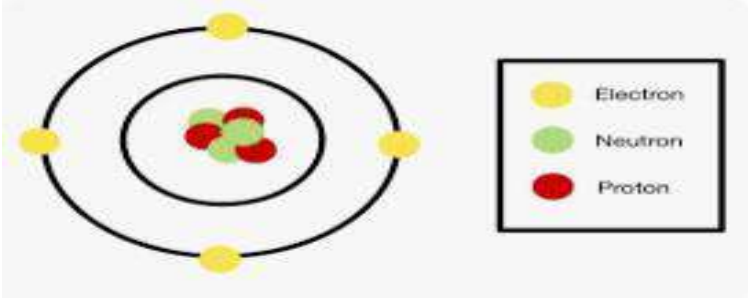
8.2.4 Neutrons

Learning Outcome: Students will be able to explain about Bohr's model of atom

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. Bohr's Atomic Model 	1. What is an atomic model? 2. Where are positively charged particles accumulated as per the nuclear model of an atom? 3. What are the drawbacks of Rutherford's atomic model of an atom? **POSTULATES OF BOHR'S ATOMIC MODEL** 1. Only certain special orbits known as discrete orbits of electrons are allowed inside the atom. 2. While revolving in discrete orbits the electrons do not radiate energy. These orbits or shells are called energy levels; these orbits are represented	1. For which book Bohr got Nobel prize? 2. The book **"Atomic Theory and description of Nature"** contains whose writings?

Main points	with K, L, M, N…etc. What do you mean by discrete orbits? What happens if electrons radiate energy while revolving round the nucleus?  BOHR'S MODEL OF AN ATOM A few energy levels in an atom, increasing order of energy of energy levels or orbits **K< L < M < N…** Q) Which forces operates between like charges? Q) What are the charged particles present in the nucleus of an atom? Q) How repulsions between Positive charged particles (Protons) are controlled in the nucleus to make it stable?	3. What is a discrete orbit? 4. "K" refers to which orbit in an atom? 5. Arrange the orbits in increasing order of their energies. 6.Make the model of an atom as per the postulates of Bohr's atomic theory? 7. Who discovered Neutron?

Neutrons	 i. Neutron is another sub atomic particle and represented by “n”, which had no charge with mass nearly equal to that of proton. Which is heavier between proton and neutron? ii. Neutrons are present in the nucleus of all atoms (except Hydrogen-zero neutrons). Why electrons and neutrons are not kept in the nucleus? What is the reason for the protons to present only in the nucleus?	8. What is the notation of Neutron? 9. Where is the position of Neutron in the atom? 10. Which charged particle has the mass nearly equal to that of the neutron? 11. Which element doesn’t contain neutron?

Summary	Factual Questions	CFU	Home Work
Students will be able to explain about Bohr’s model of atom	Why are the shells in which the electrons revolve around the nucleus of an atom called energy levels?	1. On the basis of Thomson’s model of an atom, explain how the atom is neutral as a whole. 2. On the basis of Rutherford’s model of an atom, which sub atomic particle is present in the nucleus of an atom?	1. Draw a sketch of Bohr’s model of an atom with three shells. 2. Compare all the proposed models of an atom given in this chapter.

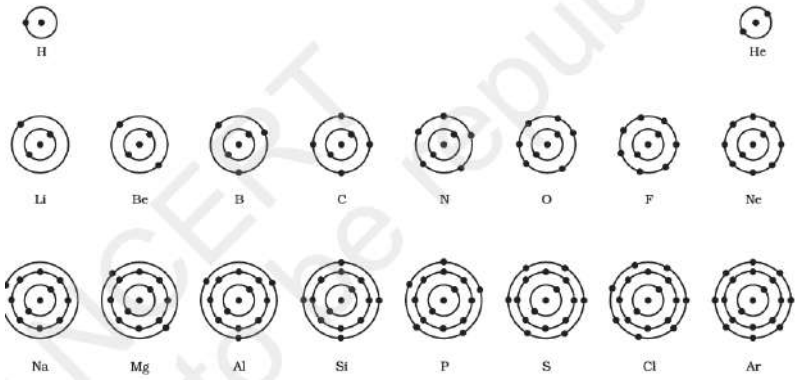
PERIOD Number: 8.5

TOPIC: 8.3 How electrons are distributed in different orbits?

Learning Outcome: Students will be able to explain the process of distribution of electrons in orbitals

Teaching Learning Process

I DO	WE DO	YOU DO
By interacting with the students to know their previous knowledge about the context of the topic. **Bohr - Bury Rules :**	1. What is an orbit? 2. What is another name for an orbit? 3. When do electrons not radiate energy? 4. How many electrons can accommodate in an orbital? 5. How electrons are distributed in different orbits/shells? 5. The maximum number of electrons present in a shell is given by the formula **$2n^2$**(where n is the orbit number or energy level index, 1, 2, 3, etc. Hence the maximum number of electrons in different shells are as follows Frist orbit or K Shell will be = $2 \times 1^2 = 2$ Second orbit or L- Shell will be = $2 \times 2^2 = 8$ 6. The maximum number of electrons that can be accommodated in the outer most orbit is 8	1)Prepare **five** questions on the content given below? **Orbit**s an energy level around the nucleus whereas **Orbital**is a three-dimensional space in which the probability of finding the electron is maximum. **Orbit** which is **closer** to the nucleus known as **core orbit** and orbit which is far away from nucleus known as valence orbit i.e., outermost orbit called as valence orbit 2) What is the formula used to calculate maximum number of electrons? 3) Make the static atomic model

	7. Electrons are not accommodated in a given shell, unless the inner shells filled. That is, the shells are filled in a step - wise manner. Do all the elements have arrangement of electrons in the shells or any difference is observed? Explain  Schematic atomic structure of first eighteen elements	displaying electronic configuration of first eighteen elements.

Summary	Factual Questions	CFU	Home Work
Students will be able to explain the process of distribution of electrons in orbitals	Can electrons be arranged in second shell or L-shell without filling first shell or K-shell?	1. Write the distribution of electrons in carbon and sodium atoms. 2. If K and L shells of an atom are full, then what would be the total number of electrons in the atom?	Summarise the rules for writing of distribution of electrons in various shells for the first eighteen elements.

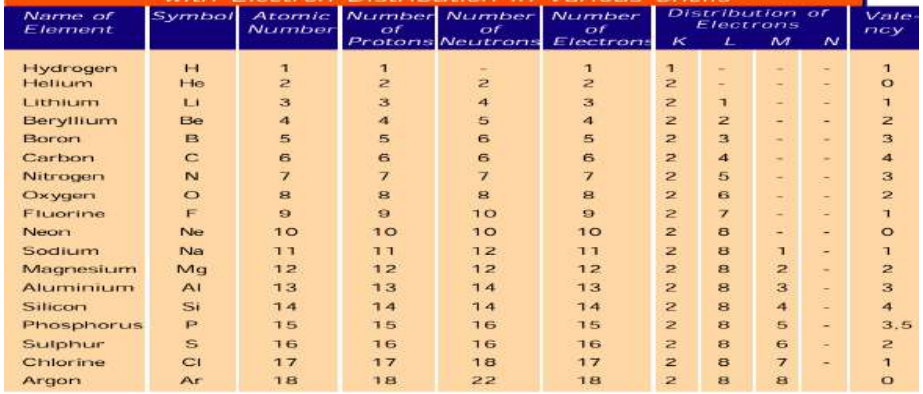
PERIOD Number: 6

TOPIC: 8.4 Valency

Learning Outcome: Students will be able to understand the concept of valency,

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding the concept of the topic. **Valence electrons:** The number of electrons present in the outer most shell of an atom are known as Valence electrons**.** **Valency:** Combining capacity of an atom with other atoms is valency. Discussion of valency, Valency electrons, Valency shell, Octet configuration	1. What are nucleons? 2. Name the sub atomic particle with negligible mass? 3. What formula is used to calculate the number of electrons present in each orbit? 4. What is the name of inner orbit of an atom? 5. What is the name of outer orbit of an atom? 6. Name the electrons in the outer-most orbit? 7. What are valency electrons? 8. What is the difference between valency and valence- electrons? In the element below can say atomic number, valence electrons and valency of an element. **Example: Cl** How to calculate the number of nucleons and protons in an atom by using atomic number(Z) and mass number(A). What is an Atomic number? What is a Mass number?	1) Elements with valency 1 are (a) always metals (b) always metalloids (c) either metals or non- metals (d) always non-metals 2) Find the no.of valence electrons in first 9 elements of the periodic table? 3) Write their valency? 4) How many valence electrons are there in Neon(Ne)?

	Table 4.1: Composition of Atoms of the First Eighteen Elements with Electron Distribution in Various Shells 	5)What is the valency of Neon (Ne)?

Name of Element	Symbol	Atomic Number	Number of Protons	Number of Neutrons	Number of Electrons	Distribution of Electrons K	L	M	N	Valency
Hydrogen	H	1	1	-	1	1	-	-	-	1
Helium	He	2	2	2	2	2	-	-	-	0
Lithium	Li	3	3	4	3	2	1	-	-	1
Beryllium	Be	4	4	5	4	2	2	-	-	2
Boron	B	5	5	6	5	2	3	-	-	3
Carbon	C	6	6	6	6	2	4	-	-	4
Nitrogen	N	7	7	7	7	2	5	-	-	3
Oxygen	O	8	8	8	8	2	6	-	-	2
Fluorine	F	9	9	10	9	2	7	-	-	1
Neon	Ne	10	10	10	10	2	8	-	-	0
Sodium	Na	11	11	12	11	2	8	1	-	1
Magnesium	Mg	12	12	12	12	2	8	2	-	2
Aluminium	Al	13	13	14	13	2	8	3	-	3
Silicon	Si	14	14	14	14	2	8	4	-	4
Phosphorus	P	15	15	16	15	2	8	5	-	3,5
Sulphur	S	16	16	16	16	2	8	6	-	2
Chlorine	Cl	17	17	18	17	2	8	7	-	1
Argon	Ar	18	18	22	18	2	8	8		0

Summary	Factual Questions	CFU	Home Work
Students will be able to understand the concept of valency,	Why do the elements Helium, Neon and Argon have zero valency?	How will you find the valency of chlorine, sulphur and magnesium?	1. Define valency by taking examples of silicon and oxygen 2. If Z = 3, what would be the valency of the element? Also, name the element.

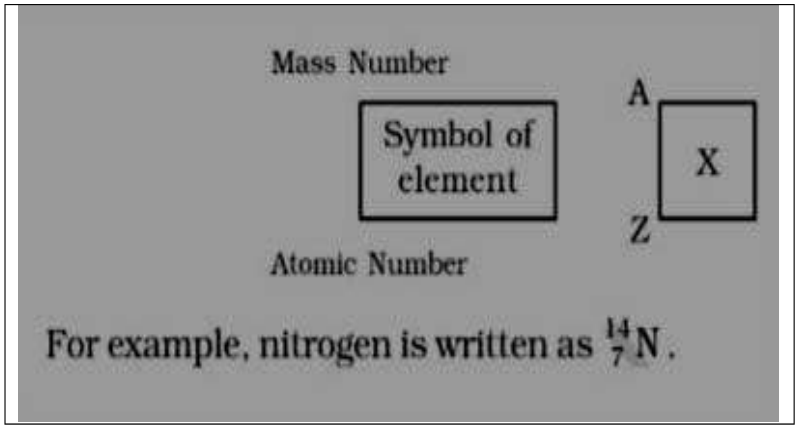
PERIOD Number: 8.7

TOPIC: 8.5 Atomic number and mass number

Learning Outcome: Students will be able to differentiate atomic number and mass number

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding the concept	What is an Atomic number? What is a Mass number? **ATOMIC NUMBER**:	1)Which of the following are true for an element?

of the topic. **ATOMIC NUMBER**: **MASS NUMBER:**	• Total number of protons present in the nucleus of an atom is defined as Atomic number. • This number is also equal to the number of electrons present in a neutral atom • It is denoted by "Z" **Example :** Let us consider Hydrogen Z=1 No of protons =1 Number of electrons =1 **MASS NUMBER:** • Protons and neutrons are present in the nucleus of an atom and the mass of an atom resides in its nucleus. "The sum of total number of protons and neutrons present in the nucleus of an atom is defined as Mass number" • It is denoted by 'A' In the notation for an atom, the atomic number mass number and symbol of element are written as:  For example, nitrogen is written as $^{14}_{7}N$.	(i) Atomic number = number of protons + number of electrons (ii) Mass number = number of protons + number of neutrons (iii) Atomic mass = number of protons = number of neutrons Atomic number = number of protons = number of electrons a. (i) and (ii) b. (i) and (iii) c. (ii) and (iii) d. (ii) and (iv) 2. The ion of an element has 3 positive charges. Mass number of the atom is 27 and the number of neutrons is 3. What is the number of electrons in the ion? a. 13 b. 10 c. 14 d. 16 4. The number of electrons in an element X is 15 and the number of neutrons is 16. Which of the following is the correct representation of the element? **(a)** $_{15}X^{31}$ **(b)** $_{16}X^{31}$ **(c)** $_{15}X^{16}$ **(d)** $_{16}X^{15}$ 5.Which of the following correctly represent the electronic distribution in the Mg atom?

		(c) 3, 8, 1 (d) 2, 8, 2 (e) 1, 8, 3 (f) 8, 2, 2

Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate atomic number and mass number	1.Which property of an element can be used for its identification?	If number of electrons in an atom i s 8 and number of protons is also 8, then (i) what is the atomic number of the atom? And (ii) what is the charge on the atom?	1. Explain with examples (i) Atomic number, (ii) Mass number

Learning Outcome: Students will be able to differentiate isotopes and isobars

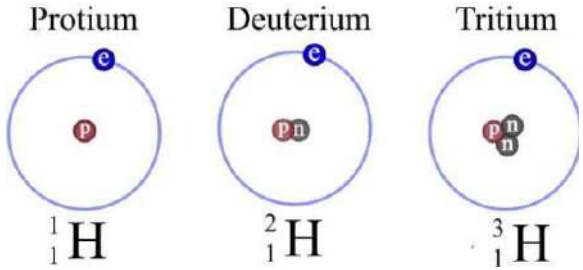
PERIOD Number: 8.8

TOPIC: 8.6 Isotopes

8.6.1 Isobars applications

Teaching Learning Process

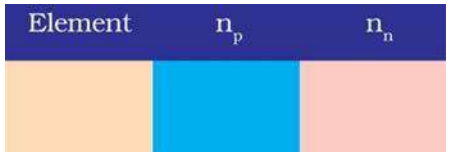
I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding the concept of the topic.	1.Write symbols for the following elements? (i)Element with 1 proton and 1 electron? (ii)Element with 1 proton , 1 neutron and 1 electron? (iii)Element with 1 proton , 2 neutrons and 1 electron ?	1. Atomic models have been improved over the years. Arrange the following atomic models in the order of their chronological order. (i) Rutherford's atomic model.

	2. Are these three atoms same or different? 3.What is the nucleus composition of these atoms? 4. What is the name of these three atoms? 5. Give the relation between atomic number and mass number of these atoms? 6.What are isotopes? "**The atoms of same element with different mass numbers are called as Isotopes"** **Examples:** Hydrogen:  Carbon: $_{6}C^{12}$ **and** $_{6}C^{14}$ Chlorine: $_{l7}Cl^{35}$ **and** $_{17}C^{37}$ The mass of chlorine is taken as 35.5 instead of 35 or 37.Why?	(ii) Thomson's atomic model. (iii) Bohr's atomic model (a) (i), (ii) and (iii) (b) (ii), (iii) and (i) (c) (ii), (i) and (iii) (d) (iii), (ii) and (i) 2. Will $_{35}Cl$ and $_{37}Cl$ have different valencies? Justify your answer? 3. The atomic number of calcium and argon are 20 and 18 respectively, but the mass number of both these elements is 40. What is the name given to such a pair of elements? 4. Complete the Table 8.1 on the basis of information available in the symbols given below (a) Cl^{35}_{17} **(b)** C^{12}_{6} **(c)** $_{35}Br^{81}$

What do you mean by average atomic mass? What is natural abundance of $_{17}Cl^{35}$ **and** $_{17}C^{37}$?

The mass of an atoms of an element is taken as the average mass of all the naturally occurring isotopes of that elements Average-Mass = $NA_1 \times A_1 + NA_2 \times A_2 + \ldots\ldots\ldots\ldots$

NA =Natural Abundance of isotope A =Mass no of isotope



$$[35X\ \frac{75}{100} + 37X\ \frac{25}{100}] = \{\frac{105}{100} + \frac{37}{4}\} = \frac{142}{4} = 35.5\ u$$

What are the uses of different isotopes?

1. An Isotope of uranium is used as fuel in nuclear reactor.
2. An isotope of Cobalt is used as in the treatment of cancer
3. An isotope of Iodine is use in the treatment of goitre. What are isobars?

Atoms of different elements with same mass number are known as Isobars .

Examples:

20 Ca^{40} ,18Ar^{40}

5. Define Isotopes ?
6. Are electrons, Protons and Neutrons same in isotopes?

7. Why mass number is different for isotopes?

8. Three elements X,Y,Z has following electrons,

protons, neutrons.

Atom	e	p	n
X	6	6	6
Y	7	7	7
Z	6	6	8

(a) Name the isotopes out of X, Y, Z ?

(b)Write the isobars out of X ,Y,Z ?

Summary	Factual Questions	CFU	Home Work

Students will be able to differentiate isotopes and isobars	Why, isotopes are importance in our life?	1.Isotopes of an element have (a) the same physical properties (b) different chemical properties (c) different number of neutrons (d) different atomic numbers 2.What are isobars? Give examples	1. Write the electronic configuration of any one pair of isotopes and isobars. 2. Write any two applications of isotopes

Teacher's Diary

Name of the Teacher:		**Name of the Month:**	
Name of the Lesson: 8. Structure of the Atom		Class: 9	

Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	**Remarks**
8.1	8.1 Charged particles in matter Activity 8.1				
8.2	8.2 The structure of an atom 8.2.1 Thomson model of an atom				
8.3	8.2.2 Rutherford's model of an atom, and the drawbacks of the model				
8.4	8.2.3 Bohr's model of an atom 8.2.4 Neutrons				

8.5	8.3 How are electrons distributed in different orbits?				
8.6	8.4 Valency				
8. 7	8.5 Atomic number and mass number				
8. 8	8.6 Isotopes 8.6.1 Isobars applications				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

February 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Sunday 1	2 9.1	3 9.2	4 9.3	5 9.4	6 NI	No Bag Day 7
Sunday 8	9 FA - 4	10	11	12	13 NI	Second Saturday 14
Sunday 15 Maha Sivaratri	16 9.5	17 9.6	18 9.7	19 9.8	20 NI	No Bag Day Cluster, PTM, SMC meeting 21
Sunday 22	23 9.9	24 9.10	25 NI	26 NI	27 NI	No Bag Day National Science Day 28

No bag Day | Cluster Complex | Teacher Resources

Week1: 9.1 – 9.4

Week 2:

Week 3: 9.5 – 9.8

Week 4: 9.9 – 9.10

Week 5:

↓ TEACHER'S NOTES ↓

Chapter: 9. Sound

I3K7H4

Learning Outcomes:

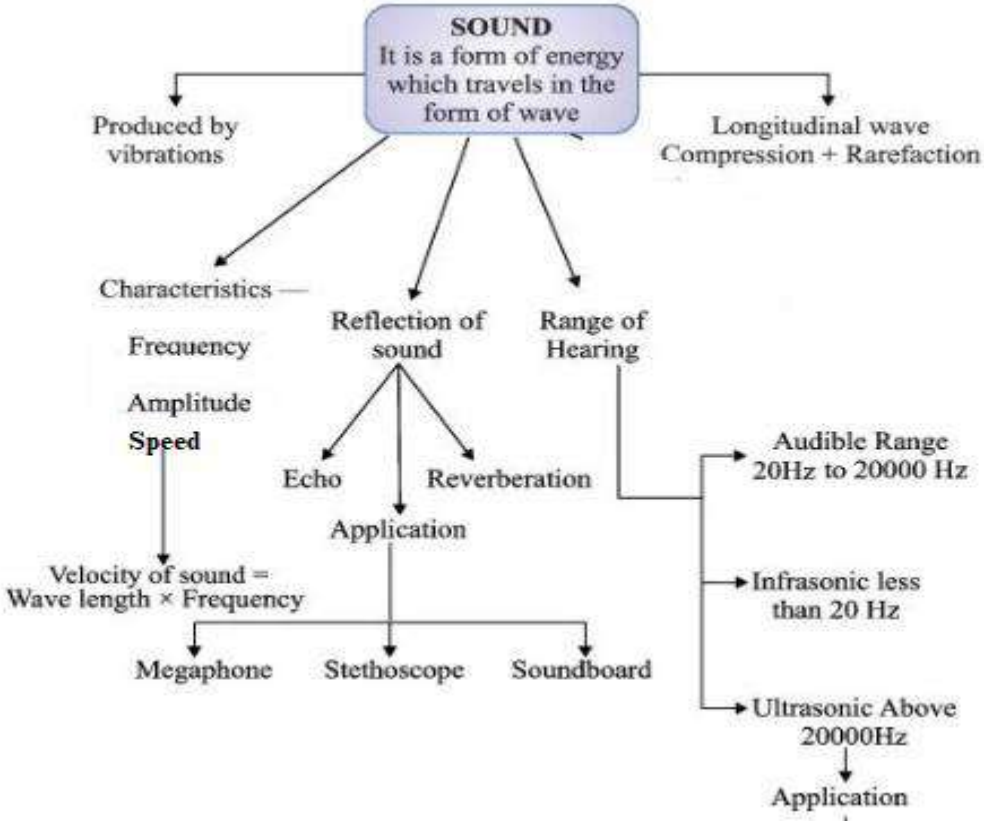
1. Explain the production of sound
2. Understand different ways of production of sound
3. Explain the propagation of sound
4. Give examples for longitudinal waves
5. Identify the relation between frequency and pitch of the sound
6. Do calculation on frequency
7. Understand that sound is different in different media
8. Differentiate echo and reverberation
9. Understand that range of hearing is different for humans and animals
10. Applies ultrasounds in day to day life.

Prior Concepts/Skills:

(Essential concepts and skills to be checked/bridged before teaching the current concept)

1. How do objects produce sound?
2. Does the sound travel if there is no medium?
3. What is the unit to measure the sound intensity?

Concept Map



Synopsis

- **Production of Sound:** Sound is produced due to vibration of different objects.
- **Propagation of sound:** Sound travels as a longitudinal wave through a material medium.
- **Types of Waves:** 1. Longitudinal wave 2.Transverse wave
- **Characteristics of sound waves:** 1.Frequency 2.Amplitude 3. Speed
- **Wave length:** The distance between two consecutive compressions or two consecutive rarefactions is called the wavelength

- **Frequency:** The number of complete oscillations per unit time is called 1 the frequency
- **Time period:** The time taken by the wave for one complete oscillation of the density or pressure of the medium is called the time period
- **Laws of Reflection of sound:** The law of reflection of sound states that the directions in which the sound is incident and reflected make equal angles with the normal to the reflecting surface at the point of incidence and the three lie in the same plane.
- **Echo:** For hearing a distinct sound, the time interval between the original sound and the reflected one must be at least 0.1 s.
- **Reverberation:** The persistence of sound in an auditorium is the result of repeated reflections of sound and is called reverberation.
- **Range of hearing:** The audible range of hearing for average human beings is in the frequency range of 20 Hz – 20 kHz.
- **Infrasonic and Ultrasonic:** Sound waves with frequencies below the audible range are termed "infrasonic" and those above the audible range are termed "ultrasonic".

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
9.1	Introduction and 9.1 Production of sound Activity 9.1		
9.2	ACTIVITY 9.2 Activity 9.3		
9.3	9.2 Propagation of sound		
9.4	9.2.1 Sound waves are longitudinal waves		
9.5	9.2.2 Characterestics of sound wave		
9.6	Explanation and example 9.1		
9.7	9.2.3 Speed of sound in different media 9.3 Reflection of sound Activity 9.5		
9.8	9.3.1 Echo 9.3.2 Reverberation		
9.9	9.3.3 Uses of multiple reflections of sound 9.4 Range of hearing		
9.10	9.5 Applications of ultrasound		

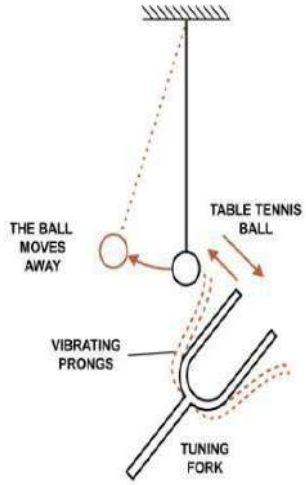
PERIOD Number: 9.1

TOPIC: Introduction : 9.1 Production of sound , Activity 9.1

Learning Outcome: Students will be able to explain the production of sound

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory activity: Take a bottle and fill bottle with some water and blow the open side of bottle with mouth, and observe the sound and take empty bottle and blow with mouth and observe the sound. **Teacher explains sound and its production**	Interaction with students to know their previous knowledge regarding to the Concept of the topic. **Activity**: Stand in front of student and clap your hand • You can heard the sound of clapping. • How do you hear that sound be cause sound travels in form of waves. <u>Example for sound object:</u> • When someone clicks the door bell and it produces the sound. Right however the sound reaches to you when they presses the bell, how does it travel to you.? • Sound can travel through solids, liquids and gases,	(Previous knowledge questions) • Which bottle made more sound empty bottle or half-filled bottle?

Students should observe, **How sound is produced from the tuning fork striking with a rubbr ball.** **By which means the SOUND will travel ?**	With this propagation sound is produced by vibrating objects. **Materials** 1. Tuning fork kit (includes tuning fork, ping- pong ball and string) 2. Tape 3. Ruler **What To Do** ● Cut a string about 1-1 ½ feet long and tape one end of it to the ping-pongball. ● Have one student hold the string away from her body. Without striking the tuning fork, move it toward the ping-pong ball. Ask students to discuss what happened. ● This time have students strike the tuning fork and move it slowly toward the ping-pongball. Discuss: what happens to the ball as the tuning fork gets closer?	 **What do you obseve from this ACTIVITY?**

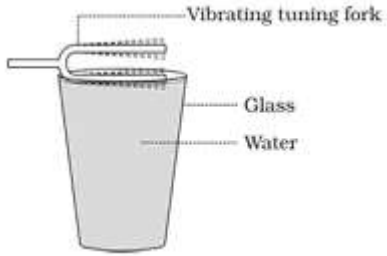
Summary	Factual Questions	CFU	Home Work
Students will be able to explain the production of sound	Are all vibrations can hear by our ear?	What do you mean sound? What is the speed of sound in air?	Explain how sound is produced?

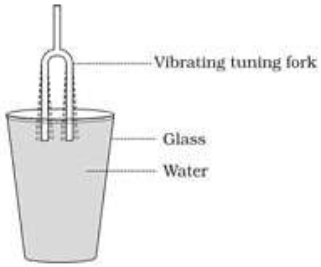
PERIOD Number: 9.2

TOPIC: ACTIVITY 9.2 , Activity 9.3

Learning Outcome: Students will be able to understand different ways of production of sound

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. **1.How do you produce SOUND?** **Activity 9.2**	1. Fill water in a beaker or a glass up to the brim. 2. Gently touch the water surface with one of the prongs of the vibrating tuning fork 3. Next dip the prongs of the vibrating tuning fork in water. 4. Observe what happens in both the cases. 5. Discuss with your friends why this happens.  	**Materials required:** 1.Beaker or glass 2. Water 3. Tuning fork **What are the observations do you observe during this activity?**

How do you observe change in by filling water into the bowl at different levels? **Is there any change observed in sound when placed tuning fork in harrizontally/ vertically to the**	 • In the above activities we have produced sound by striking the tuning fork. • We can also produce sound by plucking, scratching, rubbing, blowing or shaking different objects. • As per the above activities what do we do to the objects? • We set the objects vibrating and produce sound. • Vibration means a kind of rapid to and fro motion of an object. • The sound of the human voice is produced due to vibrations in the vocal cords. • A stretched rubber band when plucked vibrates and produces sound. • If you have never done this, then do it and observe the vibration of the stretched rubber band. Make a list of different types of musical instruments and discuss with your friends which	Explain , What happened in both cases? What type change in SOUND is observed when you put vibrating tuning fork horrizontally / vertically to the waret column?

	part of the instrument vibrates to produce sound.	

Summary	Factual Questions	CFU	Home Work
Students will be able to understand different ways of production of sound	Can a motion be periodic but not oscillatory?	1.Give examples of sound sources? 2. What is medium?	Explain sound is produced by a vibrating object?

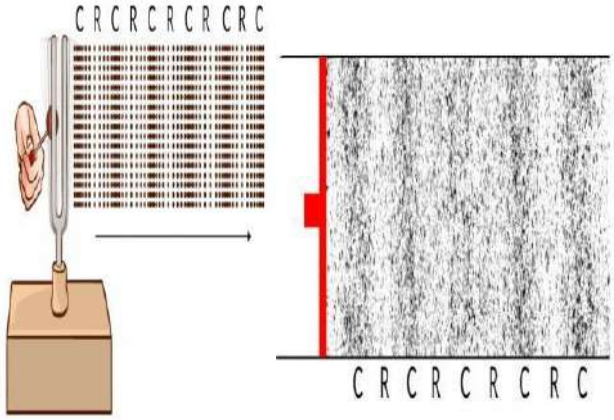
PERIOD Number: 9.3

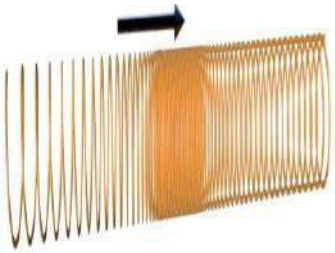
TOPIC: 9.2 Propagation of sound

Learning Outcome: Students will be able to explain the propagation of sound

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. How do the SOUND propagated through AIR? **Introductory activity:**	Sound starts from a vibrating object, propagates through a material medium and then reaches the listener's ear. While propagating through a medium the sound waves disturb the particles of the medium. In this section we will be looking at the way in which the particles are disturbed by the propagation of sound through them. Teacher explains how the sound propagates in a medium **Propagation of sound:**	(Previous knowledge questions) • What is sound? How is sound produced ? What do you know about vibration?





- What is a medium?
- How sound propagates in a medium?
- What are compressions and rare fractions of a wave?
- How density and pressure changes during the propagation of sound?

(Thus, propagation of sound can be visualized as the propagation of **density** or **pressure** variations in the medium.)

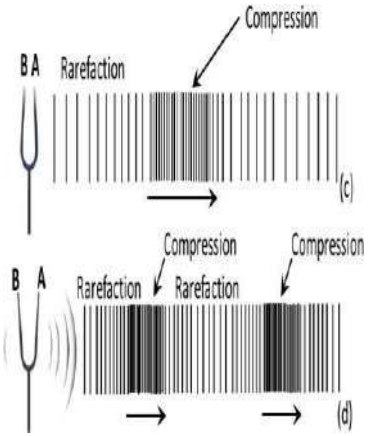
Activity:

The following activity may be conducted to visualize the way in which the sound propagates through a medium by disturbing the particles.

- For this we need a slinky and fix one of its ends to a rigid object such as a wall and its body rests on a table.
- Now, hold the free end of the slinky and push it forward.
- It will be observed that some of the turns of the slinky come closer together and

- Why are sound waves called mechanical waves?
- Suppose you and your friend are on the moon.will you be able to hear any sound produced by your friend?
- Sound is measured in decibels. The unit “bel” is named after scientist Graham Bell who invented the first telephone.

Humans can only hear sounds of frequencies between 20 Hz to 20kHz. However, animals can hear a sound higher and

What type of waves are SOUND Waves?	form a group. • This group of turns then moves forward towards the fixed end of the slinky. • The movement of the sound wave in the air takes place in a similar fashion. propagation of **density** or **pressure** variations in the medium.	lower than this frequency range 

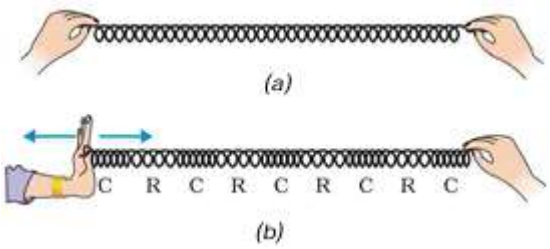
Summary	Factual Questions	CFU	Home Work
Students will be able to explain the propagation of sound	1. Why are sound waves called mechanical waves?	1. How does the sound produced by a vibrating object in a medium reach your ear? 2. Explain how sound is produced by your school bell.	1. Suppose you and your friend are on the moon. Will you be able to hear any sound produced by your friend?

PERIOD Number: 9.4

TOPIC: 9.2.1 Sound waves are longitudinal waves

Learning Outcome: Students will be able to give examples for longitudinal waves

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. What is COMPRESSION and RAREFRACTION? Introductory activity Activity 9.4  	A longitudinal wave is one where all the particles of the medium (such as gas, liquid or solid) vibrate in the same direction as the wave. Sound waves are longitudinal waves. When longitudinal waves travel through any given medium, they also include compressions and rarefactions. ✔ Take a slinky. ✔ Ask your friend to hold one end. ✔ You hold the other end. ✔ Now stretch the slinky. ✔ Then give it a sharp push towards your friend. ✔ What do you notice? ✔ If you move your hand pushing and pulling the slinky alternatively, what will you observe? ✔ If you mark a dot on the slinky, you will observe that the dot on the slinky will move back and forth parallel to the	(Previous knowledge questions) • The particles of the medium which is in contact with the vibrating object is first displaced from its equilibrium position. **Name the two types of mechanical waves.** ✔ The regions where the coils become closer are called compressions (C) ✔ The regions where the coils are further apart are called rarefactions (R). **Which of the following is true?** • The sound wave and the light

What is a LONGITUDINAL WAVE? **What are it's charectorstics?**	direction of the propagation of the disturbance. **sound propagates in the medium as a series of compressions and rarefactions. Now, we can compare the propagation of disturbance in a slinky with the sound propagation in the medium. These waves are called longitudinal waves**. In these waves the individual particles of the medium move in a direction parallel to the direction of propagation of the disturbance. The particles do not move from one place to another but they simply oscillate back and forth about their position of rest. This is exactly how a sound wave propagates, hence sound waves are longitudinal waves. ✔ There is also another type of wave, called a transverse wave. ✔ In a transverse wave particles do not oscillate along the direction of wave propagation but oscillate up and down about their mean position as the wave travels. ✔ Thus, a transverse wave is the one in which the individual particles of the medium move about their mean	waves both are longitudinal waves. ● Both the sound and light waves are transverse. ● In air, the sound waves are longitudinal as well as transverse. ● In air, the sound waves are longitudinal and the light waves are transverse.

	positions in a direction perpendicular to the direction of wave propagation	

Summary	Factual Questions	CFU	Home Work
Students will be able to give examples for longitudinal waves	What is transferred by wave motion from one place to another?	What are longitudinal waves? Define Compression and Rarefaction?	1. Describe with the help of a diagram, how compressions and rarefactions are produced in air near a source of sound. 2. Why is sound wave called a longitudinal wave?

PERIOD Number: 9.5

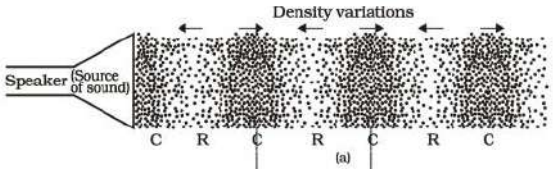
TOPIC: 9.2.2 Characteristics of sound wave

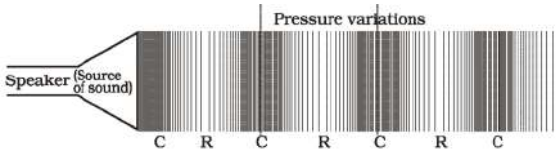
Learning Outcome: Students will be able to identify the relation between frequency and pitch of the sound

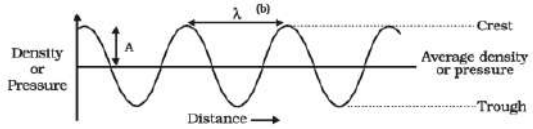
Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. What are the characteristics of SOUND wave?	• What type of wave is sound wave? • Observe the sound wave in graphic form A sound wave in graphic form is shown, which represents how density and pressure change when the sound wave moves in the medium.	

Shows different pictures







What is the wavelength of the sound wave?

Frequency

Represent the density and pressure variations, respectively, as a sound wave propagates in the medium.

Compressions are the regions where particles are crowded together and represented by the

upper portion of the curve

- The distance between two consecutive compressions (C) or two consecutive rarefactions (R) is called the wavelength.
- The wavelength is usually represented by O (Greek letter lambda).
- SI unit is metre (m)

✔ The number of complete oscillations per unit time is called the frequency.

✔ SI unit is Hertz (Hz)

✔ The SI unit of frequency was named as hertz in Heinrich Rudolph Hertz honour.

- The time taken by two consecutive compression s or rarefaction s to cross a fixed point is called the time period of the wave.

he peak represents the region of maximum compression.

Thus, compression s are regions where density as well as pressure is high.

Retractions are the regions of low pressure where particles are spread apart and are represented by the valley, that is, the lower portion of the curve.

Time period **Relation between Frequency and Time period** **What is pitch?**	• In other words, we can say that the time taken for one complete oscillation is called the time period of the sound wave. • It is represented by the symbol T. • Its SI unit is second (s). $$v = \frac{1}{T}$$ How the brain interprets the frequency of an emitted sound is called its pitch. The faster the vibration of the source, the higher is the frequency and the higher is the pitch. a high pitch sound corresponds to more number of compressions and rarefactions passing a fixed point per unit time.	A peak is called the crest and a valley is called the trough of a wave. What is FREEQUWNCY of the sound wave?

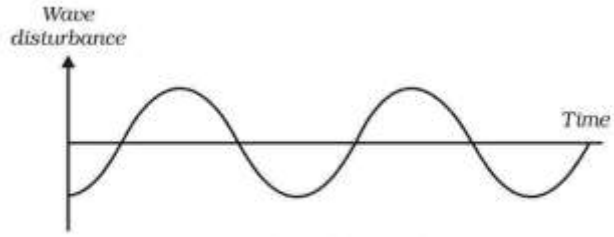
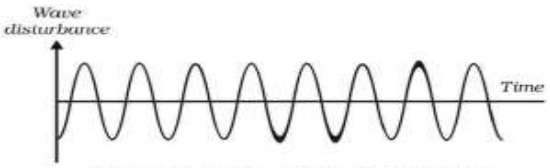
Summary	Factual Questions	CFU	Home Work
Students will be able to identify the relation between frequency and pitch of the sound	How are the wavelength and frequency of a sound wave related to its speed?	Which characteristic of the sound helps you to identify your friend by his voice while sitting with others in a dark room?	The frequency of a source of sound is 100 Hz. How many times does it vibrate in a minute?

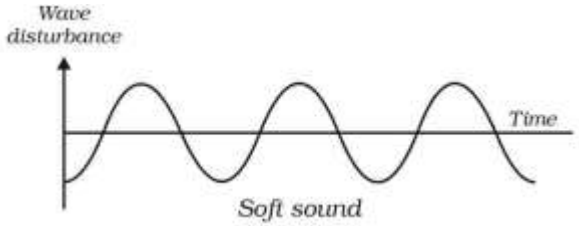
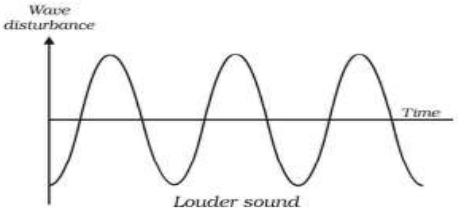
PERIOD Number: 9.6

TOPIC: Explanation and example 9.1

Learning Outcome: Students will be able to do calculation on frequency

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. **What is FREEQUand WAVELENGTH of a sound wave?** **Amplitude** The loudness or softness of a sound is determined basically by its amplitude. The amplitude of the sound wave depends upon the force with which an object is made	The magnitude of the maximum disturbance in the medium on either side of the mean value is called the amplitude of the wave. It is denoted by A The SI Unit is metre  *Wave shape for a low pitched sound*  *Wave shape for a high pitched sound*	**Whar are charectorstics of sound wave?**

to vibrate. **Tone**	  A sound wave spreads out from its source. As it moves away from the source its amplitude as well as its loudness decreases. The quality or timber of sound is that characteristic which enables us to distinguish one sound from another having the same pitch and loudness. The sound which is more pleasantis said to be of a rich quality. A sound of single frequency is called a tone. The sound which is produced due to a mixture of several frequencies is called a note and is pleasant to listen to. Noise is unpleasant to the ear! Music is pleasant to hear and is of rich quality.	Low pitch sound has low frequency high pitch of sound has high frequency **Soft sound** **Louder sound**

Note

Noise

What is Wavelength?

Relation between Speed of sound, Wavelength and frequency

Example 9.1

A sound wave has a frequency of 2 kHz and wave length 35 cm. How long will it take to travel 1.5 km?

The speed of sound is defined as the distance which a point on a wave, such as a compression or a rarefaction, travels per unit time.

speed, v = distance / time

$$= \frac{\lambda}{T}$$

$$v = \lambda \nu \left(\because \frac{1}{T} = \nu \right)$$

or $v = \lambda \nu$

Frequency, U = 2 kHz = 2000 Hz

Wavelength, λ = 35 cm = 0.35 m

The amount of sound energy passing each second through unit area is called the intensity of sound.

Loudness is a measure of the response of the ear to the sound. Even when two sounds are of equal intensity, we may hear one as louder than the other simply because our ear detects it better.

speed = wavelength u frequency.

We know that speed, v of the wave
= wavelength × frequency

$v = \lambda \nu$

= 0.35 m × 2000 Hz = 700 m/s

The time taken by the wave to travel a distance, d of 1.5 km is

$$t = \frac{d}{v} = \frac{1.5 \times 1000 \text{ m}}{700 \text{ m s}^{-1}} = \frac{15}{7} \text{ s} = 2.1 \text{ s}.$$

Thus sound will take 2.1 s to travel a distance of 1.5 km.

Summary	Factual Questions	CFU	Home Work
Students will be able to do calculation on frequency	Does the speed of sound through a medium depend on frequency?	1. Which wave property determines (a) loudness (b) pitch? 2. What are wavelength, frequency, time period and amplitude of a sound wave?	Calculate the wavelength of a sound wave whose frequency is 220 Hz and speed is 440 m/s in a given medium

PERIOD Number: 9.7

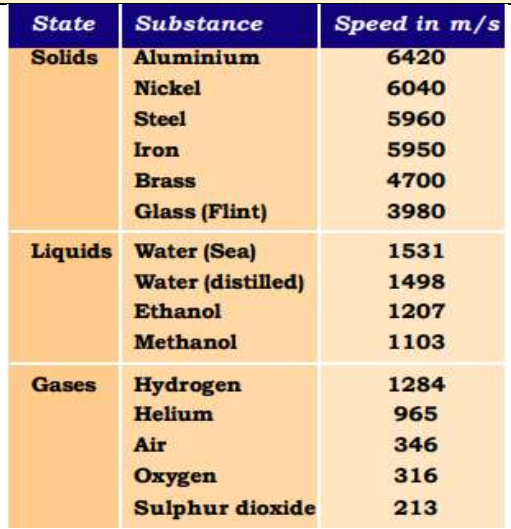
TOPIC: 9.2.3 Speed of sound in different media

9.3 Reflection of sound

Activity 9.5

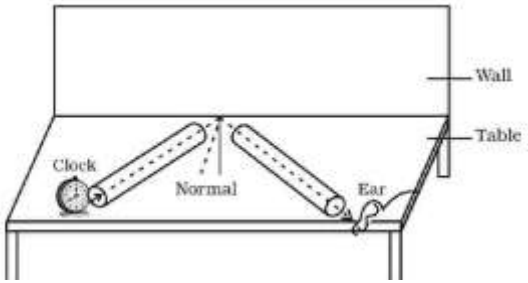
Learning Outcome: Students will be able to understand that sound is different in different media

Teaching Learning Process

I DO	WE DO	YOU DO
Sound propagates through a medium at a finite speed.The sound of a thunder is heard a little later than the flash of light is seen. So, we can make out that sound travels with a speed which is much less than the speed of light. The speed of sound depends on the properties of the medium and temperature. For example, the speed of sound in air is 331 m s^{-1} at 0ºC and 344 m s^{-1} at 22 ºC.		What type of Sound waves are ?

State	Substance	Speed in m/s
Solids	Aluminium	6420
	Nickel	6040
	Steel	5960
	Iron	5950
	Brass	4700
	Glass (Flint)	3980
Liquids	Water (Sea)	1531
	Water (distilled)	1498
	Ethanol	1207
	Methanol	1103
Gases	Hydrogen	1284
	Helium	965
	Air	346
	Oxygen	316
	Sulphur dioxide	213

Reflection of Sound: **Activity 9.5**	Sound bounces off a solid or a liquid like a rubber ball bounces off a wall. Like light, sound gets reflected at the surface of a solid or liquid and follows the same laws of reflection. The directions in which the sound is incident and is reflected make equal angles with the normal to the reflecting surface at the point of incidence, and the three are in the same plane. Activity 9.5: ● Today we are going to discuss about activity on Reflection of Sound. To understand this concept of Reflection of Sound this activity is very useful, ● Take two identical pipes, as shown in figure. ● You can make the pipes using chart paper. ● The lenth of the pipes should be sufficiently long. ● Arrange them on a table near a wall. ● Keep a clock near the open end of one of the pipes and try to hear the sound of the clock through the other pipe. ● Adjust the position of the pipes so that you can best hear the sound of clock. ● Now measure the angles of incidence and reflection and see the relationship between the angles. ● Lift the pipe on the right vertically to a	● What is the apparatus taken ? (Two identical pipes made up of chart paper, table clock, ptotractor.) How one of the pipe is arranged? (Making 30^0 with the Normal) ● What is the angle of reflection for the angle of incidence of



What do you observe from this activity?

Is sound waves reflected?

small height and observe what happens.

- In place of clock, a mobile phone on vibrating mode may also be used.
- Let us arrange the apparatus to do activity on Reflection of Sound. (15 min)
- Arrange two identical plastic pipes of length approximately 1m and of diameter 10 cm on atable.
- Using a chalk piece draw a line ON on the table.
- Now draw a line OC making an angle 30^0 with the ON.
- Put one of the plastic pipes along this line all OC such that the end P of the olastic pipe is very clse to the point O on the wall.
- Now put the second plastic pipe RS on the table, keeping its end R towards the wall on the other side of the normal ON.
- Keep the table clock close to the open end of pipe PQ.
- Adjust the the position of pipe RS on the table to hear the sound of the table clock.
- Mark the position of end S of the pipe RS where you can hear Maximum Sound.

30^0? (Angle of Reflection is also 30^0)

- What are the approximate length of pipes that are taken for the experiment?
- What is the angle made by the Normal with the wall?
- Where we have to fix the second pipe in the experiment?
- What is meant by angle of Incidence?
- What is meant by angle of Reflection?
- What is the relation between

All the LAWS OF REFLECTION are apllied to the SOUND waves?	• We can prove the following laws of Reflection of Sound. • (i) The incident sound wave and the reflected sound wave and Normal at the point of incidence lie in the same plane. • (ii) Angle of reflection of sound is always equal to the angle of incidence of sound.	angle of incidence and angle of Reflection? • How will be the Laws of Reflection of sound in solids and liquids?

Summary	Factual Questions	CFU	Home Work
Students will be able to understand that sound is different in different media	Flash and thunder are produced simultaneously. But thunder is heard a few seconds after the flash is seen, why?	In which of the three media, air, water or iron, does sound travel the fastest at a particular temperature?	1. Does sound follow the same laws of reflection as light does? Explain. 2. Give two practical applications of reflection of sound waves

PERIOD Number: 9.8

TOPIC: 9.3.1 Echo , 9.3.2 Reverberation

Learning Outcome: Students will be able to differentiate echo and reverberation

Teaching Learning Process

I DO	WE DO	YOU DO
Introductory activity: • When you shout near a tall building then what will you hear?	activity: • Today we are going to discuss about Reflected Sound which is known as Echo.	(previous knowledge questions) • When you make a sound in a empty room then what will you hear?

	To understand this concept I wil give some more examples. ● What happens when we shout our names in a Well? ● Echoes heard due to reflection of Sound. ● From the near by reflecting surface. ● From the surface of water in a well. ● From the surface of a mountain or cliff ● From far off building ● Let us discuss about persistence of hearing and Echo. ● The effect of sound in the ear should last for about 0.1 seconds to identify. If another sound is received with in this time, it cannot be distinguished by the ear. ● Echo is produced when sound is reflected from a surface plane or curved. ● The sound produced by the source is heard by listener by two ways. ● The first is due to direct sound wave sand the due to reflected waves.	● When you clap in between the two mountains then what will you hear? Why the echoes are heard? ● What is the working phenomena in resound? ● Name the two incidents where the echoes are formed?
● When you clap near a Mountain then what will you hear? Teacher's statement: This sound is called an Echo. A reflected sound arriving at the listener is called an Echo.		

Say any two DAILY LIFE OBSERVATIONS based on Reflection on SOUND?	• So to hear a distinct echo the time interval between the original sound and the reflected sound must be atleast 0.1 seconds. • If we take speed of sound to be 344m/s at a given temperature 22^0C in air. • The sound must go to the obstacle and reach back the ear of the listener on reflection after 0.1 seconds. • Hence the total distance covered by the sound from the point of generation to the reflecting surface and back should be atleast 34.4m • Hence for hearing distinct echoes, the minimum distance of the obstacle from the source of sound must be 17.2m • The rolling thunder is due to successive reflections of the sound from a number of reflecting surfaces. • What is the minimum distance of the obstacle to hear an echo? Introductory activity: • When you shout near a tall building then what will you hear?	• What is meant by persistence of hearing? • What is the value of persistence of hearing? • What are the conditions for echoes to take place? • What is the velocity of sound in air? • What is the minimum distance of the obstacle to hear an echo ? • Why the rolling thunder occurs?

What is an ECHO? Teacher's activity: ● Today we are going to discuss about Reverberation. To understand this concept I will give some more examples. ● What happens when we shout our names in a Well?	● When you clap near a Mountain then what will you hear? ● Repeated reflection of sound from the walls until it is reduced to a value where it is no longer audible. ● The repeated reflection of sound that results in the persistence of sound is called Reverberation. ● In an auditorium or big hall excessive reverberation is highly undesirable. ● ● Let us discuss about how to reduce Reverberation. ● In an auditorium or big hall excessive reverberation is highly undesirable. ● In an auditorium, cinema theatres and buildings the speech and music should be	(previous knowledge questions) ● When you make a sound in a empty room then what will you hear? ● When you clap in between the two mountains then what will you hear? ● Why the reverberation takes place? ● What is the working phenomena of ReSound in Mic set? ● Name the two incidents where the reverberations are formed?

Teacher's statement: Even though the source of sound is turned off, the listener hears it for sometime. This is called Reverberation.	heard clearly. ● To reduce reverberation the enclosure should be designed with absorbing materials like wood, Carpet, card-board, mats, etc at the side walls, top and bottom. Materials like compressed fibreboard, rough plaster or draperies are also used. ● The seat materials are also selected on the basis of their sound absorbing properties. ● Sound produced by footsteps of walking people can be reduced by the use of matting on the floor . ● The distribution of sound energy should be uniform in the room. For that the wall at the back of speaker should have parabolic reflectors so that large audience can hear the sound. ● What is meant by Reverberation?	● Which material is used to reduce Reverberation? ● What are the examples for sound absorbent materials ? ● What are the conditions for reverberation to take place? ● Which phenomena plays important role in the construction of good auditoria?

Summary	Factual Questions	CFU	Home Work
Students will be able to differentiate echo and	What causes the rolling	1. What is an echo?	1. An echo is heard in 3 s. What is the distance of the

reverberation	sound of thunder?	2. What is reverberation? 3. Distinguish between echo and reverberation?	reflecting surface from the source, given that the speed of sound is 342 m s^{-1}? 2. What is reverberation? How can it be reduced

PERIOD Number: 9.9

TOPIC: 9.3.3 Uses of multiple reflections of sound, 9.4 Range of hearing

Learning Outcome: Students will be able to understand that range of hearing is different for humans and animals

Teaching Learning Process

I DO	WE DO	YOU DO
image on IFP 	Introductory activity: ● When a ball hits the wall then what will happen? ● When light falls on mirror then what happens? Teacher's activity: ● Today we are going to discuss about Uses of Mulitiple Reflections of Sound. To understand this concept, I will give some	Is SOUND Reflection possible?

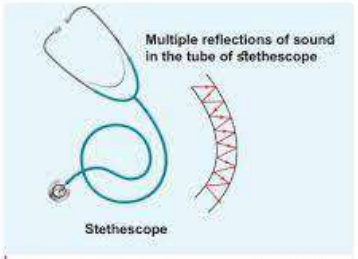
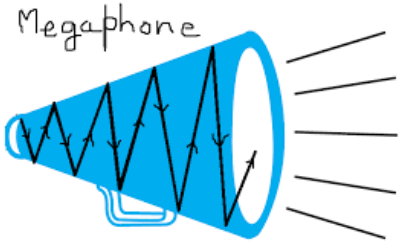


Figure 11.22 Stethoscope and multiple reflection of signal in a rubber tube



Teacher's statement: Megaphones Loudhailers, horns, musical instruments such as trumpets and shanghais are designed to send sound in a particular direction without spreading it in all directions.

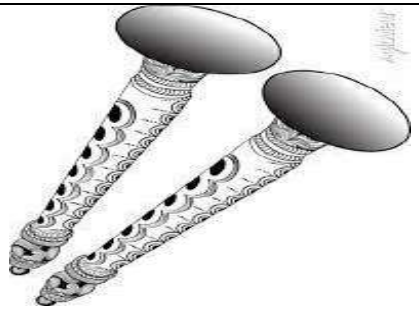


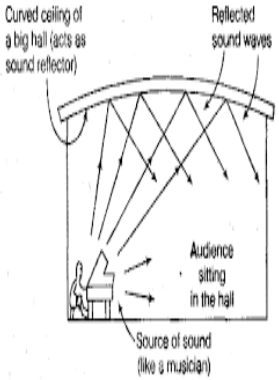
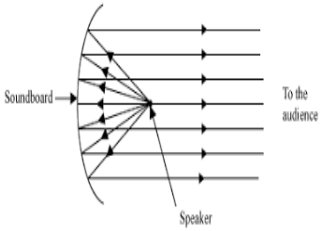
more examples.

- When you are not feeling well with breathing problem then with instrument the doctor will check?
- Let's watch this picture on IFP and give response.
-
- (Teacher's statement:Stethoscope is a medical instrument used for listening to sounds produced with in the body, mainly in the heart or lungs. In stethoscopes the sound of the patient's heartbeat reaches the doctor's ears by multiple reflections of Sound.)
  - Sometimes in mini public gatherings we wil use Megaphone…..
  - What is the shape of the Megaphone?
  - What is the use of Megaphone?
  - (Student response: To amplify person's voice)
  - Stethascope and Megaphone works depending upon the property of sound i.e

Write some DAILY LIFE APPLICATIONS of Sound Reflection?

How SOUND is propagated through a STHETHASCOPE?

 	Reflection of Sound. Let us discuss about Use of Reflection of Sound in the construction of good auditoria. To understand this concept, I will give some more examples. ● Where will you see a movie? ● How the ceilingsof cinema halls constructed? ● (Teacher's statement:Generally the ceilings of concreate halls, conference halls and cinema halls are curved so that sound after reflection reaches all corners of the hall. Sometimes a curved sound board may be placed behind the stage so that the sound after after reflecting from the sound board, spreads evenly across the width of the hall.) ● What are the applications of Reflection of Sound? ● How the ceilings of concert halls, conference halls and cinema halls are constructed?	How Sound is passed through , the pipes and organs when we blow them?

 Curved sound board behind the speaker or the stage in big hall 	(student's response: The ceilings are constructed in curved shape such that reflected sound reaches all the corners of the hall.) ● Reflection of sound plays important role in the working of Stethoscope, Megaphone and in designing concert halls. ● Can human beings hear all the sounds? ● What frequencies of sound can human being hear? ● What are audible range of frequencies? ● What is frequency of Ultra sonic sounds? ● is the frequency of infrasonic sounds greater or lesser than ultra sonic sounds? ● The human beings can hear sounds of frequencies from 20 Hz to 20000 Hz. This is called Audible range. ● One Hz = one cycle/second ● 1kHz = 1000Hz ● Children under the age of five and some animals like dogs can hear upto 25 kHz. ● Let's watch this picture on IFP and give response. ● Sounds of frequencies below 20 Hz are called Infrasonic s or Infra sound.	Differentiate the terms - ECHO and REVERBERRATION? How to reduce ECHO / REVERBERRATIONS? What is an AUDIABLE RANGE of sound to

What are ULTRASONICS and INFRASONICS?	• Whales and Elephants produce sound in the Infrared Range. • Earthquakes produce low frequency Infra sound. • Rhinoceroses communicate using Infra sound of frequency 5Hz. • Frequencies higher than 20kHz are called Ultrasonic sound. • Ultrasound is produced by animals such as Dolphins, Bats and Porpoises. • Rats also play games by producing Ultrasound.	HUMANS? What is ULTRASONICS? What is INFRASONICS? What are their ranges?

Summary	Factual Questions	CFU	Home Work
Students will be able to understand that range of hearing is different for humans and animals	Why do the animals behaves in a peculiar manner before an earthquake?	1. Give two practical applications of reflection of sound waves. 2. What is the audible of frequency?	1. A sound wave travels at a speed of 339 m s^{-1}. If its wavelength is 1.5 cm, what is the frequency of the wave? Will it be audible? 2. What is the range of frequencies associated with (a) Infrasound? (b) Ultrasound?

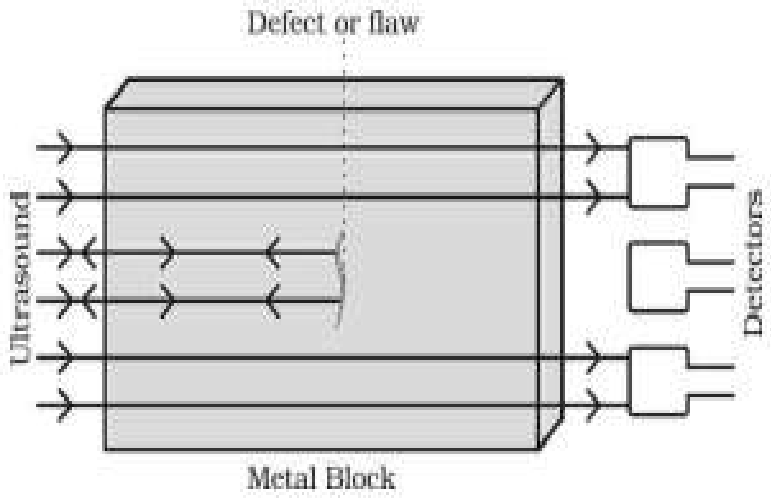
PERIOD Number: 9.10

TOPIC: 9.5 Applications of ultrasound

Learning Outcome: Students will be able to applies ultrasounds in day to day life.

Teaching Learning Process

I DO	WE DO	YOU DO
Interaction with students to know their previous knowledge regarding to the Concept of the topic. What is E.C.G, ULTRA SOUND SCANNING? Let us discuss about Applications of Ultrasounds.	Ultra sounds are high frequency waves. • Ultrasounds are used to clean spiral tubes, odd shaped parts, electronic components etc. Due to the high frequency, the particles of dust, grease and dirt get detached and drop out. • . Ultrasounds can be used to detect cracks and flaws in metal blocks, in big structures like buildings, bridges, machines and also scientific equipment. • Ultrasonic waves are made to reflect from various parts of the heart and form the image of the heart. This technique is called Echocardiography. • Ultrasound scanner is an instrument which uses ultrasonic waves for getting images of internal organs of the human body. • Ultra sound scanner is used to form the image of patients organs such as the liver, gall bladder, uterus, kidney etc .	• Which waves are used in echo cardiography? • What is the use of Ultra sound scanning? • What is the use of ultrasonography?



- What is the Audible range?
- What are Infrasonic?
- What are the uses of Ultrasonic?
- What are the uses of Ultrasounds scanning?
- Ultrasounds plays important role in taking the electrocardiography, electro sonography.

- What will happen when a Sound wave strikes a hard surface?
- What is the importance of these E.C.G , and ULTRA SOUND SCANNING in DIAGNOSING of probmem of a patient in theMEDICAL field?

Summary	Factual Questions	CFU	Home Work
Students will be able to applies ultrasounds in day to day life.	Why cannot ordinary sound be used in detection of defects or flaws in a metallic block?	1. What is ultrasound? 2. Explain some important applications of ultrasound in industry and medicine.	3. How is ultrasound used for cleaning? 4. Explain how defects in a metal block can be detected using ultrasound

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson: 9. Sound		Class: 9	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
9.1	Introduction and 9.1 Production of sound Activity 9.1				
9.2	Activity 9.2 Activity 9.3				
9.3	9.2 Propagation of sound				
9.4	9.2.1 Sound waves are longitudinal waves				
9.5	9.2.2 Characteristics of a sound wave				
9.6	Explanation and example 9.1				
9.7	9.2.3 Speed of sound in different media 9.3 Reflection of sound Activity 9.5				

9.8	9.3.1 Echo 9.3.2 Reverberation				
9.9	9.3.3 Uses of multiple reflections of sound 9.4 Range of hearing				
9. 10	**9.5 Applications of ultrasound**				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Head Teacher's Signature Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

TLM Required

Activity Number	Activity Name	Required Material	Required Chemical
CHAPTER -1 : Motion			
1.1	Rest and Motion	Discussion with students	
1.2	experienced that the train in which you are sitting appears to move while it is at rest?	Discussion with students	
1.3	Measure the distance and magnitude of displacement	Meter scale , Rope	

1.4	Speedometer	Discussion on speedometer	
1.5	Analysis the motion of object	Information given in the table 7.1	
1.6	Average speed	Assessment of speed	
1.7	Difference between speed of lightning and thunder	Discussion with students	
1.8	Terminology in motion of a body	Discussion on acceleration	
1.9	Plotting distance – time graph	Information given in table	
1.10	Plotting distance – time graph	Information given in table	
CHAPTER- 2: MATTER AROUND US			
2.1	Nature of matter – Is it Continuous or particulate?	100 mL beaker Marker , Glass road / starrer	Sugar / salt Water
2.2	How small are these Particles of a matter?	100 mL beakers - 4 , Test tube – 1 Dropper	Potassium Permanganate
2.3	Are particles of matter are continuously Moving?	Incense stick , Match box	
2.4	Observe the diffusion property of matter	Beakers -2 Blue / Red ink , Dropper	Water Honey
2.5	Observe the motion of particles	Beakers – 2	Potassium permanganate Hot Water , Cold water
2.6	Play this game in the field— make four groups and form human chains as suggested:	No need – Group activity	
2.7	Particles have force of attraction between them in a solid -1	Iron nail, Piece of chalk Rubber band	

2.8	Particles have force of attraction between them in a solid -2	Beaker / bucket	Water
2.9	Particles have force of attraction between them in a solid -3	Pen , book . wood , etc.	
2.10	Shape and volume of liquid	Different shapes of container	Water , Oil , Milk , Cold drink
2.11	observe the compressibility nature of solids , liquids & gases	100 mL syringes – 3	Sand Water
2.12	Find out the malting pint of ice / Observe the phase change of matter	Bunsen burner Tripod , Wire gauge , Stand , Clamp Thermometer , Glass starrer	Ice
2.13	observe the sublimation process	Bunsen burner , Tripod , Wire gauge China dish , Inverted funnel , Cotton plug	Ammonium Chloride (NH4Cl)
2.14	Observe the factors affecting rate of evaporation	China dish, Dropper	Water
CHAPTER – 3 : LAWS OF MOTION			
3.1	Observe static inertia	Carom table , Coins	
3.2	Observe static inertia	Glass (beaker) , Cardboard Coin (1rupee / 5 rupee)	
3.3	Observe static inertia	Tumbler , Tray	
3.4	Newton third law	Carts (Rolling) -2	
CHAPTER- 4 : IS MATTER AROUND IS PURE ?			
4.1	Types of mixtures	250 mL beakers -4 Spatula	Copper sulphate Potassium permanganate

4.2	observe solution, suspension and colloids	250 mL beakers -4 , Spatula Glass rod , Torch light / mobile flash Filter paper	Copper sulphate crystals Chalk powder, Ink /milk Water
4.3	solubilities of a solution vs temperature	250 mL beakers -2 Bunsen Burner , Lighter / match box Wire gauge , Thermometer	Salt , Sugar Water , Barium chloride
4.4		Bar magnet , China dish Bunsen Burner , Lighter / match box Wire gauge	Sulphur powder Iron fillings
CHAPTER – 5: GRAVITATION			
5.1	A stone describing a circular path with a velocity of constant magnitude	Thick thread , Small stone	
5.2	Thrown upward body	Stone / ball / any object	
5.3	Motion of object under influence of Gravitational force	Stone , White paper	
5.4	Influence of Gravitational force	Plastic bottle Plastic bucket	Water (to fill bucket)
5.5	Water upward pressure	Beaker (500 mL) Iron nail , Cork	Water
5.6	Why some bodies floats and some bodies sinks ?	Beaker Cork & Iron nail (same mass)	Water
5.7	Verification of Archimedes principal	Stone , Thread , Beaker Spring balance	Water

CHAPTER – 6 : ATOMS AND MOLECULES			
6.1	verification of law of conservation of mass	Conical flask Rubber cork Test tubes – 2 Ignition tube (glass tube) Thread	Copper sulphate Sodium carbonate Barium chloride Sodium sulfate Lead nitrate Sodium chloride
6.2	Ratio of atoms in a molecule	No need (Mental activity)	
CHAPTER -7 : WORK AND ENERGY			
7.1	Not much work in spite of work hard	Discussion on work (general & scientific meaning)	
7.2	Work done	Discussion on daily life work done	
7.3	Work done – object not displaced	Discussion on daily life work not done	
7.4	Positive and negative work	Discussion on positive & negative work	
7.5	Sources of Energy	List out energy sources	
7.6	Kinetic energy	Heavy ball Bed / sand	
7.7	A moving object can do work	Trolly Wooden block (known mass) Pan	
7.8	Potential Energy -1	Rubber band	
7.9	Potential energy – 2	Slinky (long spring)	

7.10	Potential Energy - 3	Toy car (should have key)	
7.11	Potential Energy - 4	Heavy object	
7.12	Arrow and stretched string	Bamboo stick (bow) Arrow	
7.13	Energy conversation	Discussion on energy conversion	
7.14	Collecting gadgets – energy Conservation	Identification of energy conversion gadgets	
7.15	Verification of Law of energy Conservation	Heavy objects	
7.16	Rate of doing work	Discussion on power	
7.17	Estimate Current bill	Students performed at home	
CHAPTER -08 : STRUCTURE OF ATOM			
8.1	Charged Particle's in matter	Comb, Glass rod , Silk cloth , Ballon	
8.2	Atomic model preparation	No need(students work at home)	
CHAPTER -09 : SOUND			
9.1	Production of sound	Tuning fork , Thread Light weight ball , stand	
9.2	Vibrating tuning force touching the water surface	Glass / beaker, Tuning fork	Water
9.3	Musical Instruments	Different musical instruments	
9.4	Longitudinal Waves	Slinky (long spring)	
9.5	Reflection of Sound	Two identical pipes , Clock / stop watch	